AI Infrastructure - China (H/A)

China AI Infrastructure Handbook: Materials, Power & Equipment

Industry Overview

China's advantages in US$1.4tn AI buildout during 15FYP

Our global team expects the AIDC system TAM to grow from US$278bn in 2025 to US$2.2tn by 2030. We expect China's AI capex to rise from US$110bn in 2025 to US$350bn in 2030. Total capex during 15FYP could reach US$1.4tn, funded by internet firms, teleco, and govt support. China's advantages in global AI buildout: 1) dominant role in AI materials, producing ~50% of global refined copper, 70% of rare earth/indium; and 50% of drill bits; 2) ample power with 1/3rd of global power supply and c.50% of solar, wind & ESS capacity; 3) power tariffs 30-60% lower vs. US/Europe; 4) a younger grid, below 20 years vs. US/EU's 40 years; 5) an integrated industrial base that enables AI equipment suppliers to expand capacity rapidly following technology breakthrough. This handbook presents key charts, themes, FAQs on AI materials, power and equipment.

Materials: Copper, copper foil, E-glass, InP, optical fiber…

Copper: Copper use in China data centers may rise from 340kt in '25 (2.1% of demand) to 1.3mnt by 2030 (6.8%), led by power delivery & interconnects (Exhibit18). With supply constrained, copper is our top pick. **PCB material**: Upgrades and limited scope to switch capacity between products, due to equipment bottleneck, should lead to high-end copper foil shortage (400-600t in YE26) & glass fiber, supporting price hike. **Optical materials**: Upgrades to 1.6T/3.2T and likely CPO/NPO adoption long-term may lift optical intensity and drive a surge in demand for InP (critical material for lasers in OT) and optical fiber. A key China supplier expects fiber demand to double in '27. We also discuss AI applications for MLCC, specialty gases, tungsten and magnets. Stocks: Zijin, CMOC and JXC for copper; Defu for copper foil; Sinoma/Jushi for glass fiber; Yunnan Germanium for InP.

Global AIDC power: from 2025's 100GW to 2030's 316GW

We expect global DC capacity additions to grow from 13GW in 2025 to 56GW by 2030; leading to cumulative DC power at 316GW (100GW in 2025) with rising rack level density and per chip consumption. By 2030, AIDC is expected to a/c for 4%/2.5% of global/China power demand. **Power** is likely to a/c for 10-15% of China's AI capex during 15FYP. Themes: 1) RMB4tn of China **grid** capex during 15FYP (7% CAGR); 2) tight transformer supply in FY26-28E & continued export demand; 3) **gas turbine** tight &diesel engines as backup power; 4) ~20% CAGR in BESS installation, 27% CAGR in power supply; 5) 25% CAGR in China cooling. Buy Nari/Sieyuan/Jereh/Weichai/Megmeet.

AI equipment: cherry picking

China is not yet a leader in AI equipment, but some areas are developing rapidly. We forecast **OT equipment TAM to reach RMB55bn** by 2028 (67% CAGR), given 1) OT shipment CAGR of 46% in 2025-30E; 2) ASP hikes amid upgrades to 1.6T/3.2T. We expect CPO testing equipment demand to surge in 2027-28. Buy RoboTechnik, a global leader in OT & CPO equipment. PCB equipment and drill bits: HLC and HDI PCBs used in AI are driving demand for laser-drilling equipment and consumable drill bits. Buy Dtech.

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Refer to important disclosures on page 77 to 80. Analyst Certification on page 69. Price Objective Basis/Risk on page 62.

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Exhibit 1: Key stocks
We present the key stocks involved below

Stock	Subsector	27E PE
AI Materials		
Zijin	Cooper	8x
Jiangxi Copper	Copper	6x
CMOC	Cooper	7x
Defu Tech	Copper foil	25x
Sinoma S&T	Glass fiber	15x
Jushi	Glass fiber	11x
Yunnan Germanium	Germanium	55x
Powering AI		
NARI	Grid	19x
Sieyuan	Transformer	16x
Jereh	Gas turbine	14x
Weichai	Diesel engine	9x
Megmeet	Power supply	21x
AI equipment		
RoboTechnik	Optical Equipment	43x
Dtech	Drill bit	20x

Source: BofA Global Research

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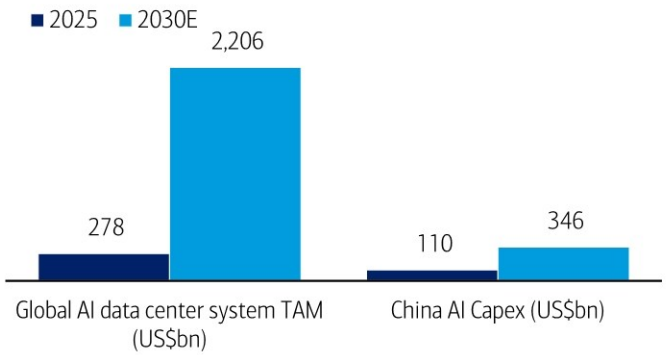
FYP: Five-year plan/ DC: data center/ PCB: Printed circuit board/ CPO: Co-Packaged optics/ MLCC: Multilayer ceramic capacitor/ InP: Indium phosphide/ TAM: Total addressable market/ OT: Optical transceivers

Key charts

Exhibit 2: Global vs. China AI Spending
Our global team estimates global AIDC system TAM to reach US$2.2tn in 2030E and we see China AI Capex to reach US$350bn in 2030E.



Source: Company data, BofA Global Research estimates

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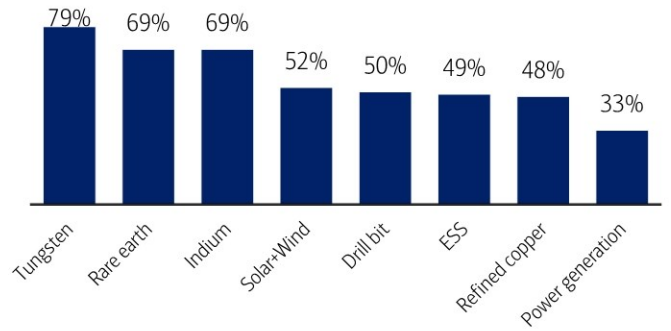
Exhibit 3: China's dominant role in AI materials & power
China holds a dominant role in tungsten, rare earth, indium, renewable energy, PCB drill bit, ESS, refined copper and power generation.



Note: Materials: China's output as % of global in 2025; Power: Cumulative installation for solar, wind, ESS as of end 2025. 2025 annual power generation for power generation
Source: USBS, China NEA, Global Wind Energy Council, IEA, IRENE, BofA Global Research

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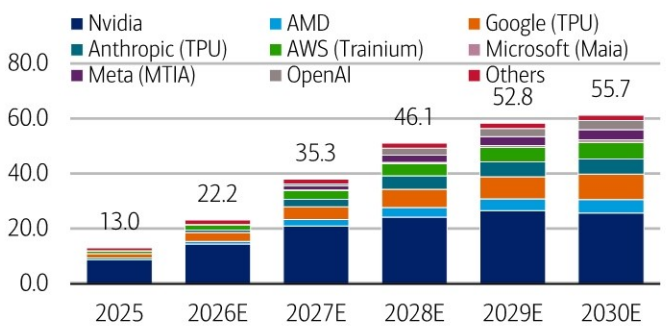
Exhibit 4: Bottom-up GW analysis: ~212GW incremental capacity from 2026 to 2030
Estimated annual GW data center deployment based on bottom-up XPU vendor sales



Source: Company data, BofA Global Research

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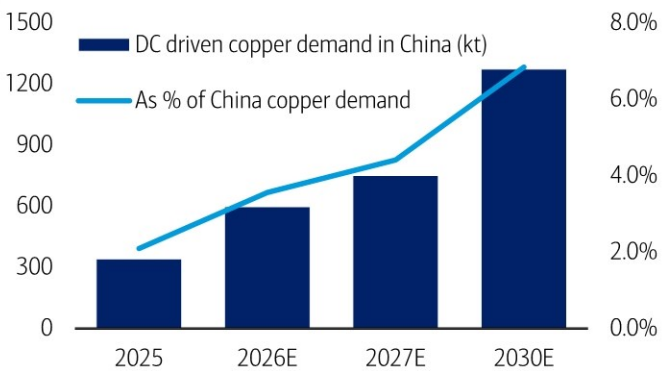
Exhibit 5: DC-driven copper demand to see 30% CAGR in 2025-30E
China DC-driven copper demand (kt)



Source: SMM, Wind, BofA Global Research

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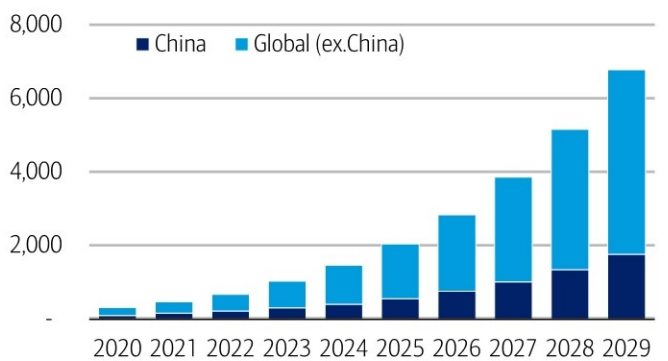
Exhibit 6: AI driven PCB copper foil market size forecast (RMBmn)
Frost & Sullivan forecast global AI PCB copper foil market to grow 36% CAGR in 2024-29E, and China market to grow 35% CAGR



Source: Frost & Sullivan, BofA Global Research

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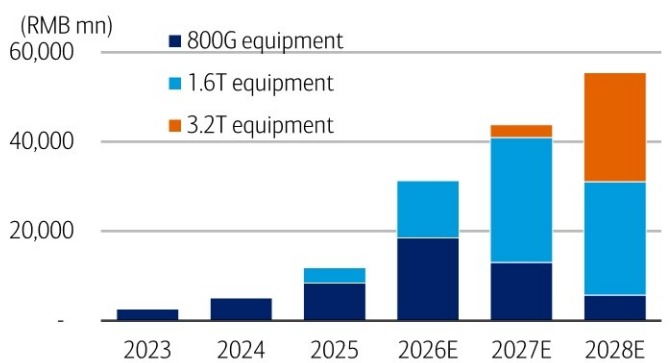
Exhibit 7: Optical transceiver equipment market size to reach RMB30bn+ in 2026E, with 67% CAGR over 2025-28E
TAM of optical transceiver manufacturing equipment.



Source: BofA Global Research estimates

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Exhibit 8: What materials feed into AIDC
We present key AI materials companies across metals, PCB materials, thermal materials, optical materials, and electronic/semiconductor materials.

Source: BofA Global Research

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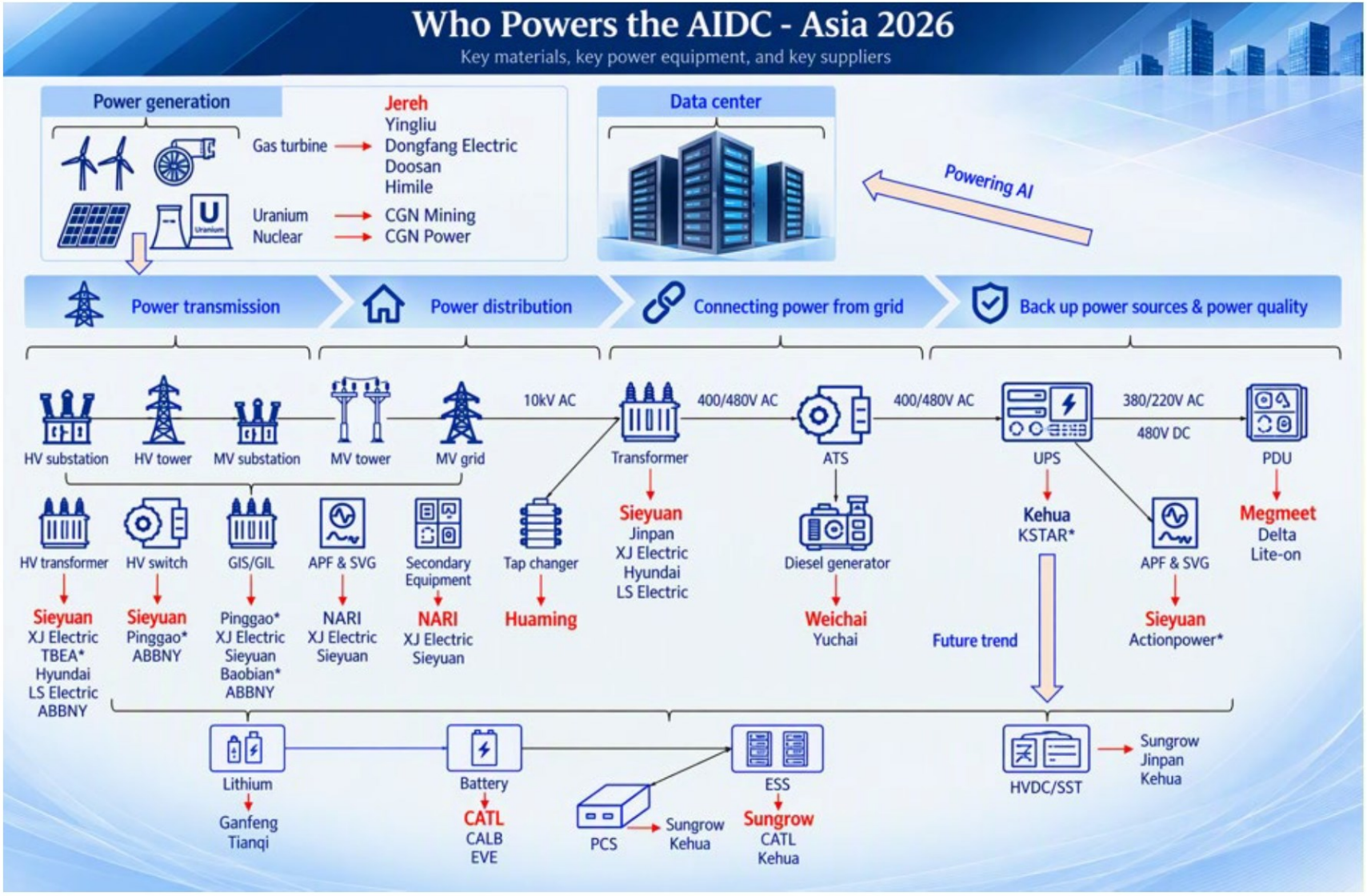
Exhibit 9: Who powers AIDC
Key materials, key power equipment, and suppliers for AIDC



Source: BofA Global Research; Note: Companies highlighted in red are our picks; * companies with * are not covered; ATS: automated transforming switch; UPS: uninterruptible power system; APF: active power filter; AC: alternative current; DC: direct current; MV: medium voltage; HV: high voltage; GIS: gas insulated switchgear; GIL: gas insulated line; ESS: energy storage system; HVDC: high voltage direct current; SST :solid-state transformers

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Stock summary

Exhibit 10: Stock summary

We present stocks involved in the report below, including AI power, materials and machinery.

Company	Subsector	BBG Ticker	Share price (LCY)	Mkt cap US$M	PE (x) 27E	PE (x) 28E	PB (x) 27E	PB (x) 28E	EV/EBITDA (x) 27E	EV/EBITDA (x) 28E	ROE % 27E	EPS CAGR 26-28E	PEG 27E
Materials													
Zijin Mining	Copper	2899 HK	31	112,850	8x	9x	2.7x	2.3x	6x	6x	31%	4%	2.12
CMOC Group Ltd	Copper	3993 HK	14	44,365	7x	7x	2.2x	1.9x	4x	4x	29%	13%	0.54
Jiangxi Copper	Copper	358 HK	30	17,190	6x	6x	1.0x	1.0x	4x	4x	16%	2%	2.41
MMG Limited	Copper	1208 HK	8	12,285	6x	6x	11.5x	9.2x	3x	3x	27%	12%	0.53
Defu Tech	Copper foil	301511 CH	77	7,267	24x	18x	8.3x	6.4x	17x	13x	38%	88%	0.28
Sinoma Sci & Tech	Glass fiber	002080 CH	42	11,016	15x	13x	2.8x	2.5x	10x	9x	20%	30%	0.50
China Jushi	Glass fiber	600176 CH	35	20,774	11x	10x	2.9x	2.5x	8x	7x	30%	35%	0.31
JL Mag	Magnet	6680 HK	15	4,571	13x	11x	2.3x	2.1x	2x	2x	16%	33%	0.39
Jiangsu Zhongtian Technology	Optical fiber	600522 CH	28	14,430	7x	6x	1.8x	1.5x	6x	5x	27%	30%	0.24
Yunnan Germanium	Germanium	002428 CH	79	7,717	58x	37x	22.4x	15.5x	39x	26x	47%	119%	0.49
Power & Power Equipment													
Sieyuan Electric	Transformer	002028 CH	113	13,192	16x	13x	3.7x	3.0x	12x	9x	25%	29%	0.55
Hainan Jinpan	Transformer	688676 CH	53	3,633	22x	17x	3.7x	3.2x	16x	13x	19%	37%	0.59
Huaming Equipment	Transformer	002270 CH	19	2,550	18x	16x	4.4x	3.9x	13x	12x	26%	16%	1.10
NARI Technology	Grid	600406 CH	23	27,658	19x	18x	3.0x	2.8x	15x	14x	16%	8%	2.26
CATL	ESS	3750 HK	412	182,539	15x	12x	4.3x	3.7x	9x	7x	24%	17%	0.88
Sungrow Power	ESS	300274 CH	76	23,603	8x	7x	2.1x	1.7x	6x	5x	29%	21%	0.36
Jereh	Gas turbine	002353 CH	87	13,275	15x	11x	2.8x	2.2x	9x	7x	22%	71%	0.20
Dongfang Electric	Gas turbine	1072 HK	16	10,937	8x	7x	1.1x	1.0x	0x	0x	12%	14%	0.59
Anhui Yingliu	Gas turbine	603308 CH	33	3,573	29x	20x	3.9x	3.5x	25x	19x	14%	58%	0.50
Weichai Power Co. Ltd	Diesel engine	2338 HK	25	27,730	9x	7x	1.6x	1.4x	1x	1x	18%	32%	0.27
Yuchai International	Diesel engine	CYD US	25	949	6x	4x	0.1x	0.1x	2x	2x	11%	30%	0.19
Kehua Data	Power supply	002335 CH	26	2,957	28x	22x	2.7x	2.4x	15x	14x	10%	19%	1.48
Shenzhen Megmeet	Power supply	002851 CH	95	8,329	22x	11x	4.7x	3.4x	18x	9x	24%	165%	0.13
Cooling													
Envicool	Cooling	002837 CH	45	8,574	51x	41x	12.1x	10.1x	40x	32x	26%	49%	1.03
Shenling	Cooling	301018 CH	74	4,147	40x	29x	7.9x	6.6x	30x	23x	22%	74%	0.53
AI equipment													
RoboTechnik	Optical Equipment	300757 CH	435	11,435	44x	20x	9.9x	6.8x	31x	15x	26%	310%	0.14
Han's Laser	PCB Equipment	002008 CH	65	9,958	16x	15x	2.8x	2.5x	12x	11x	19%	26%	0.64
Dtech Technology	Drill bit	1377 HK	192	16,163	20x	12x	8.0x	6.1x	15x	9x	39%	90%	0.22

Source: BofA Global Research

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China AI Infra Handbook: Summary

Global/China AI capex: US$2.2tn/US$350bn in 2030E

Our global team estimates that the AI data center systems TAM will grow from c.US$280bn in 2025 to US$2.2tn by 2030E, implying a 51% CAGR over 2025-30E. Growth will be driven by the rapid adoption of agentic AI, accelerating investment by hyperscalers and sovereigns, and higher infrastructure content per AI cluster. We expect China's AI capex to increase from US$110bn in 2025 to US$350bn by 2030E, implying a 26% CAGR over the same period and cumulative spending of c.US$1.4tn during the 15th Five-Year Plan period. Internet firms are expected to account for 65-70% of total AI capex, followed by policy-driven investment at 20-25% and telecom operators at mid-single digit %.

China's Unique Advantages in Global AI Infrastructure

China is a key player in global AI development, underpinned by unique advantages across critical materials, power infrastructure and manufacturing scale. China accounted for c.79% of global tungsten mine production, c.69% of rare-earth mine production, c.69% of indium production and c.48% of refined copper production in 2025, while its integrated processing capabilities, mature specialty-gas supply chain and accelerating import substitution in advanced materials such as PCB copper foil and E-glass enhance supply security and cost competitiveness. China offers ample, low-cost power and a younger grid, with a c.30% effective reserve margin (vs. <25%/c.15% in the US/Europe), 30–60% lower industrial power prices, and grid infrastructure averaging <20 years old (vs. >40 years in the US/Europe). Meanwhile, China's integrated industrial base, extensive customer ecosystem and rapid capacity expansion enable domestic AI equipment suppliers to shorten delivery times, accelerate localization and gain global market share following technological breakthroughs. Together, these advantages position China to play an increasingly important role in scaling global AI infrastructure.

AI materials: structural supply tightness emerging

#1 Copper: AI to Drive 6.8% of China's Copper Demand by 2030E

AI-driven data center expansion is emerging as a key driver of copper demand. China DC demand is set to rise from 341kt in 2025 to 1,272kt by 2030E, a 30% CAGR, lifting its share from 2.1% to 6.8%. Despite concerns about "optics over copper," copper remains essential for power delivery and short-distance interconnects, while rising power demand further supports grid-related use. Combined with ongoing supply constraints and projected deficits into 2026–27E, these factors should continue to support copper demand and pricing. Buy **Zijin Mining (2899 HK)**, whose 1H26 copper and gold output ranked No.4 globally, with strong volume growth and leverage to our constructive copper price outlook. Buy **Jiangxi Copper (358 HK)** and **CMOC (3993 HK)**, leading copper suppliers in China.

#2 PCB materials: AI Drives High-End Copper Foil and E-glass Demand

Copper foil: AI-driven PCB upgrades are accelerating the shift from RTF and earlier-generation HVLP copper foil to HVLP3/4/5. Industry experts forecast a monthly HVLP4 shortage of c.400–600t by end-2026, with supply constrained by equipment bottlenecks, low yields, and lengthy qualification cycles. We expect shortages to widen in 2027E, supporting high-end copper foil pricing and margins. Buy **Defu Tech (301511 CH)** on its market-leading HVLP shipments in China and capacity expansion supported by Mifune and self-developed surface-treatment equipment.

E-glass: AI-driven upgrades in PCBs and advanced packaging are accelerating demand for low-Dk/Df and low-CTE special e-glass, which enables faster data transmission, lower signal loss and improved thermal reliability. High technical barriers and limited qualified suppliers have created a favorable supply-demand backdrop, supporting strong pricing and margins. Buy **Sinoma Sci & Tech (002080 CH)** on its leadership in special e-glass commercialization and aggressive capacity expansion, and **China Jushi (600176**

CH) on rising e-glass profitability and its underappreciated AI materials growth opportunity.

#3 Optical materials: AI Accelerates InP and Fiber Growth

InP as AI-driven optical upgrades accelerate the transition from 400G to 800G/1.6T and, longer term, NPO/CPO architectures increase optical intensity. As an indispensable material for lasers, photodetectors and photonic integrated circuits, InP demand should outpace optical module shipment growth. Meanwhile, high technical barriers, high market concentration, 18–36-month capacity expansion cycles and lengthy customer qualification constrain supply elasticity. Buy **Yunnan Germanium (002428 CH)** on positive pricing outlook.

Optical fiber: AIDC is emerging as a key growth driver for high-end optical fiber. According to a key optical fiber supplier, North American data-center-related demand could double from 150–200mn fiber-km in 2026 to 300–400mn in 2027, while China demand is expected to increase from 50mn to 100mn fiber-km. Supply expansion remains constrained by technical barriers, raw-material bottlenecks, and lengthy qualification and construction cycles, supporting higher prices and margins. Buy **Zhongtian (600522 CH)** on fiber price upside, subsea recovery, and grid growth.

#4 Minor metals: AI Tightens Tungsten and Magnet Markets

Tungsten demand is shifting toward AI-related applications. Higher-layer-count server PCBs are increasing tungsten carbide drill-bit consumption, while advanced logic and HBM production is driving demand for high-purity WF6. Weak traditional demand, higher recycled supply, and imports should keep China's tungsten market broadly balanced in 2026E. SMM estimates that China's tungsten deficit could widen to 7.3kt in 2027E, providing renewed support for tungsten prices.

NdFeB magnet: AI data centers and humanoid robots are emerging as new growth drivers for high-performance NdFeB magnets, which support cooling and power systems as well as high-torque, precision joint control. Global demand is expected to grow at a 9% CAGR over 2025–28E, while China's controls on rare-earth mining, refining quotas, and exports should keep supply tight and support pricing. Buy **JL Mag (6680 HK)**, the world's leading supplier of high-performance permanent magnets.

#4 Electronic Materials: AI Upgrades Drive Import Substitution

MLCC: Rising compute and power density are increasing MLCC content in AI servers and shifting demand toward high-capacitance, low-ESR/ESL, and high-reliability products, while commodity MLCC supply remains broadly balanced. Tighter overseas supply could accelerate qualification of Chinese suppliers, extending localization from high-end MLCCs to upstream $BaTiO_3$ powder, ultrafine nickel powder, and manufacturing equipment. However, material consistency, yields, and reliability remain key barriers.

Specialty gases: AI-driven demand for GPUs, HBM, and other advanced chips is increasing requirements for ultra-high-purity specialty gases used in deposition, etching, cleaning, and doping. China's large semiconductor manufacturing base and established fluorochemical supply chain should support further import substitution, building on progress in NF_3 and WF_6. Purity, consistency, delivery capabilities, and lengthy customer qualification cycles remain the key constraints.

Global AI power: from 100GW in 2025 to 316GW in 2030E

Our global team expects annual DC capacity additions to rise 4x, from ~13GW in 2025 to ~56GW by 2030. We see cumulative DC capacity to growth from ~100GW in 2025 to 316GW in 2030 and expect global DC electricity consumption to rise from below 500TWh in 2025 to 1,322TWh by 2030, implying a 24% CAGR. In China, DC power demand is expected to increase from 29GW to 78GW over the same period, with consumption rising to 323TWh. Three factors are driving the sharp increase in power demand: 1) rapid growth in AIDC workloads ; 2) higher power consumption per chip as

GPUs replace CPUs; and 3) rising rack-level power density (~100x Increase in Rack Power Density by 2030E vs 2022 see exhibit 44).

Power remains the key constraint on AIDC deployment. By 2030, AIDC could account for 4.0%/2.5% of global/China power demand, while power infrastructure represents c.10–15% of DC capex. We estimate China's AI power infrastructure investment will grow from US$16bn in 2025 to US$43bn by 2030, a 22% CAGR. China's strengths in power capacity, renewables, grid infrastructure, and equipment manufacturing position it well to benefit from AI-driven demand. Meanwhile, aging grids and power constraints overseas create opportunities for Asian power equipment companies. **5 key themes**:

#1 7% CAGR of China grid capex during 2025-30E: State Grid Corporation of China (SGCC)/China Southern Grid (CSG) investment were +12.6%/+14.8% YoY in 1H26. SGCC and CSG announced their 15th FYP capex target of RMB4tn/RMB1tn and we see grid investment to reach RMB4tn during 2026-30E. We expect China grid investment to grow 12%/7% in 2026/27E and a 7% CAGR over 2025-30E. Buy **Nari (600406 CH)** on China grid capex.

#2 Transformer tightness, export surges: Per our power equipment expert in Asia conference(ex-Schneider) (see our report, 2026 APAC Conference & Tour Takeaways: Metals, AI Materials, Power & Energy), HV transformers remain the most constrained segment, with lead times of 2–3 years, positioning Asian players to benefit. China transformer exports grew 28% YoY in 9M26, and we forecast 2026E export growth of 25-30% YoY. Buy **Sieyuan (002028 CH)** on transformer export.

#3 Gas turbines supply tightness to till 30E; Opportunities for Chinese suppliers: Chinese companies across the gas turbine value chain should benefit from the global power shortage and tight gas turbine supply. Chinese diesel engine and gas genset suppliers are also gaining traction in the US. Buy **Jereh (002353 CH)** and **Weichai (2338 HK)** on US power demand.

#4 23% '25-30E CAGR for BESS installation: Energy transition and AIDC power demand drive ESS deployment. We forecast global BESS/AIDC-related BESS demand to grow at c.23%/27% CAGRs over 2025–30E. Buy **Sungrow(300274 CH)** on ESS export.

#5 Power supply: We forecast China's AIDC power supply market to grow at a c.27% CAGR over 2025-30E, driven by accelerating AIDC deployments and the transition toward next-generation power architectures. Buy **Megmeet (002851 CH)**

China cooling market: 25% CAGR in 2025–30E

We expect China's data center cooling market to reach RMB46bn by 2030E, driven by rising AIDC power density and efficiency requirements as capacity expands from 29GW to 78GW. Liquid cooling is the key growth driver: demand is set to rise from 1.4GW to 10GW, a 49% CAGR, while penetration increases from 30% to 70%. By contrast, air cooling is expected to grow at a modest 6% CAGR. Buy **Envicool (002837 CH)** on liquid-cooling demand and its global positioning.

AI equipment: cherry picking

Compared with AI power and materials, Chinese companies have not yet established a leading position in the broader AI equipment market. However, selected domestic suppliers are gaining traction in niche segments through strong execution, faster delivery and rapid capacity expansion. As technologies mature and customer validation progresses, these advantages could accelerate import substitution and market-share gains. We highlight opportunities in optical transceiver assembly and testing equipment, CPO testing equipment, PCB drill bits and PCB laser-processing equipment.

#1 Optical equipment: We expect the transition from 800G to 1.6T/3.2T optical transceivers, alongside CPO/NPO/OCS adoption, to drive both volume and ASP growth for precision assembly and testing equipment. We estimate that the global optical-transceiver equipment TAM will reach RMB55bn by 2028E, implying a 67% CAGR in

2025–28E. We expect CPO adoption to accelerate from 2028E, with testing equipment demand front-loaded to 2026–27E. Buy **RoboTechnik (300757 CH)**, a global leader in CPO testing and optical transceiver assembly equipment.

#2 PCB equipment: We expect increasing PCB complexity and drilling intensity in AI server racks to drive demand for drilling equipment and consumable drill bits. We estimate that the Nvidia-related AI PCB drill-bit market will reach RMB14bn by 2028E, representing a 96% CAGR in 2026–28E, while Chinese laser-drilling equipment suppliers gain share through technology upgrades. Buy **Dtech (1377 HK)**, the world's largest PCB drill-bit manufacturer.

China's US$1.4tn AI buildout in 15th FYP

Global AIDC system TAM to reach US$2.2tn by 2030E

In our June 2026 AI infrastructure report, Traditional & transitional: AI materials & Power, we highlighted the strong link between rising AI capex and demand for power and AI materials. Our global team now expects the data center systems TAM to increase from US$278bn in 2025 to US$2.2tn by 2030E, as discussed in Accelerating the Frontier: Agentic race expands AI TAM to $2.2Tn.

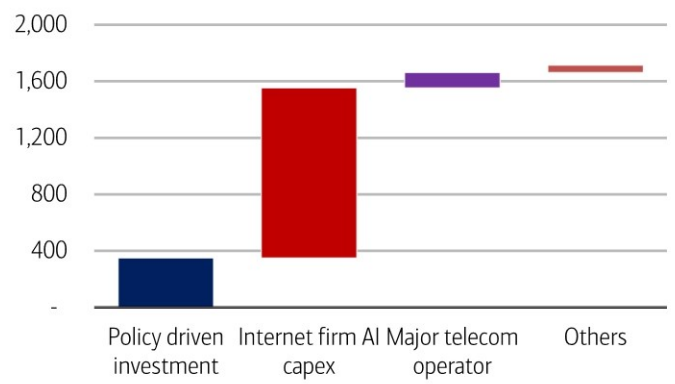
The higher forecast reflects four structural drivers: 1) rapid adoption of agentic AI, which is substantially increasing token consumption and inference compute requirements; 2) accelerating AI infrastructure investment by hyperscalers, neoclouds and frontier AI labs; 3) continued expansion of sovereign AI infrastructure; and 4) higher infrastructure content per AI cluster, including accelerators, CPUs, memory, networking and connectivity.

China AI capex to reach US$350bn in 2030E

We expect China's AI capex to increase from US$110bn (RMB785bn) in 2025 to US$346bn (RMB2.3tn) by 2030E, implying a 26% CAGR. This would translate into cumulative AI capex of c.US$1.4tn during the 15th Five-Year Plan period (2026-30E).

Growth should be supported by diversified funding sources. We estimate that Internet firms will account for 68% of total AI capex during the 15th Five-Year Plan period, assuming 80% of their announced capex is allocated to AI. Telecom operators are expected to contribute another 6%, while policy-driven investment should account for 22%. According to the NDRC, relevant institutions estimate that China's computing network development will generate RMB4tn in incremental direct investment during the 15th FYP. Given the overlap between policy-driven spending and enterprise AI capex, we include c.RMB2tn of policy-driven investment in our forecast, broadly in line with the assumption in our previous AI infrastructure report.

Exhibit 11: China AI capex funding mix in 2027E
Major Internet firms to lead China AI capex growth



Source: Company data, BofA Global Research estimates

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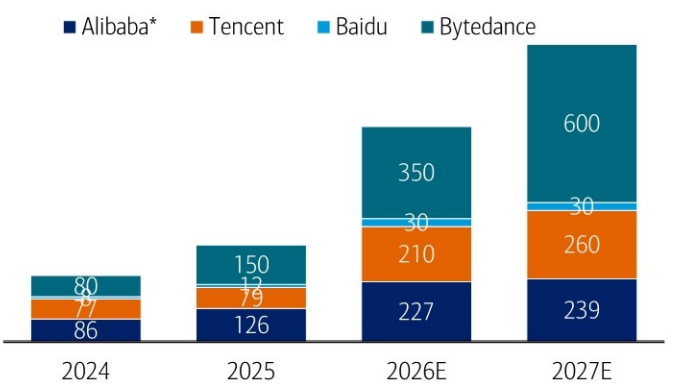
Exhibit 12: Major Chinese internet companies' capex to rise 123% YoY in 2026E
Annual CAPEX trend from 2024 to 2027E (RMBbn)



Source: company reports, SCMP, Reuters, Financial Times, BofA Global Research
*Alibaba 2024/2025E/2026E/2027E used FY2025/FY2026E/FY2027E/FY2028E

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China's unique advantages in AI development

China's critical metals advantage in AI

China's dominant position in critical materials and industrial inputs provides a strong foundation for AI infrastructure expansion. According to the USGS, China accounted for c.79% of tungsten mine production, c.69% of rare-earth mine production, c.69% of indium production, and c.48% of refined copper production in 2025. These materials are essential across AI hardware, power equipment, optical components and semiconductor manufacturing. Beyond upstream raw materials, China is also accelerating import substitution in AI-related advanced materials, including PCB copper foil and electronic fiberglass (E-glass), further strengthening the resilience of its domestic supply chain. China also benefits from integrated processing capabilities and a mature specialty-gas supply chain. Together, these advantages enhance supply security, improve cost competitiveness and support the rapid scaling of domestic AI infrastructure.

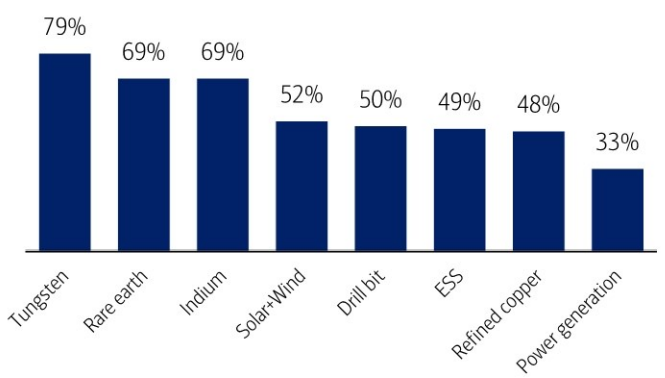
China offers ample and low-cost power with young grid infrastructure

1) China's ~30% effective reserve margin, versus <25% in the U.S. and ~15% in Europe, coupled with 3.9TW of installed capacity (~50% of global renewable capacity), a larger generation base and lower electricity costs, leaves it better positioned to accommodate rising power demand from electrification, data centers and AI. 2)Industrial power prices are also 30-60% lower than in the U.S. and Europe, providing a meaningful cost advantage for energy-intensive industries. 3) A younger grid (<20 years in China vs. >40 years in the U.S./Europe), backed by ~RMB4trn of grid investment during 2025-30E, together with its extensive transmission network and mature supply chain.

China's integrated manufacturing system supports AI equipment development

China's integrated industrial base enables domestic AI equipment suppliers to expand capacity and gain market share more rapidly once technology breakthrough. We estimate Chinese optical transceivers manufacturers account for majority of global production capacity, providing domestic equipment suppliers with a large customer base and opportunities. We believe faster delivery, capacity expansion and rising equipment localization should enable Chinese optical equipment suppliers to capture further market share. Similar advantages are emerging in AI PCB equipment, i.e. Han's Laser offers lead times of around three months versus around one year for global peers, while Dtech has established a leading position in PCB drill bits and continues to expand capacity.

Exhibit 13: China's dominant role in AI materials & power
China holds a dominant role in tungsten, rare earth, indium, renewable energy, PCB drill bit, ESS, refined copper and power generation.



Note: Materials: China's output as % of global in 2025; Power: Cumulative installation for solar, wind, ESS as of end 2025. 2025 annual power generation for power generation
Source: USBS, China NEA, Global Wind Energy Council, IEA, IRENE, BofA Global Research

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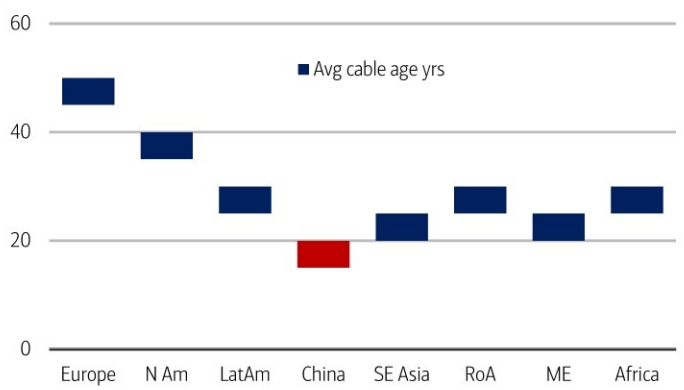
Exhibit 14: China has less cable average age compared with EU and US
Age of global grid assets (years) in 2020.



Source: Nexans

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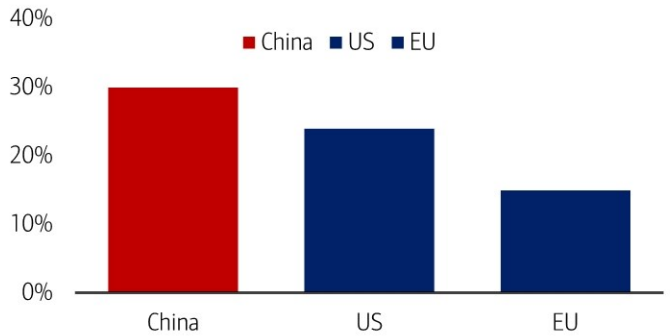
Exhibit 15: China has higher reserve margin of 30%
Estimated effective power reserve margin by area



Source: BofA Global Research estimates, Korea Power exchange. Note: US data from BofA estimates. Reserve margin calculated as (total eligible capacity – max load)/max load

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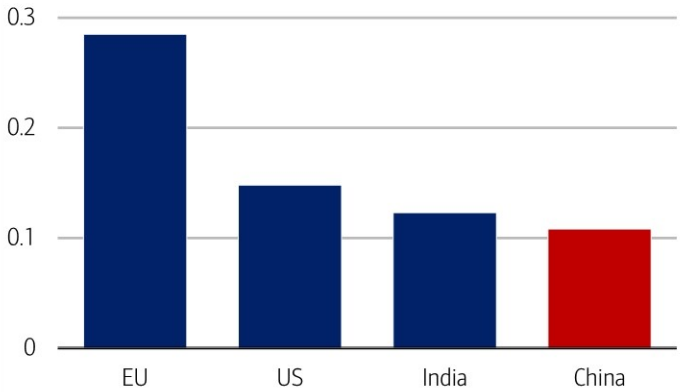
Exhibit 16: China has relatively lower business electricity price
Electricity price for business (US$/MWh) comparison (2023–2026 average)



Source: BofA Global Research estimates, Global Petrol Prices.

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AI Metals: copper & minor metals

Copper: Growing demand from AIDC amid tight supply

DC to account for 6.8% of copper demand in 2030E, vs. 2.1% in 2025

Our detailed calculations indicate that China's DC-driven copper demand stood at 341kt in 2025 and will reach 1,272kt by 2030E, implying a 30% CAGR over 2025–30. This comprises 704kt from data center operations, 530kt from the grid and 39kt from power station construction. Data center-driven demand is expected to account for c.6.8% of China's total copper demand in 2030E, up from 2.1% in 2025. We conclude that AIDCs are becoming a key growth driver for China's copper demand.

Demand shifts to AIDC and AIDC-driven power distribution

Copper has long been widely used in power infrastructure, construction, home appliances and transportation, primarily in wiring, cables and electrical equipment. Rapid AIDC expansion is creating new demand for high-speed data transmission, power delivery, thermal management and semiconductor manufacturing. At the same time, rising power density is increasing copper intensity per data center.

AI development is also driving substantial electricity demand, requiring additional generation capacity, grid upgrades, and transmission and distribution infrastructure, including power stations, transformers and cables. These upstream investments further support copper demand and reinforce the metal's critical role across the AI ecosystem.

Exhibit 17: Copper's usage in AI
We present copper's usage in AI supply chain below.

Source: BofA Global Research

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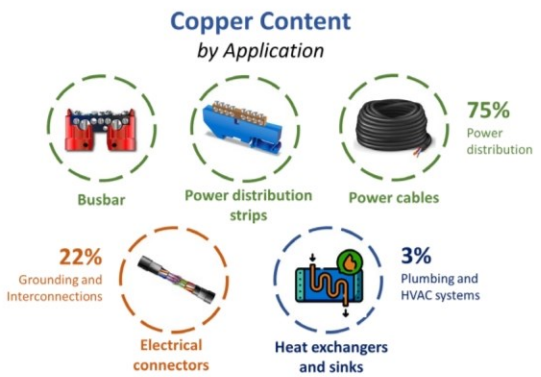
"Optics over copper" concerns overstated; AI to lift copper demand

Concerns that optical interconnects will reduce copper use in AI infrastructure appear overstated. Optics is gaining share in high-speed, long-distance transmission, but copper remains essential for power delivery, server-level interconnects and short-reach connections, where it retains cost, efficiency and reliability advantages.

More broadly, AIDC expansion is increasing power consumption, hardware intensity and power density, all of which are copper-intensive. Despite some substitution by optics, aggregate copper demand across the AI value chain should therefore continue to rise.

Exhibit 18: Copper is the key material in building AIDC
Copper applications in data centers



Source: BlueWeave Consulting, BofA Global Research

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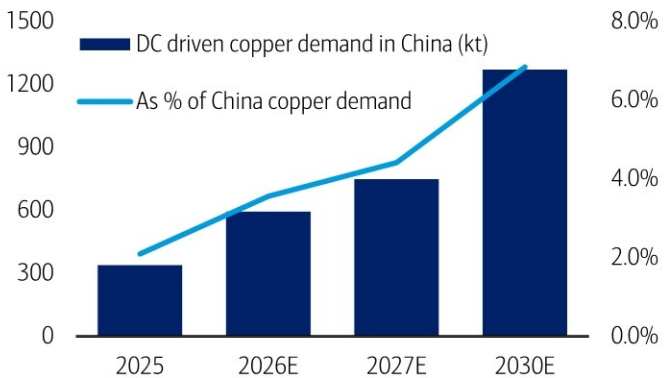
Exhibit 19: DC-driven copper demand to see 30% CAGR in 2025-30E
China DC-driven copper demand (kt)



Source: BofA Global Research

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China copper demand to grow 3%/2% YoY in 2026/27E

We estimate China's copper demand will grow 3%/2% YoY in 2026/27E, driven primarily by the power sector, which accounts for over 50% of total demand. By contrast, property is expected to remain a drag.

Exhibit 20: China copper demand to grow by 3%/2% in 2026/27E
China copper supply-demand balance

(kt)	2022	2023	2024	2025E	2026E	2027E
Production	10,282	11,440	12,061	13,397	13,799	14,075
% of YoY	*3%*	*11%*	*5%*	*11%*	*3%*	*2%*
Net import	3,653	3,458	3,585	3,039	3039	3039
% of YoY	*11%*	*-5%*	*4%*	*-15%*	*0%*	*0%*
Total Supply	**13,935**	**14,898**	**15,645**	**16,437**	**16,839**	**17,115**
% of YoY	*5%*	*7%*	*5%*	*5%*	*2%*	*2%*
Total demand	**14,015**	**15,021**	**15,560**	**16,299**	**16,757**	**17,024**
% of YoY	*2%*	*7%*	*4%*	*5%*	*3%*	*2%*
Supply-Demand surplus/(deficit)	(80)	(123)	85	138	82	90

Source: SMM, Wind, BofA Global Research

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Exhibit 21: 1,272kt copper will be used in China DC in 2030E
Summary of copper consumption in DC

(Unit: kt)	2025	2026E	2027E	2030E
Copper in DC operation	191	343	402	704
Copper in grid	139	232	322	530
Copper in power station	11	21	27	39
Total copper demand in China DC	341	596	751	1,272

Source: BofA Global Research

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Global copper supply deficit in 2026

Copper supply remains constrained by declining ore grades and operational disruptions at major mines, limiting near-term growth. Our global team expects supply to be broadly flat in 2026, followed by a modest 4.6% recovery in 2027. Upstream copper concentrate tightness is also evident, with spot TC/RCs remaining below –USD200/t, a historical low. With AI-driven demand continuing to rise, however, the market is likely to remain in deficit, which we estimate at 397kt in 2026.

Exhibit 22: Copper is set to be in deficit this year
Copper supply and demand balance

'000 tonnes	2023	2024	2025	2026E	2027E
Global production	26,152	26,652	27,953	27,955	29,229
YoY change	6.9%	1.9%	4.9%	0.0%	4.6%
Global consumption	25,846	26,776	27,775	28,352	29,201
YoY change	2.6%	3.6%	3.7%	2.1%	3.0%
Balance	**305**	**-125**	**178**	**-397**	**28**
Market inventories	1,019	1,440	2,206	1,809	1,837
Weeks of world demand	2.1	2.8	4.1	3.3	3.3
LME Cash ($/t)	**8,484**	**9,150**	**9,947**	**13,278**	**15,250**
LME Cash (c/lb)	**385**	**415**	**451**	**602**	**692**

Source: Platts, Woodmac, CRU, Bloomberg, company reports, ICSG, BofA Global Research

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Exhibit 23: Copper social inventory
China copper social inventory is now in very low level.

Source: Bloomberg, BofA Global Research

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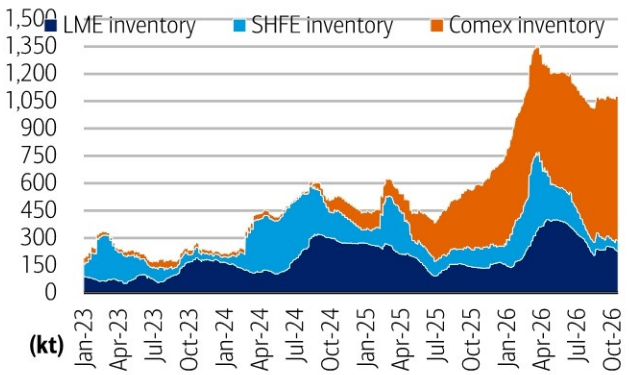
Exhibit 24: Copper inventory at exchanges
COMEX inventory has kept building



Source: Bloomberg, BofA Global Research

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Minor metals

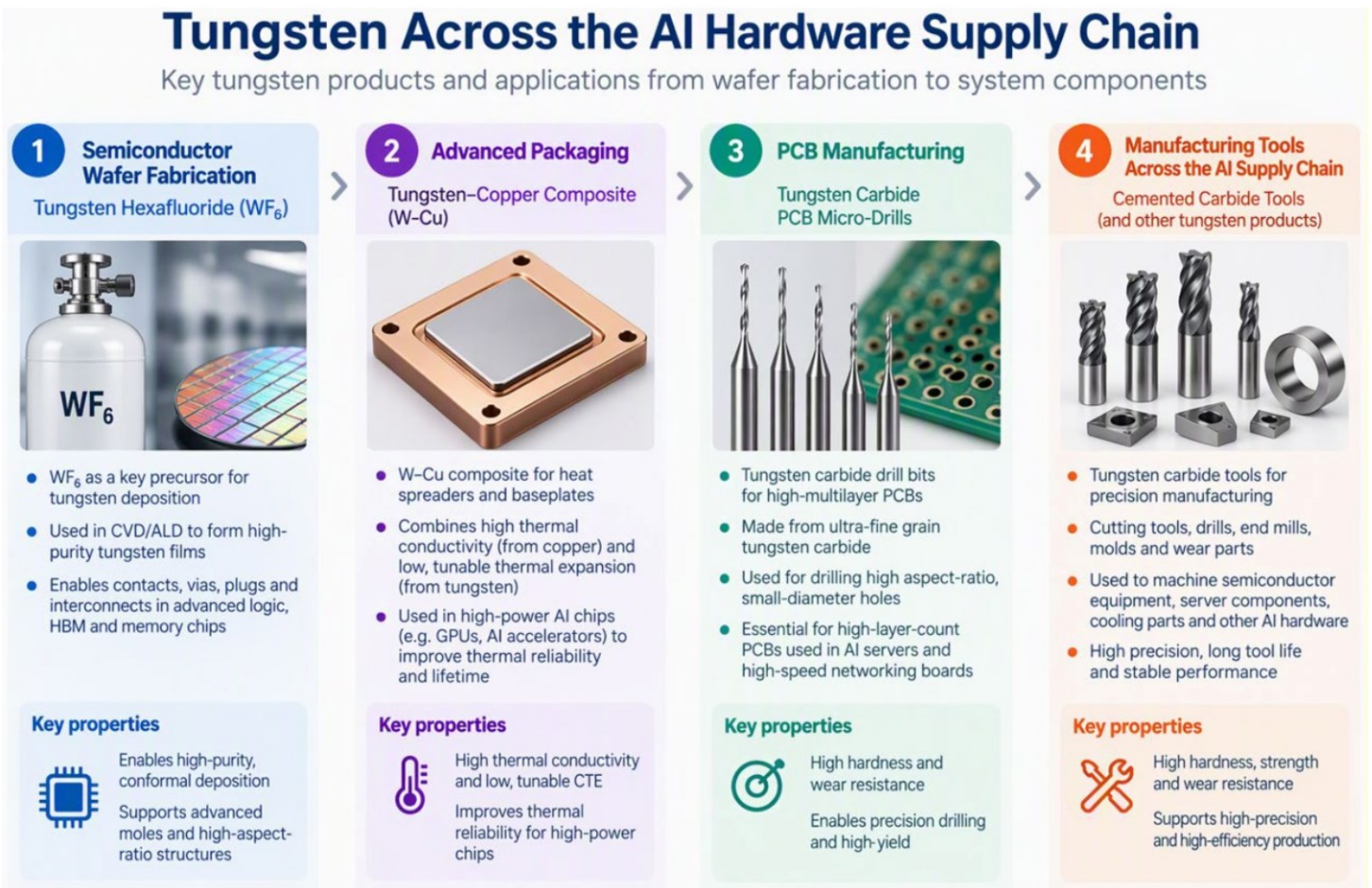
Tungsten: emerging AI demand amid tight supply

Tungsten is a strategic minor metal traditionally used in cemented carbide tools because of its exceptional hardness, wear resistance and thermal stability. In the AI era, its applications are expanding into advanced electronics. Tungsten carbide is used in precision printed circuit board (PCB) drill bits, while tungsten hexafluoride (WF6) is used to form contacts, vias and interconnects in semiconductor manufacturing.

According to SMM, China's tungsten consumption is expected to grow by only 1.72% YoY to 84.5kt in 2026, reflecting subdued traditional demand. However, AI is creating structural growth opportunities. Increasingly complex server PCBs with higher layer counts require more tungsten carbide drill bits, while advanced logic and high-bandwidth memory (HBM) production supports demand for high-purity WF6. Demand for cemented carbide and tungsten materials is expected to improve in 2027.

SMM expects China's total tungsten supply to rise 7.7% YoY to 83.7kt in 2026, as higher recycled-tungsten supply and imports offset a c.6.4% decline in primary concentrate output. The market should remain broadly balanced in 2026. However, slowing growth in mine and scrap supply, combined with improving demand, could widen the deficit to 7.3kt in 2027 and renew support for tungsten prices.

Exhibit 25: Tungsten in AI
We present tungsten's usage in AI below.



Source: BofA Global Research

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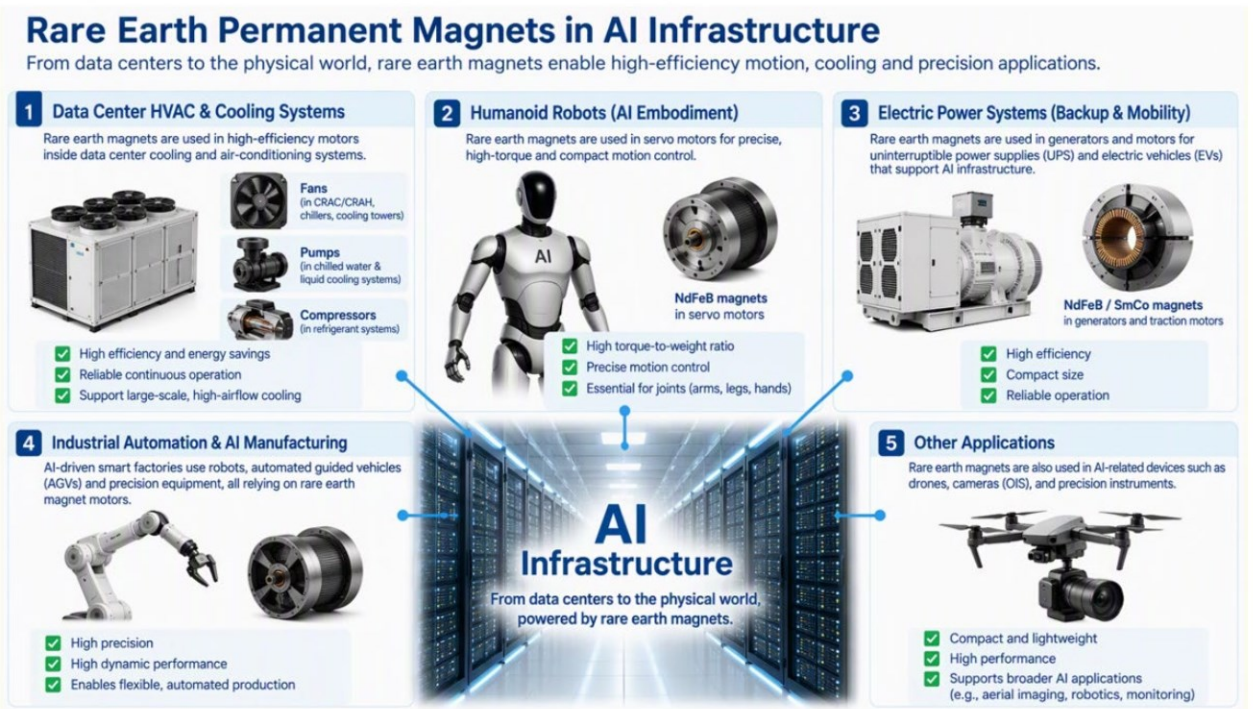
High-performance magnets: AI and robotics growth

High-performance permanent magnets, mainly NdFeB-based, are widely used in air conditioning, wind power and EVs. AI is shifting demand toward AIDCs and humanoid robots, where magnets support cooling, power and automation systems, as well as precise, high-torque joint control. Global demand is expected to grow at a 9% CAGR over 2025–28. Meanwhile, China's controls on rare earth mining, refining quotas and exports should keep supply tight.

Exhibit 26: Permanent magnets in AI

We present permanent magnets' usage in AI below.



Source: BofA Global Research

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Exhibit 27: Global high-performance magnet demand forecast

We estimate that global high-performance magnet demand will deliver 9% CAGR in 2025-28E.

	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
High performance mag demand (kt)	54	65	79	85	103	114	131	141	156	172
YoY growth	*14%*	*20%*	*21%*	*9%*	*20%*	*12%*	*15%*	*7%*	*11%*	*10%*
EV driven demand (kt)	7	10	21	33	43	53	65	74	87	101
	6%	*44%*	*116%*	*53%*	*33%*	*23%*	*22%*	*14%*	*18%*	*16%*
Wind power driven demand (kt)	8	13	16	11	13	13	19	20	20	21
YoY growth	*25%*	*54%*	*22%*	*-33%*	*22%*	*4%*	*41%*	*4%*	*5%*	*4%*
Inverter AC driven demand (kt)	8	9	11	11	13	14	14	14	14	14
	19%	*21%*	*24%*	*0%*	*14%*	*10%*	*1%*	*-5%*	*0%*	*0%*
Humanoid robots driven demand (t)	-	-	-	-	-	11	77	381	1,351	2,562
YoY growth	-	-	-	-	-	-	*633%*	*395%*	*254%*	*90%*
Others (kt)	32	33	30	31	34	34	34	34	34	34

Source: Frost & Sullivan, BofA Global Research

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FAQ on AI Metals

Question #1: China copper demand outlook

Weakness in property and home appliances has raised concerns about the outlook for China's copper demand.

Our view: We expect China's copper demand to grow 3%/2% in 2026/27E, led by the power sector, which accounts for over 50% of total demand. Automotive, representing 13% of demand, is expected to grow 2-3% in both years, while home appliances, accounting for 13%, are expected to decline 2-4% annually. Property remains the key drag, with demand expected to fall 9% in 2026E and continue contracting in 2027.

Question #2: Global copper supply outlook

Potential supply resumptions at major mines have raised expectations of stronger global copper production growth in 2027.

Our view: According to our global commodities team, global copper production is expected to remain broadly flat in 2026, resulting in a c.397kt deficit. Production is then expected to increase by c.4.6% in 2027, bringing the market broadly back into balance. Key swing factors include First Quantum, which could contribute c.100-150kt, and Grasberg, where more than 100kt of contained copper production could potentially resume, according to SMM. However, projects in Tibet are unlikely to contribute meaningful volumes in 2027, with a more visible contribution expected from 2028. Scrap supply should also remain tight following China's reverse-invoicing policy, with limited scope for improvement over the next three to six months.

Question #3: US copper inventories and tariffs

The build-up of US copper inventories has raised questions about potential re-exports and the impact of the eventual tariff decision on global copper flows.

Our view: The US has become the global sink for refined copper, accumulating more than 1.4mnt. Although tariff expectations initially drew copper into the US, steep CME contangos now make storage sufficiently profitable to keep the metal tied up. Meaningful re-exports would likely require either stronger nearby US demand to flatten the CME curve or higher LME prices to compensate traders for moving copper out of the US. The eventual tariff decision may therefore have limited sustained impact, while a recovery in Chinese demand could matter more for global copper flows.

Question #4: TC/RC outlook and smelter production

Deepening smelter losses have intensified market discussions about potential production cuts in China.

Our view: TC/RCs are likely to remain under pressure. Chinese copper smelters are currently operating at an estimated loss of c.RMB5,000/t due to declining TC/RCs and weaker sulfuric acid prices. Despite the severe margin pressure, the industry has yet to see widespread voluntary production cuts, suggesting that meaningful supply discipline may take time to emerge. Negotiations for 2027 term contracts are also expected to be particularly challenging, as more smelters move away from the traditional annual benchmark and adopt pricing linked to spot TC/RC indices.

PCB/Optical Materials: Cu foil, E-glass, InP

Copper foil: high-end demand outpacing supply

PCB copper foil is a key CCL material that forms the conductive layers of PCBs. In high-speed applications, lower surface roughness reduces conductor loss and signal attenuation, making advanced low-profile products such as HVLP critical to signal integrity.

Exhibit 28: The upgrade to M8/M9/M10 CCL is characterized by progressive improvements in glass fiber cloth, copper foil, and resin systems
CCL material evolution takes place across M6-M9 grade products

Grade	Typical Application	Glass Type	Resin System	Copper Foil	Note
FR-4	General PCB, consumer electronics	E-glass	Epoxy/ modified epoxy	Standard Rz copper	Cost-driven, signal loss less critical
M6	Networking / mid-speed server PCB	E-glass/low-Dk1	Epoxy-PPO blend	VLP	Signal integrity starts to matter
M7	AI server / high-speed PCB	Low-Dk1/low-Dk2	PPO/PPE	HVLP3	High-speed signal transmission, supply tightness
M8/M8.5	AI server/ high-end switching	Low-Dk1/low-Dk2	PPO/PPE/BMI/hydrocarbon	HVLP3/HVLP4	Skin effect, supply tightness
M9 / M10	AI server/ substrate-like PCB / packaging	Low-Dk2/Q-glass	Hydrocarbon/PTFE/PPO/PPE	HVLP4/HVLP5	Warpage + chip-package mismatch, supply tightness

Source: BofA Global Research

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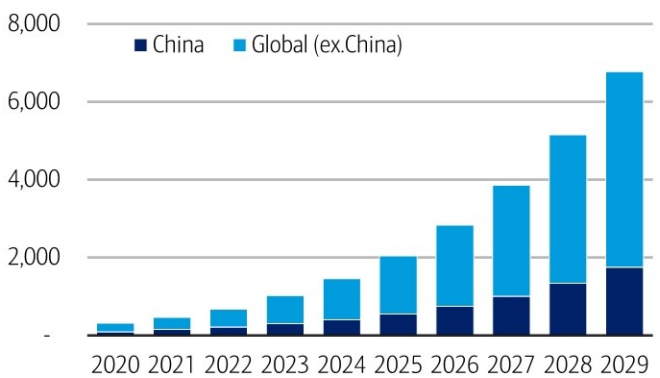
Demand for high-end copper foil is accelerating as AI servers and high-speed networking require faster transmission, more PCB layers and tighter signal-integrity control. This is driving a shift from RTF and lower-generation HVLP products toward HVLP3/4/5. Industry experts expect HVLP4 demand to reach 1,600-1,800t per month by end-2026, versus supply of only c.1,200t, implying a monthly shortage of c.400-600t. Demand should accelerate further in 2027.

Supply is unlikely to keep pace, given limited surface-treatment equipment, low yields and lengthy qualification cycles. Average HVLP yields are only c.60%, versus c.90% for RTF, and are even lower for HVLP4; customer qualification takes one to three years. We therefore expect shortages to widen in 2027, particularly for HVLP3-5.

Exhibit 29: AI-driven PCB copper foil market size forecast (RMBmn)
Frost & Sullivan forecast global AI PCB copper foil market to grow 36% CAGR in 2024-29E, and China market to grow 35% CAGR.



Source: Frost & Sullivan, BofA Global Research

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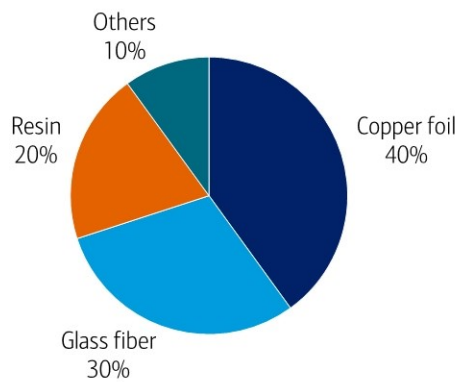
Exhibit 30: CCL cost breakdown
Copper foil accounts for 40% of CCL costs, followed by glass fiber at 30% and resin at 20%.



Source: BofA Global Research

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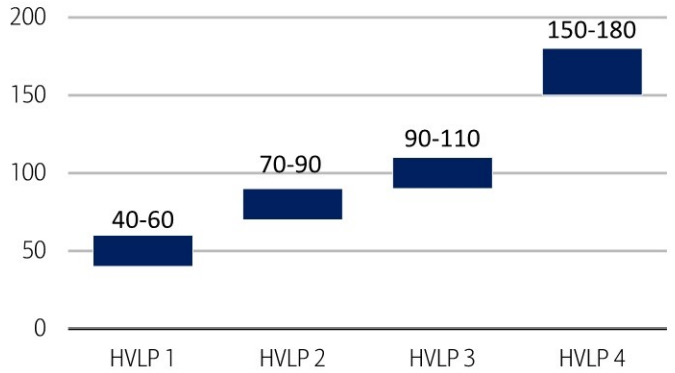
Exhibit 31: HVLP processing fee (RMB '000/t)
Pricing gaps are significant across HVLP generations.



Source: BofA Global Research

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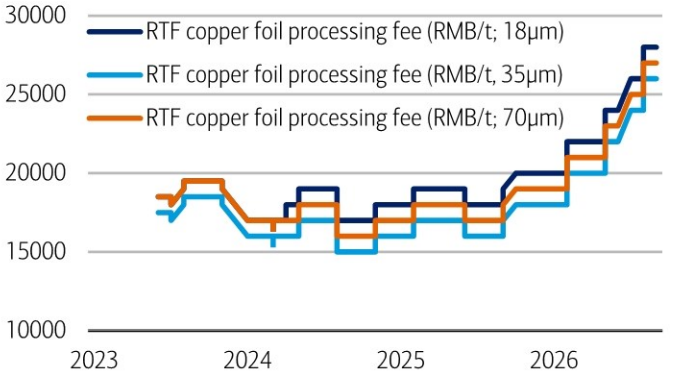
Exhibit 32: RTF copper foil processing fee
RTF copper foil processing fees have continued to rise in 2026.



Source: SMM, BofA Global Research

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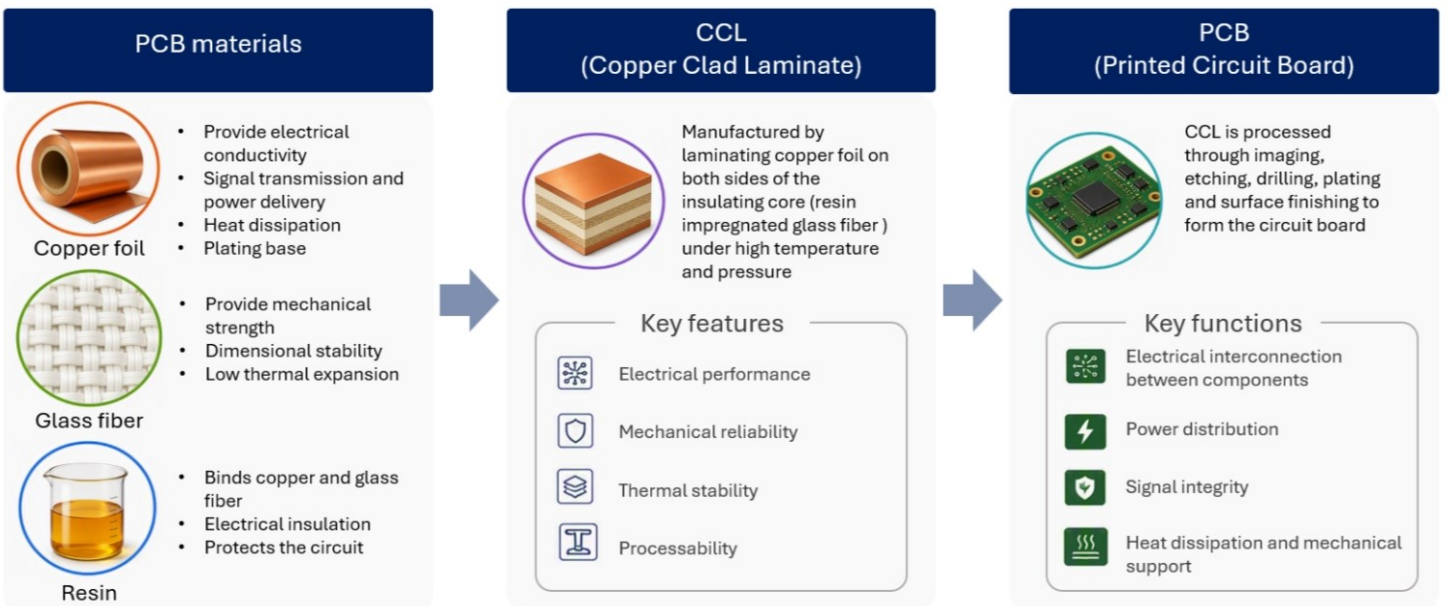
Exhibit 33: PCB materials supply chain
We present PCB materials supply chain below.



Source: BofA Global Research

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Exhibit 34: Copper foil by types
We present main copper foil types below.

Product	Category	Model	Key features	Direct applications	End applications
Electrolytic copper foil	LiB copper foil	Ultra-thin copper foil (Thickness ≤5μm) *ASP: RMB25-35k/t*	• Ultra-thin LiB copper foil with smooth surfaces, ultra-low grammage, low profile, and high strength/elongation. • Enables higher anode loading, reduces fracture risk, and improves LiB energy density.	Lithium battery	NEV High-end consumer electronics
		Extra-thin copper foil (Thickness 6-12μm) *ASP: RMB15-25k/t*	• Smooth double-sided foil with high strength and oxidation resistance. • Designed for the longer cycle-life requirements of ESS.	Lithium battery	NEV E-bike Consumer electronics ESS
	PCB-grade copper foil	Hyper very low profile (HVLP) foil (Rz <1μm) *ASP: RMB40-180k/t*	• Ultra-low roughness with high peel strength and resin adhesion. • Enables low-loss, high-speed signal transmission.	High-frequency & high-speed substrates	High-end networking devices Antenna PCBs Millimeter-wave radars AI accelerators
		Very low profile (VLP) Foil (Rz 1-2μm) *ASP: RMB30-40k/t*	• Low roughness with strong resin adhesion and pattern stability. • High electrical and thermal conductivity.	Encapsulated PCBs, high-density multilayer PCBs	High-frequency PCBs /Microelectronics
		Reverse treated foil (RTF) (Rz 2-5μm) *ASP: RMB25-30k/t*	• Cost-efficient roughened glossy surface with excellent high-frequency performance. • Ultra-low roughness, high oxidation resistance, and enhanced flexibility.	Multilayer PCBs, high-frequency signal transmission boards	Base stations / servers
		High temperature Elongation (HTE) Foil (Rz 5-10μm) *ASP: RMB20-25k/t*	• Low roughness with high peel strength. • High heat resistance and elongation for crack prevention.	Multilayer substrates, high-density multilayer wiring substrates, encapsulated PCBs	Electronic home appliances
		Flexible copper foil (FCF) *ASP: RMB150-180k/t*	• High density with excellent flexural and etching performance. • Ultra-low roughness and oxidation resistance.	Flexible substrates	Flexible electronics / Flexible circuit boards Flexible screen
	Carrier foil	Carrier-attached ultra-thin copper foil (With peelable carrier layer) *ASP: RMB90-120/m²*	• Supports ultra-thin foil and ultra-fine circuits. • Excellent peel strength and stability for advanced packaging.	Semiconductor packaging substrates	AI chips / Advanced semiconductors

Source: Company data, BofA Global Research

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Special e-glass for high-end PCBs

Explosive growth in AI computing demand and higher performance requirements for PCBs and packaging substrates are driving upgrades in upstream materials. These include electronic cloth manufactured from special yarns with a low dielectric constant (low-Dk), low dielectric loss (low-Df) and low coefficient of thermal expansion (low-CTE). These materials increase data-transmission speed, reduce transmission loss and improve stability, making them core inputs for high-end PCBs and packaging substrates. Few suppliers can currently produce these special yarns, resulting in a supply-demand imbalance that is keeping prices and margins high. Market leader Nittobo has recorded explosive volume growth since 2020, driven by emerging demand from AI and network communications, although Chinese players are rapidly catching up.

What are low-Dk/Df and low-CTE electronic yarn/cloth?

Low-Dk/Df electronic fabric is woven from electronic yarns with a low dielectric constant and low dielectric loss using advanced processes. A material's dielectric properties describe its ability to store and dissipate electrostatic energy under an electric field and are typically expressed as dielectric constant (Dk) and dielectric loss (Df). Dk indicates a material's ability to store electrical energy; the lower the Dk, the faster and more efficiently signals travel through the dielectric. Df indicates the amount of energy lost in a dielectric material; the lower the Df, the better the integrity of signals transmitted through it. Low-Dk/Df electronic fabric reduces signal loss in boards and increases transmission speed. It is widely used in applications requiring fast, low-loss signal transmission, such as AI high-speed servers and 5G communications.

Low-CTE electronic fabric is woven from electronic yarns with a low coefficient of thermal expansion (CTE), which measures the rate at which a material's dimensions change with temperature. Low-CTE electronic fabric can effectively reduce the CTE of boards, lower thermal stress and improve reliability. It is used mainly in advanced IC substrates and other applications that must match the extremely low thermal expansion of chips.

Exhibit 35: Glass fiber performance requirement for different applications

Demand for special glass fiber has been growing due to expanding more downstream demand applications across telecom, datacenter, AI server, edge equipment, as well as EV.

Application	Device/component	Substrate type		Required performance	Glass fiber type	
					High-end	Middle-end
5G+ cellular network Data center AI Server	Processor / Controller	Semiconductor package substrate	CPU/GPU/ASIC	Low CTE	Low-CTE	Ordinary E-glass
			NAND memory	Low CTE	Low-CTE	Ordinary E-glass
			LPDDR memory	Low CTE	Low-CTE	
			DDR memory	Low dielectric tangnet	1st & 2nd gen low-Dk	Ordinary E-glass
	Motherboard	Servers motherboard (AI, General)		Low dielectric tangnet	1st & 2nd gen low-Dk	Ordinary E-glass
		Switch motherboard (Core, AI)		Low dielectric tangnet	1st, 2nd & 3rd gen low-Dk	Ordinary E-glass
		Network card motherboard (NIC)		Low dielectric tangnet	1st & 2nd gen low-Dk	Ordinary E-glass
Smartphone Tablet Mobile PC Other compute-intensive edge devices	Processor / Advanced SoC	Semiconductor package substrate	AP·CPU/NPU	Low CTE	Ultra-thin Low-CTE, Low-CTE	Ultra-thin E-glass
	Non-volatile memory		NAND memory	Low CTE	Ultra-thin Low-CTE	Super ultra-thin E-glass
	Volatile memory		DDR memory	Low CTE	Ultra-thin Low-CTE (Smartphone)	
	Volatile memory		DDR memory	Low dielectric tangnet	1st gen low-Dk (PC)	
	Motherboard	Motherboard		Low dielectric tangnet	Ultra-thin 1st gen low-Dk	Ultra-thin E-glass
	Wireless communication	RF package substrate		Low dielectric tangnet	Ultra-thin 1st gen low-Dk	Ultra-thin E-glass
Desktop Laptop PC	CPU / Memory	Semiconductor package substrate	CPU/GPU/ASIC	Low CTE	Low-CTE	Ordinary E-glass
	Motherboard		DDR memory	Low dielectric tangnet	1st gen low-Dk (PC)	Ordinary E-glass
Electric Vehicles with Advanced Driver Assistance Systems	Advanced SoC	Semiconductor package substrate		Low CTE	Low-CTE	Ordinary E-glass
	Millmeter wave radar	Module board		Low dielectric tangnet	Ultra-thin 1st gen low-Dk	Ordinary E-glass

Source: Nittobo, company data, BofA Global Research

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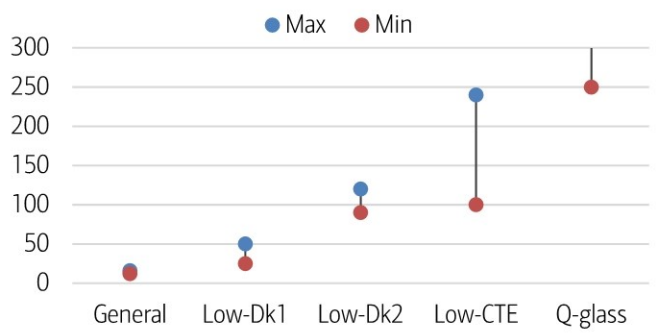
General e-glass supply increasingly tight

AI hardware demand is driving multifold growth not only in special e-glass with low-Dk or low-CTE properties, but also in general e-glass used in peripherals. Grace Fabric's projections illustrate the strength of this demand. The global PCB market is expected to reach USD118.5bn by 2030, representing a 6.4% CAGR over 2025-30, with growth concentrated in high-layer-count PCBs, high-build-up HDIs and IC packaging substrates. The global e-glass market, including both general and special products, is expected to reach USD6,012.0mn by 2030, representing a 15.4% CAGR over 2025-30. The global market for general e-glass cloth alone is expected to reach USD3,660.8mn by 2030, representing a 9.6% CAGR over the same period.

As a result, general e-glass prices have increased by more than 10% month on month since September 2025. Cumulatively, prices for ordinary grades such as 7628 have tripled, while those for premium grades, including thin and ultra-thin variants, have risen even more sharply, highlighting the extent of market tightness.

Exhibit 36: Price range of different e-glass
AI special e-glass is far more expensive than general e-glass.



Source: BofA Global Research, company data

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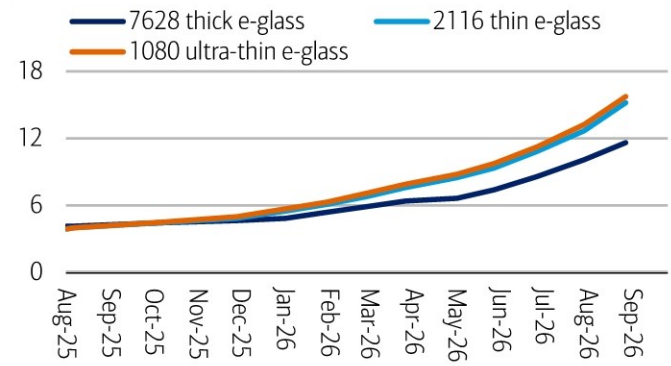
Exhibit 37: General e-glass price hikes
General e-glass prices have been surging on supply shortage.



Source: BofA Global Research, company data

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FAQ on PCB Materials

Question #1: PCB copper foil demand outlook

Adequate supply of lower-end products and uncertainty over the pace of AI-related PCB upgrades have raised questions about the strength of copper foil demand.

Our view: The market is increasingly bifurcated, with lower-end products such as HTE remaining adequately supplied while shortages are concentrated in high-end HVLP foil, particularly HVLP4. Industry experts' end-2026 HVLP4 demand forecasts range from c.1,600-1,800t per month under a conservative scenario to c.2,500t under a more bullish scenario, versus expected monthly output of only c.1,200t. Under the bullish scenario, demand could rise further to c.3,000t per month by mid-2027. We therefore expect HVLP4 to remain in structural undersupply under either scenario.

Question #2: HVLP supply and key bottlenecks

Planned HVLP3/4 capacity expansion has raised questions about whether supply could catch up with demand over the next few years.

Our view: Supply growth should remain constrained, as only a limited number of copper foil suppliers plan to expand HVLP3/4 capacity. Surface-treatment equipment remains the key bottleneck. According to industry experts, HVLP4 equipment is available only from Japanese suppliers, which face long lead times, largely secured order books and capacity constraints. Suppliers that have secured equipment orders, or can develop and manufacture the equipment in-house, should gain market share over the next few years.

Question #3: Further HVLP copper foil price upside

Several price increases YTD have raised questions about whether Chinese HVLP producers have further room to raise prices.

Our view: We expect Chinese HVLP producers to raise prices again in 4Q26 and 1Q27. Chinese HVLP foil remains c.15-20% cheaper than Japanese products, but this gap should narrow as tight Japanese supply redirects demand to Chinese suppliers and domestic CCL manufacturers accelerate import substitution. Industry experts expect Chinese HVLP prices to broadly catch up with Japanese prices following the potential 1Q27 increase.

Question #4: General e-glass price outlook

Jushi's measured pricing approach has raised questions about the magnitude and sustainability of further general e-glass price increases.

Our view: The market currently expects grade 7628 prices to increase by RMB1.5/m per month, reaching RMB16/m by end-2026. Jushi is not seeking larger increases, as it aims to minimize downstream disruption and concerns about shortages. However, if CCL manufacturers implement further price increases before year-end, Jushi would likely raise prices again in 2027, as tight FR4 CCL supply has been driven mainly by the e-glass shortage.

Question #5: Threat from alternative looms

The volume production of 7628 e-glass cloth on Chinese looms has raised concerns about a potential increase in supply and competition.

Our view: The threat remains limited. Chinese looms are already using qualified G75 e-glass yarns to produce 7628 cloth, but their yields are generally c.10% lower than Toyota's. Meanwhile, yarn prices are rising alongside cloth prices, significantly increasing production costs for Chinese looms and limiting their ability to disrupt the market.

Question #6: Implications of e-glass-free M10 CCL

Potential substitution of e-glass with PTFE or hydrocarbon in M10 CCL has raised concerns about the longer-term outlook for e-glass demand.

Our view: The impact on e-glass demand should be limited. E-glass-free designs would replace only certain PCB layers, as e-glass cloth would still be required to provide structural integrity and prevent warpage. Given PCB processing challenges and low yields, M10 is likely to be deployed only in the most frequency-demanding trays, such as 3.2T networking switches. Other compute boards, switch boards, midplanes and LPUs in Nvidia's VR200 would continue to use M8+/M9 CCL.

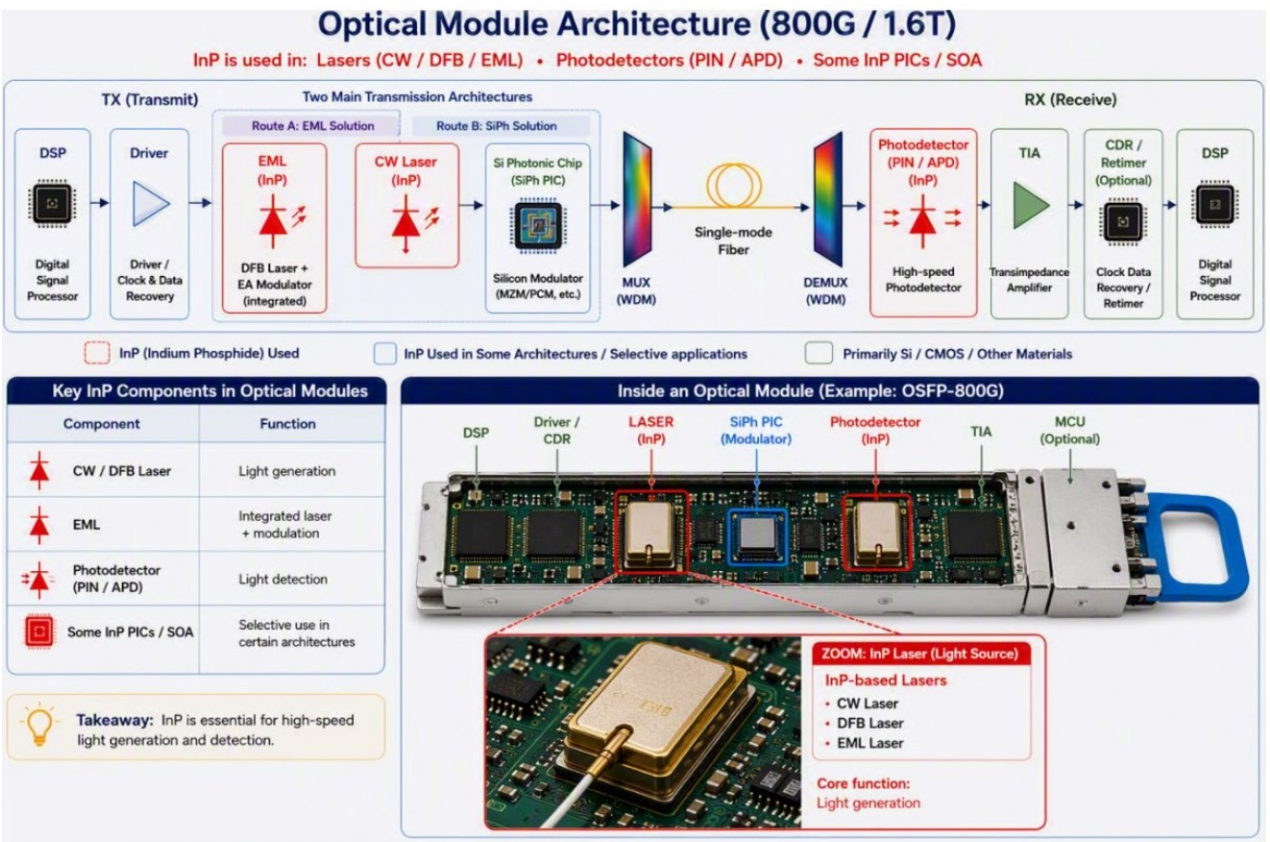
Indium Phosphide: AI's Critical Material Bottleneck

View: AI optical interconnect upgrades are driving InP into a supply-constrained upcycle

We believe indium phosphide (InP) is becoming one of the most strategically important materials in the AI optical value chain, with structural demand growth increasingly underpinning its investment case. Optical modules are migrating from 400G to 800G and 1.6T, while future architectures such as near-packaged optics (NPO) and co-packaged optics (CPO) are expected to increase optical intensity further. Unlike silicon, InP remains indispensable for light generation and detection, supporting critical components such as EML lasers, CW lasers, photodetectors and photonic integrated circuits. As a result, InP demand should grow significantly faster than optical module shipments.

Importantly, the industry is entering a classic supply-constrained growth cycle. InP substrate manufacturing requires high-purity raw materials, crystal-growth expertise, extended yield ramp-up periods and lengthy customer qualification. Capacity expansion typically takes 18-36 months, and customer certification can require a further 8-12 months. Supply elasticity is therefore substantially lower than in conventional semiconductor manufacturing. At the same time, global supply remains concentrated among a handful of players, creating a significant mismatch between AI-driven demand growth and industry capacity additions. We believe industry tightness is likely to persist through 2027-28, supporting pricing and profitability across the InP value chain.

Exhibit 38: AI optical module architectures may evolve, but InP remains the critical enabling material
Optical Module Architecture (800G/ 1.6T) – InP Usage



Source: BofA Global Research

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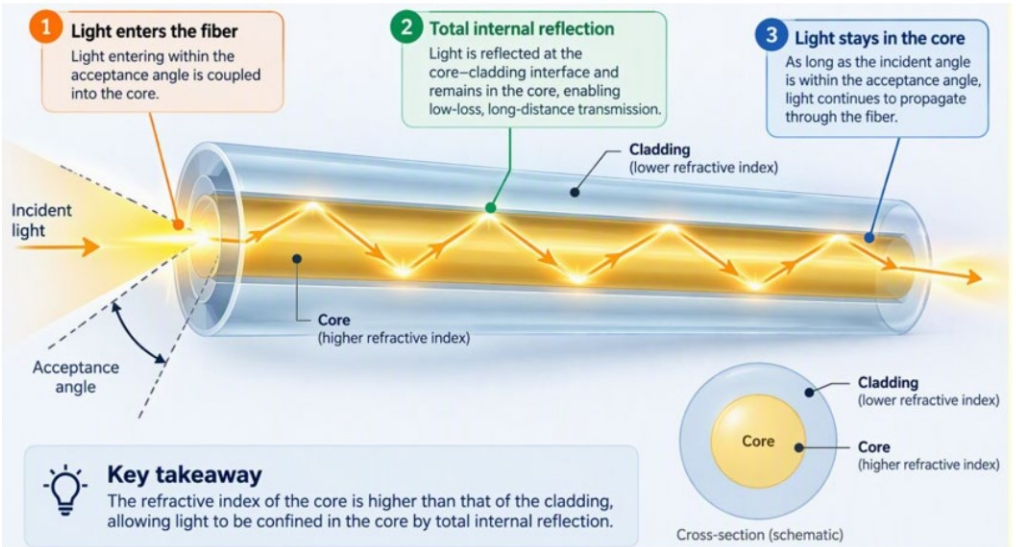
Optical fiber: core communications medium

Optical fiber enables high-bandwidth, low-loss data transmission over long distances. Traditionally used in telecom networks, it is becoming increasingly critical to AIDC interconnects and long-distance transmission between computing clusters.

Exhibit 39: Light propagation in an optical fiber
Light is confined within the core by total internal reflection and travels long distances with low loss.



Source: Coherent, BofA Global Research

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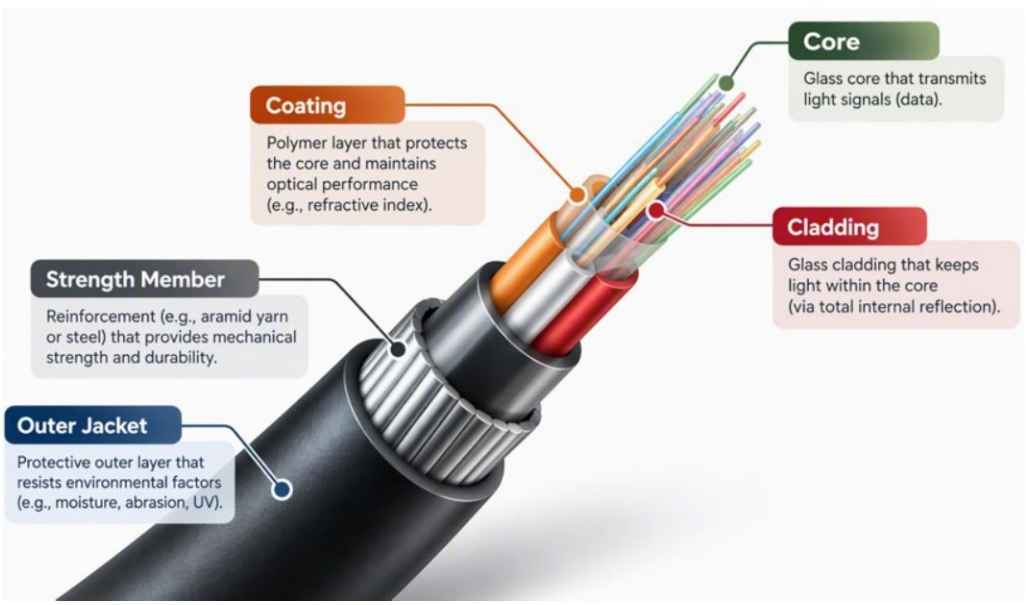
Exhibit 40: An example structure of optical fiber cable
Contains ultra-thin glass/plastic core, cladding, coating, strength materials and outer jacker to protect the inside



Source: Vcelink, BofA Global Research

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AIDC drives structural demand for high-end optical fiber

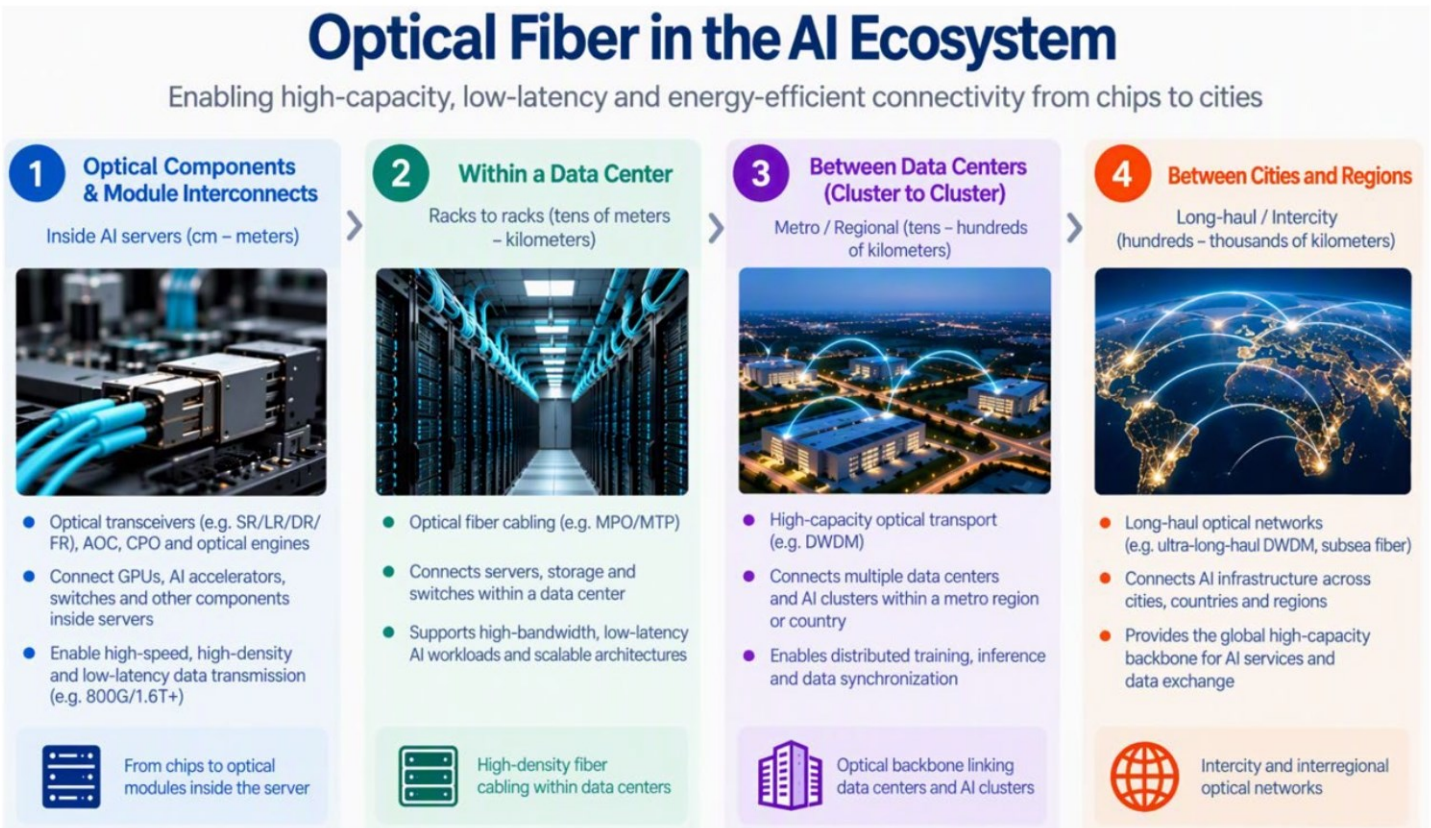
According to a key optical fiber supplier, North American data-center-related fiber demand could double from 150–200mn fiber-km in 2026 to 300–400mn fiber-km in 2027. China demand is expected to rise from 50mn to 100mn fiber-km, while

incremental demand from domestic and overseas cloud service providers could reach 200–250mn fiber-km in 2027. Telecom demand remains broadly stable. Drone-related demand has also surprised to the upside, rising from an initial estimate of c.50mn to 150mn fiber-km.

Structural tightness in high-end fiber supply

Overall industry capacity remains sufficient, but AIDC demand is creating structural tightness in high-specification fiber and optical preforms. Our channel checks indicate that YOFC, Hengtong and ZTT each plan meaningful preform capacity additions, although project construction typically takes 1-2 years. Overseas expansion is progressing more slowly amid raw-material constraints, while technical barriers and lengthy qualification cycles are limiting new supply. The tighter supply-demand balance is supporting substantial price hikes and margin expansion for optical fiber makers.

Exhibit 41: Optical fiber in AI
We present optical fiber's usage in AI ecosystem below.



Source: Company data, BofA Global Research

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FAQ on optical materials

Question #1: Optical fiber demand outlook

Rapid AIDC deployment has raised expectations for strong optical fiber demand growth in North America and China.

Our view: Our discussions with Chinese optical fiber manufacturers indicate that data-center-related fiber demand is rising rapidly in North America and China. North American demand is estimated at c.150-200mn fiber-km in 2026 and could double in 2027, while China demand is projected to rise from c.50mn to c.100mn fiber-km. Incremental demand from domestic and overseas cloud providers could reach c.200-250mn fiber-km in 2027, exceeding available capacity at leading suppliers.

According to a top optical fiber supplier in China, telecom demand remains broadly stable, but procurement prices have risen sharply. China Mobile's 2026 volume is largely flat YoY, while its tender price increased from RMB16.8/fiber-km in 2025 to RMB85 in 2026. China Telecom's demand is up YoY, and China Unicom's delayed tender is expected to exceed its 2025 volume. Drone-related demand has also surprised to the upside, with industry estimates rising from c.50mn fiber-km at the start of 2026 to c.150mn currently.

Question #2: Optical fiber supply and key bottlenecks

Capacity-expansion plans among leading Chinese manufacturers have raised questions about whether new supply could keep pace with rapidly growing AIDC demand.

Our view: According to our channel check, capacity-expansion plans are relatively visible among China's four leading manufacturers: YOFC, Hengtong Optic-Electric, FiberHome and ZTT. YOFC and ZTT each plan to add c.800-1,000t of optical fiber preform capacity. Hengtong plans to add c.800t to its existing c.3,600t, with commissioning expected in 2026 or early 2027.

Overseas expansion remains slow because of constraints in specialty gases and other raw materials, while high technical barriers and lengthy qualification cycles limit new supply. Expansion typically takes c.1-1.5 years for leading producers and c.1.5-2 years for smaller players. We therefore expect supply to respond gradually despite strong demand.

Question #3: Potential substitution of InP by silicon photonics

The increasing adoption of silicon photonics has raised concerns that InP could eventually be displaced in advanced optical architectures.

Our view: Silicon photonics primarily addresses optical modulation and integration, whereas silicon is inherently inefficient at generating and detecting light. High-performance lasers and photodetectors therefore continue to rely on InP-based materials. Even in advanced silicon photonics architectures, InP remains the core platform for optical sources. As transmission speeds increase, demand for higher-performance lasers is likely to rise, suggesting that silicon photonics will complement rather than displace InP demand.

Question #4: Risk of InP oversupply

Capacity-expansion announcements by major InP suppliers have raised concerns about whether the market could quickly move into oversupply.

Our view: We believe the risk of near-term oversupply is low. The key constraint is not announced capacity but qualified capacity. High-end optical applications require stringent defect-control standards, stable crystal quality and extensive customer certification. Although major suppliers have announced expansion plans, most new capacity is unlikely to contribute meaningfully before 2027-2028. Meanwhile, downstream customers are increasingly signing long-term supply agreements and securing capacity strategically, reflecting growing concerns about material availability. We therefore expect InP pricing to remain well supported over the next two years.

Question #5: InP import-substitution opportunity in China

China's leading position in optical module manufacturing and refined indium production has increased attention on the localization potential of InP substrates.

Over view: The global InP substrate market remains dominated by overseas suppliers, while Chinese suppliers collectively hold a relatively small market share. However, China has become the world's largest optical module manufacturing base and controls a significant share of global refined indium production. Against a backdrop of export controls, supply-chain security concerns and localization initiatives, Chinese optical component vendors are increasingly evaluating and qualifying domestic InP suppliers. We believe local substitution represents a multi-year structural opportunity, creating substantial growth potential for domestic substrate manufacturers.

Semi Materials

MLCCs: small components, critical enablers of AI compute

Multilayer ceramic capacitors (MLCCs) consist of alternating layers of ceramic dielectric and metal electrodes. They stabilize voltage, filter high-frequency noise and maintain signal integrity across GPUs, HBM, server motherboards, voltage-regulation modules, power supplies and networking equipment. As compute and power density rise, MLCC content is increasing, driving demand for high-capacitance, low-ESR/ESL and high-reliability products.

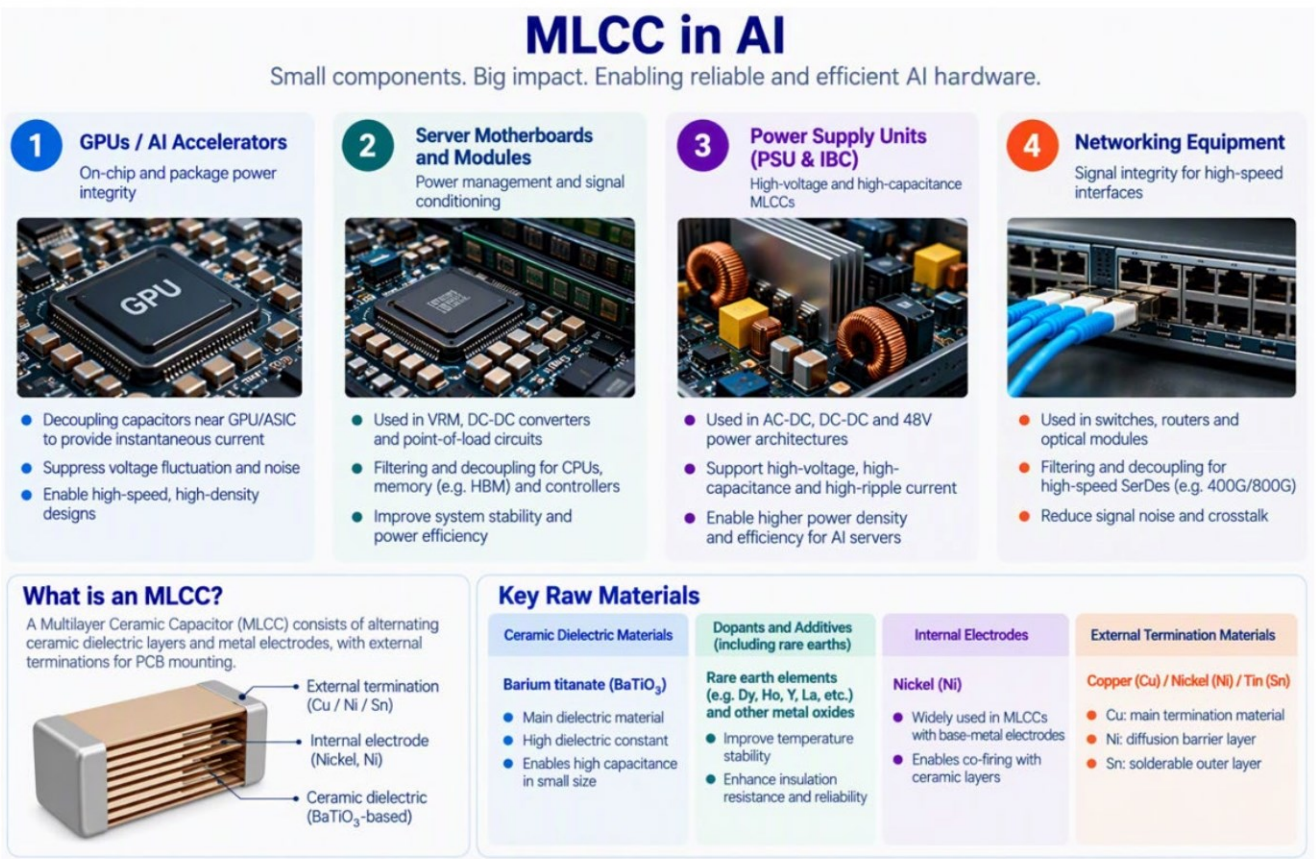
AI drives selective tightness in high-end MLCCs

Key materials include $BaTiO_3$ dielectric powder; ultrafine nickel powder for internal electrodes; and copper, nickel and tin for external terminations. Rare earths such as Dy, Ho, Y and La serve as functional dopants that improve stability and reliability, rather than as primary materials. Supply tightness is concentrated in high-capacitance, high-voltage and high-reliability MLCCs, while commodity products remain relatively balanced.

China: localization opportunity extending from components to materials

Tighter overseas supply could accelerate domestic AI hardware customers' qualification of Chinese MLCC suppliers, supporting high-end import substitution. As domestic production scales, localization should extend upstream to $BaTiO_3$ powders, ultrafine nickel powders and manufacturing equipment. Key barriers remain material consistency, ultrathin-layer processing, yields and long-term reliability.

Exhibit 42: MLCC in AI
We present MLCC's usage in AI below.



Source: Company data, BofA Global Research

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Specialty gases: AI-driven demand and import substitution

What are electronic specialty gases?

Electronic specialty gases are ultra-high-purity gases used directly in semiconductor and electronics manufacturing. They support deposition, etching, cleaning and doping processes. Key products include silane, WF_6, NF_3, CF_4, phosphine and arsine.

AI exposure mainly through advanced chip manufacturing

AI exposure comes primarily from the production of GPUs, HBM and other advanced chips, rather than from direct use in data centers. Increasingly complex, three-dimensional chip structures require greater gas purity and consistency, making specialty gases more critical to chip performance and manufacturing yields.

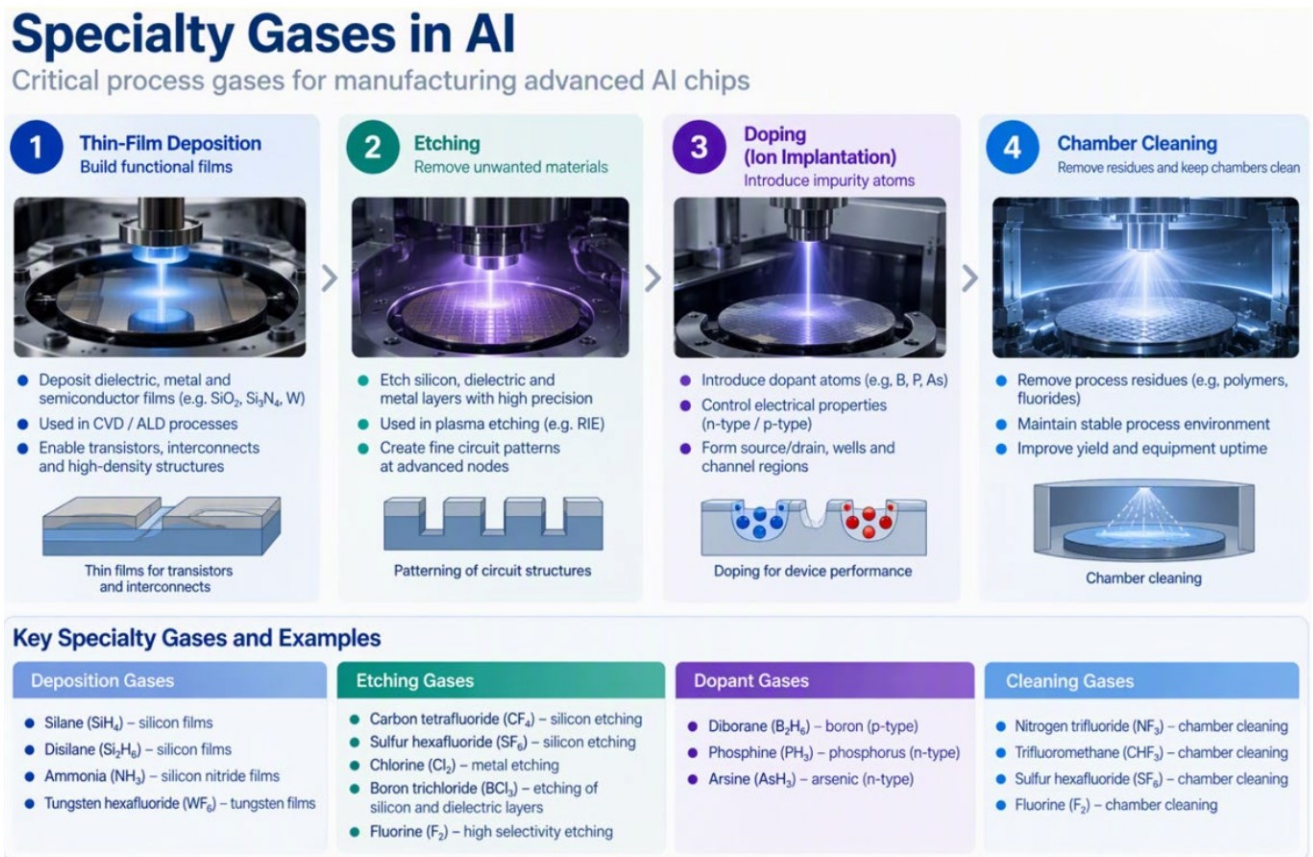
China's bottlenecks: purity, consistency and delivery capabilities

China's main bottlenecks lie in ultra-high-purity and advanced-process products. Key challenges include purification, testing, product consistency, cylinder treatment, transportation and recycling, compounded by lengthy customer-qualification cycles.

China's opportunity: accelerating import substitution

China benefits from a large semiconductor manufacturing base and an established fluorochemical supply chain. Domestic suppliers have achieved import substitution in products such as NF_3 and WF_6. Meanwhile, Peric Special Gases, Huate Gas and Nata Opto-electronic Material are expanding their product portfolios and increasing their penetration of domestic semiconductor supply chains.

Exhibit 43: Specialty gases in AI
We present specialty gases' usage in AI below.



Source: Company data, BofA Global Research

BofA GLOBAL RESEARCH

Powering AI: transformers, turbines, ESS

Our global team expects annual DC capacity additions to rise 4x, from ~13GW in 2025 to ~56GW by 2030. We see cumulative DC capacity to growth from ~100GW in 2025 to 316GW in 2030 if the pipeline is fully realized. According to the IEA, global data center electricity consumption is expected to grow at a 24% CAGR, from below 500TWh in 2025 (1.6% of power demand) to 1,322TWh by 2030 (4.0% of power demand). In China, data center power demand is projected to rise from 29GW in 2025 to 78GW by 2030, while electricity consumption increases from 121TWh to 323TWh, equivalent to ~2.5% of national power demand.

Higher compute density is driving greater power demand: 1) AI workloads are translating compute demand into power demand; 2) GPU adoption is increasing chip-level power consumption; 3) rack power density has risen from 10-15kW in traditional cloud data centers to 100-120kW today, with >1MW racks on the horizon; and 4) power and cooling infrastructure must scale accordingly, supporting investment in power delivery, backup power and thermal management systems.

According to the AIDC expert at our Asia conference, power availability and grid connectivity remain the key bottlenecks to AIDC deployment. (see our report, 2026 APAC Conference & Tour Takeaways: Metals, AI Materials, Power & Energy) China benefits from abundant power capacity, lower power costs, dominance in renewable supply, a younger and well-established grid network, and a strong power equipment value chain. Overseas power shortages and aging grid infrastructure, exacerbated by AIDC demand, create opportunities for Asian power equipment players.

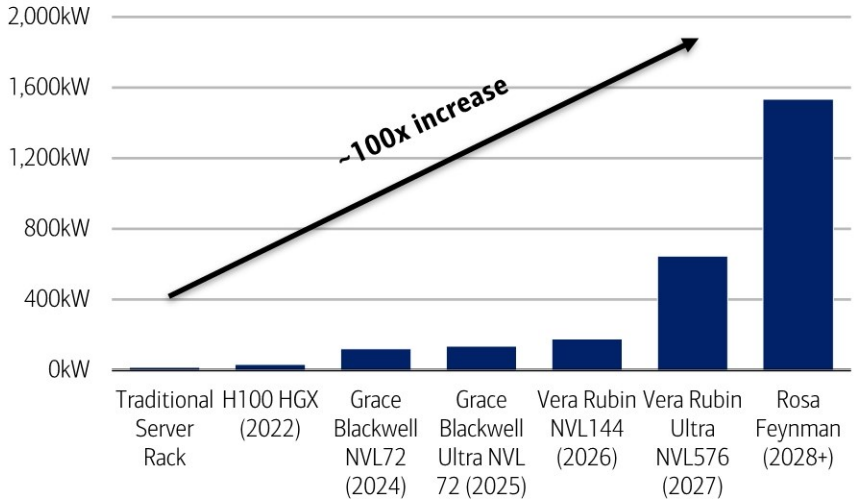
Power-related capex accounts for 10-15% of total DC capex. We forecast China's AI power-related investment to reach US$43bn in 2030E, up from US$16bn in 2025, implying a 22% CAGR.

We expect China's data center cooling market to reach RMB46bn by 2030E, driven by rising AIDC power density and efficiency requirements as capacity expands from 29GW to 78GW. Liquid cooling is the key growth driver: demand is set to rise from 1.4GW to 10GW, a 49% CAGR, while penetration increases from 30% to 70%. By contrast, air cooling is expected to grow at a modest 6% CAGR.

Key powering AI themes include:

- 7% CAGR of China grid capex during 2025-30E
- HV transformers remain the tightest segment
- Gas turbine supply remains tight; Chinese players are catching up
- 23% BESS CAGR through '30E despite near-term concerns
- China's AIDC power supply system to grow at a 27% CAGR over 2025-30E

Exhibit 44: By end-2030, we think Nvidia's roadmap could lead to >1.5 megawatt racks for the Feynman era, nearly 100x higher than a standard server CPU rack at 10-15 kilowatts
Power capacity per rack (kilowatts) across various generations of Nvidia platforms vs. traditional server rack



Source: BofA Global Research estimates, Nvidia, company reports

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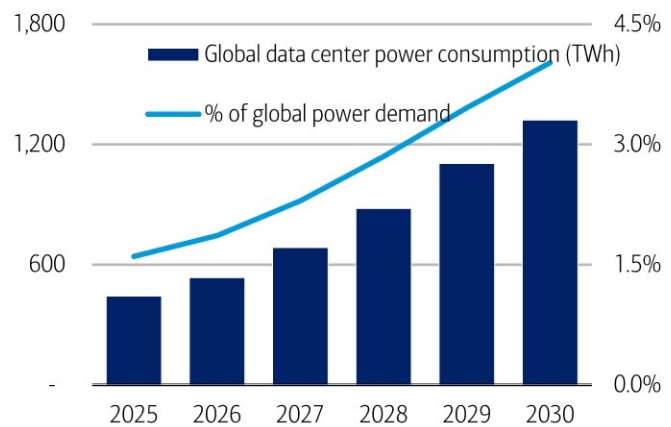
Exhibit 45: Global DC power consumption to see 24% 2025-30E CAGR
Global data center is estimated to consume 1,322TWh in 2030E



Source: IEA, Gartner, BofA Global Research

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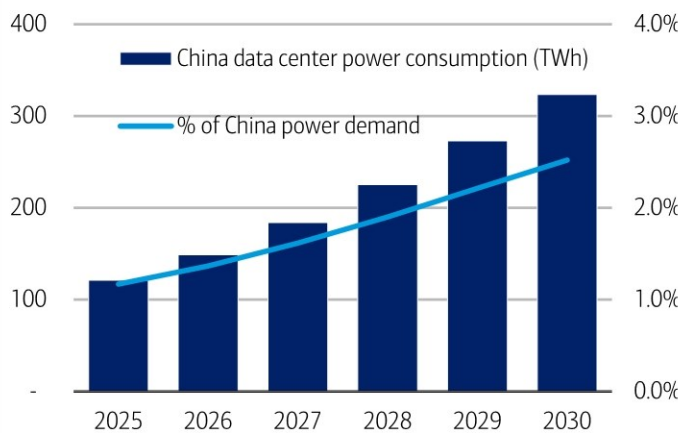
Exhibit 46: China DC power consumption to see 22% 2025-30E CAGR
China data center is estimated to consume 323TWh in 2030E



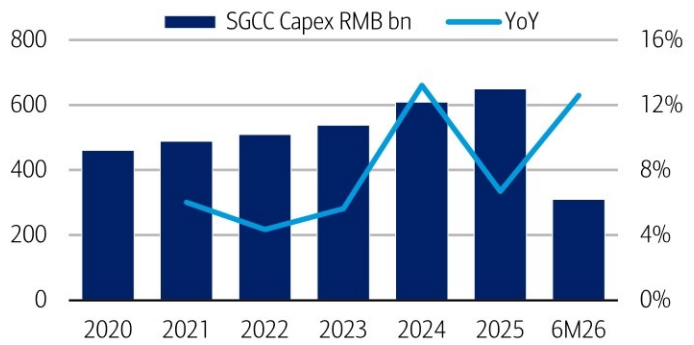
Source: IEA, Gartner, BofA Global Research

BofA GLOBAL RESEARCH

Theme 1: 7% CAGR of China grid capex during 2025-30E

In 1H26, investment by State Grid Corporation of China (SGCC) and China Southern Power Grid (CSG) reached RMB310bn and RMB89.2bn, up 12.6% and 14.8% YoY, respectively. Looking ahead, SGCC and CSG have outlined investment plans of approximately RMB4tn and RMB1tn, respectively, over the 2026-30 period. Against this backdrop, we forecast China's grid investment to grow by 12%/7% in 2026/27E and a 7% CAGR during 2025-30E.

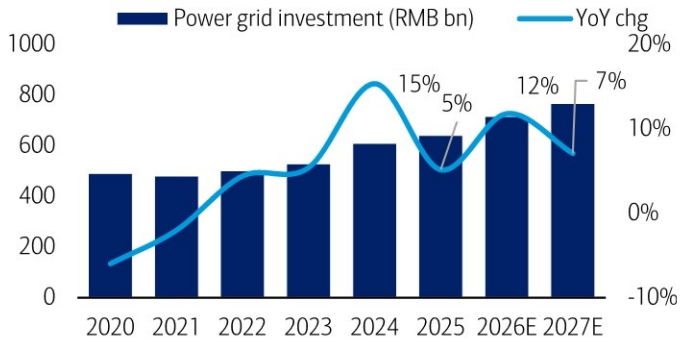
Exhibit 47: 6M26 SGCC Capex +12.6% YoY
State Grid Corporation of China (SGCC) capex



Source: SGCC, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 48: We estimate grid investment will grow 12% YoY in 2026
Grid investment forecast in China



Source: SGCC, BofA Global Research estimates

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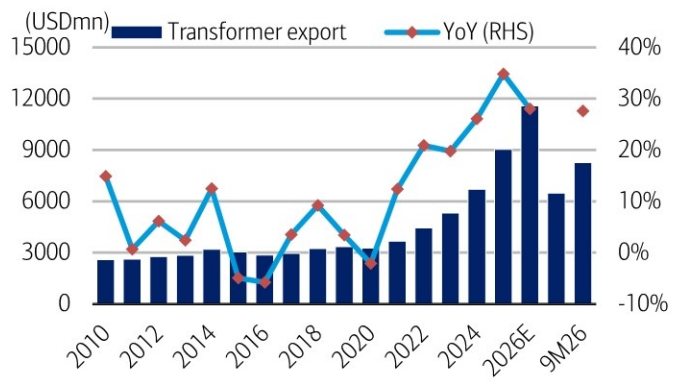
Theme 2: HV transformers remain the tightest segment

We remain positive on power equipment, supported by renewables, electrification, data centers and grid upgrades. High-voltage (HV) transformers remain the most constrained segment, with lead times of 2-3 years, according to our power equipment expert (see our report, 2026 APAC Conference & Tour Takeaways: Metals, AI Materials, Power & Energy). China continues to benefit from its capacity advantage and comprehensive supply chain. China transformer exports were up 28% YoY in 9M26 and are expected to grow 25-30% in 2026. However, U.S. Executive Order 14420 and European localization policies have increased regulatory uncertainty for Chinese suppliers.

Exhibit 49: China transformer export +28% in 9M26
China transformer export (USDmn)



Source: China Customs, BofA Global Research

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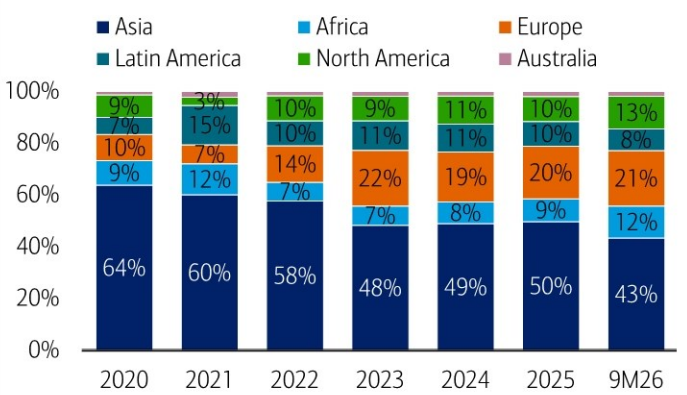
Exhibit 50: Asia accounts for c.40% of China's transformer export
China's transformer export value by destination



Source: China Customs, BofA Global Research

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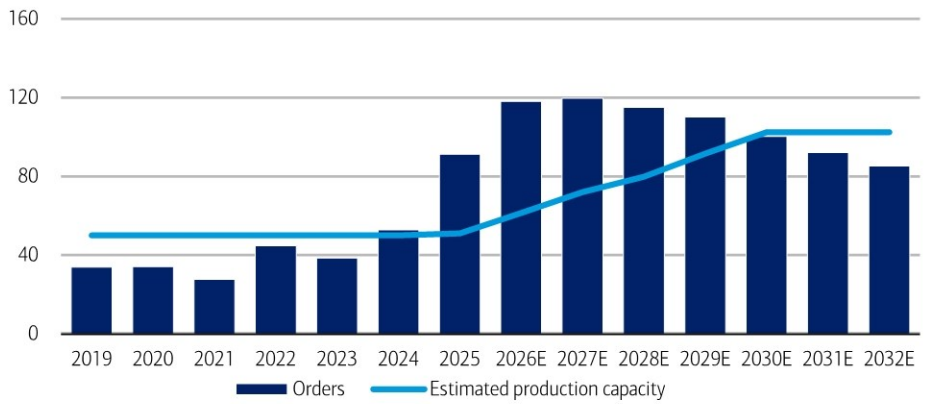
Theme 3: Gas turbine supply tight; CN players catching up

BofA Global Research's industrial research team forecasts c.120GW pa of gas turbine orders over 2026-28E, with lead times exceeding three years. See our report: Siemens Energy AG: 25 for 2026: Turning Heat into Repeat 06 March 2026. Globally, Siemens Energy, GE Vernova and Mitsubishi Heavy Industries dominate the supply of large- and medium-sized gas turbines, reflecting high technological barriers. See our report,: Cooking with gas.

Global gas turbine shortages are creating opportunities for Chinese suppliers across the value chain. Companies such as Dongfang, Jereh and Yingliu are using shorter lead times and cost advantages to gain share, although their product performance still trails that of global leaders. This theme has driven a re-rating of traditional power equipment names, supported by Dongfang Electric's gas turbine exports to Canada and capacity expansion plans, Shanghai Electric's overseas order wins, and Harbin Electric's export-oriented R&D efforts. Meanwhile, Jereh continues to benefit as a packager of U.S. gas turbine engines and a provider of integrated power solutions for AIDC.

Exhibit 51: Gas turbine supply tightness to till 2030E
Global gas turbine order forecast (bars) and estimated production capacity (line) (GW, simplcycle basis)



Source: BofA Global Research estimates, McCoy, company data. Note: shows orders >10MW, excluding oil & gas

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Exhibit 52: Gas turbine value chain
Gas turbine value chain and key players

Source: Company data, BofA Global Research

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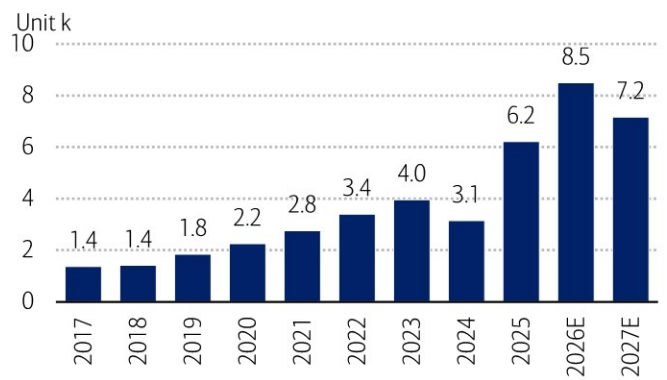
Large-bore diesel engines are emerging as a key beneficiary of AIDC expansion. Supported by strong data center buildout, demand for backup power engines is expected to remain robust through 2026 before normalizing. Chinese suppliers such as Weichai and Yuchai are expanding capacity aggressively and are well positioned to gain market share from international peers, whose capacity additions remain more gradual. We expect supply tightness to ease from 2027, with overseas expansion becoming the next growth driver. Following Weichai's successful entry into North America, Yuchai is targeting exports from 2027 after completing UL/EPA certifications, supporting further global share gains.

Reciprocating gas gensets are gaining traction as an alternative to gas turbines for data center power, supported by lower capex and maintenance costs, as well as persistent gas turbine supply constraints. Global leaders such as Caterpillar and Cummins are expanding capacity aggressively to capture growing AIDC demand, while Chinese suppliers are following suit. Weichai plans to begin North American exports in late 2026, while Yuchai is targeting 2028 after completing UL/EPA certifications. We see capacity expansion and overseas market penetration as key drivers of global share gains for Chinese engine makers in the data center power market.

Exhibit 53: China AIDC backup-power diesel engines demand
We expect AIDC diesel engines demand to post strong growth in 2026



Source: BofA Global Research estimates, MIIT

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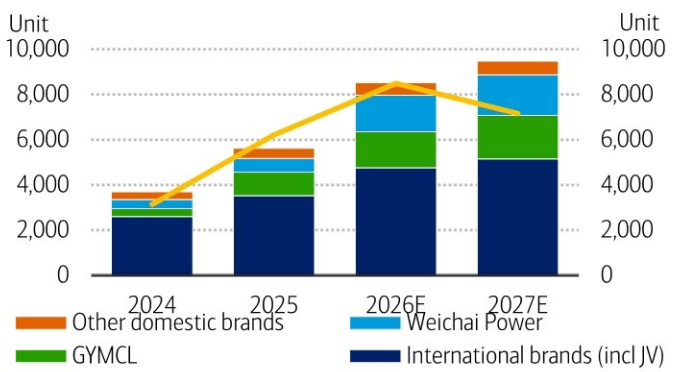
Exhibit 54: China AIDC backup diesel engines demand & capacity
The supply shortage will be relieved in 2027 on capacity expansion



Source: BofA Global Research estimates, company data, GYMCL: Guangxi Yuchai Machinery Company Limited

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Exhibit 55: Cost and performance comparison of multiple equipment for AIDC prime power
We see gas turbines have lowest CAPEX and LCOE, long life cycle, but SOFC has highest efficiency, low redundancy, and shortest construction time

Type of equipment	CAPEX	LCOE	Construction period	Life cycle	Startup time	Efficiency	Redundancy	O&M costs
Heavy-duty GT (Single cycle)	Lowest	Low	Long	Long	Long	Low	High	Medium
Heavy-duty GT (Combined cycle)	High	Lowest	Longest		Longest	High		
Aero-derivative GT	Medium	Medium	Shortest	Medium	Short	Low	Medium	Medium
Reciprocating gas gensets	Low	High	Medium	Medium	Shortest	Medium	Medium	Low
SOFC (before ITC)	Highest	Highest	Short	Short	Longest	High	Low	High
SOFC (with ITC)	Low	High						

Source: BofA Global Research estimates, RaboBank, IEA, Bloom Energy, BCG, LCOE: Levelized Cost of Electricity, US Federal Government provides 30% Investment Tax Credit (ITC) subsidy to SOFC solutions

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Theme 4: 23% 25-30E BESS CAGR despite N-T concerns

BESS is becoming increasingly important for AI data centers, supporting both power quality management and backup power. We forecast China/global BESS installations of 264/425GWh in 2026 and 320/533GWh in 2027. Over the longer term, we forecast 2025-30 CAGRs of 22%/23% for China/global BESS installations, supported by renewable curtailment pressures, global electrification and AIDC demand. Driven by rising demand for AI computing power, AIDC-related BESS installations are expected to

reach 70GWh by 2030, representing a 27% CAGR and c.8% of incremental global BESS demand.

Since the year's start, we have observed a notable mismatch between ESS installation volumes and battery production schedules. China's ESS battery shipments reached 797GWh, up 95% YoY in 10M26, while China's ESS installations reached 99GWh, up 21% YoY in 9M26. China's non-EV battery exports reached 103GWh, up 46% YoY in 9M26. This mismatch raises concerns about demand sustainability, particularly in China. We believe the shipment-installation gap mainly reflects project lead times and inventory build-up, particularly overseas, as well as system redundancy, battery replacements and the exclusion of some off-grid projects from official statistics.

Exhibit 56: ESS value chain
ESS value chain

Source: Company data, BofA Global Research

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Exhibit 57: Forecast a 2025-30 CAGR of 22%/23% for China and global BESS installation
Global BESS installation forecast

	2020	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E
China new installation in GWh	3	5	16	47	102	183	264	320	385	442	490
Europe new installation in GWh	2	4	10	19	22	28	43	61	75	85	95
US new installation in GWh	4	11	17	24	35	54	57	64	74	81	89
ROW new installation in GWh	1	1	1	14	29	42	61	87	120	154	193
Total new BESS installation in GWh	**10**	**21**	**44**	**104**	**187**	**307**	**425**	**533**	**654**	**762**	**867**
YoY											
China new installation YoY		81%	224%	193%	118%	80%	44%	21%	20%	15%	11%
Europe new installation YoY		83%	148%	84%	15%	27%	54%	44%	23%	13%	12%
US new installation YoY		203%	53%	41%	47%	55%	6%	12%	16%	9%	9%
ROW new installation YoY		0%	-6%	1421%	103%	45%	44%	43%	37%	29%	25%
Total new installation YoY		**119%**	**109%**	**136%**	**81%**	**64%**	**39%**	**25%**	**23%**	**17%**	**14%**
Global ESS battery shipment in GWh	**27**	**44**	**121**	**185**	**301**	**550**	**850**	**1,060**	**1,260**	**1,380**	**1,525**
Global ESS battery shipment YoY		63%	175%	53%	63%	83%	55%	25%	19%	10%	11%

Source: SPE, Woodmac, CNESA, EESA, Pvinfolink, Bloomberg, BofA Global Research estimates

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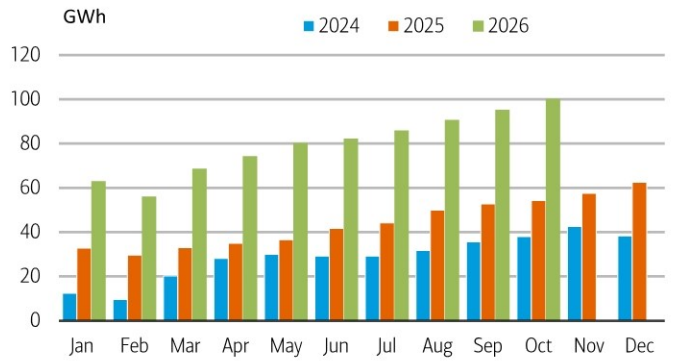
Exhibit 58: ESS battery cell production to +95% YoY in 10M26
ESS battery cell production in GWh



Source: SMM, BofA Global Research

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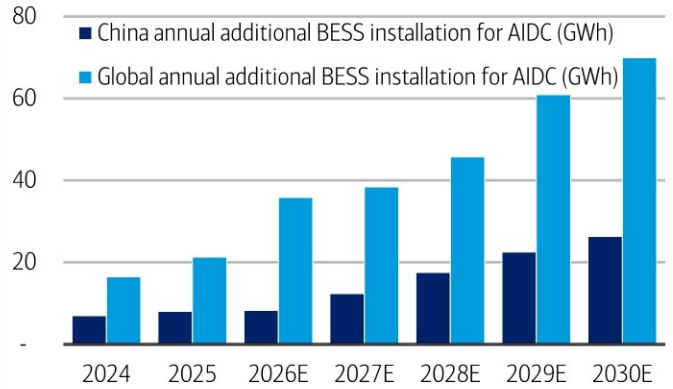
Exhibit 59: BESS additional installation for AIDC in 2030
BESS additional installation for AIDC to reach 70GWh in 2030



Source: BofA Global Research

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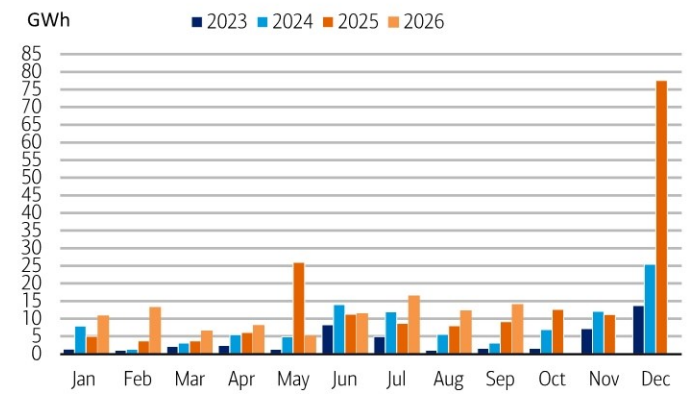
Exhibit 60: China ESS installation +21% YoY in 9M26
Monthly China ESS installation in GWh



Source: CNESA, BofA Global Research

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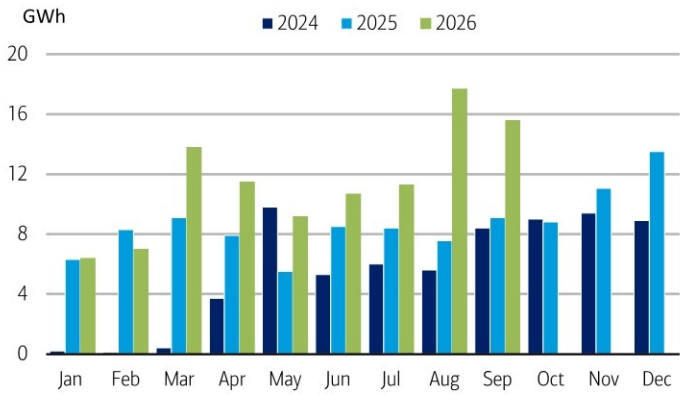
Exhibit 61: China non-EV battery export +46% YoY in 9M26
Monthly China Non-EV battery export in GWh



Source: CABIA, BofA Global Research

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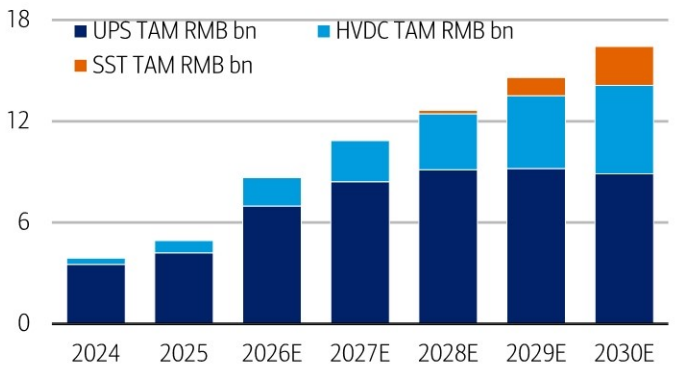
Theme 5: CN's power supply system: 27% 2025-30E CAGR

The power supply system is critical to AIDC because it ensures stable, continuous power for AI operations. Supported by continued AI spending by internet firms and government-led initiatives, we forecast the China AIDC power supply system market (UPS + HVDC + SST) to achieve a 27% CAGR over 2025-30E.

Uninterruptible power supply (UPS) remains the dominant data center power architecture, with an 85-90% share, while high-voltage direct current (HVDC) is gaining adoption because of its higher efficiency, lower costs and better renewable integration. Looking ahead, Nvidia's 800VDC roadmap and emerging solid-state transformer (SST) technology should drive demand for next-generation power infrastructure, including customized power systems, supercapacitors, BBUs and SiC/GaN semiconductors.

Exhibit 62: China AIDC power supply market (UPS+HVDC+SST) to deliver 27% 2025-30E CAGR
China AIDC power supply equipment TAM



Source: Companies, BofA Global Research

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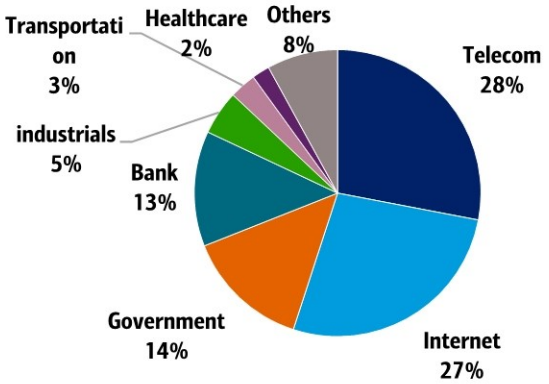
Exhibit 63: Internet and telecom account for 55% UPS application
Applications of UPS in China



Source: Companies, BofA Global Research

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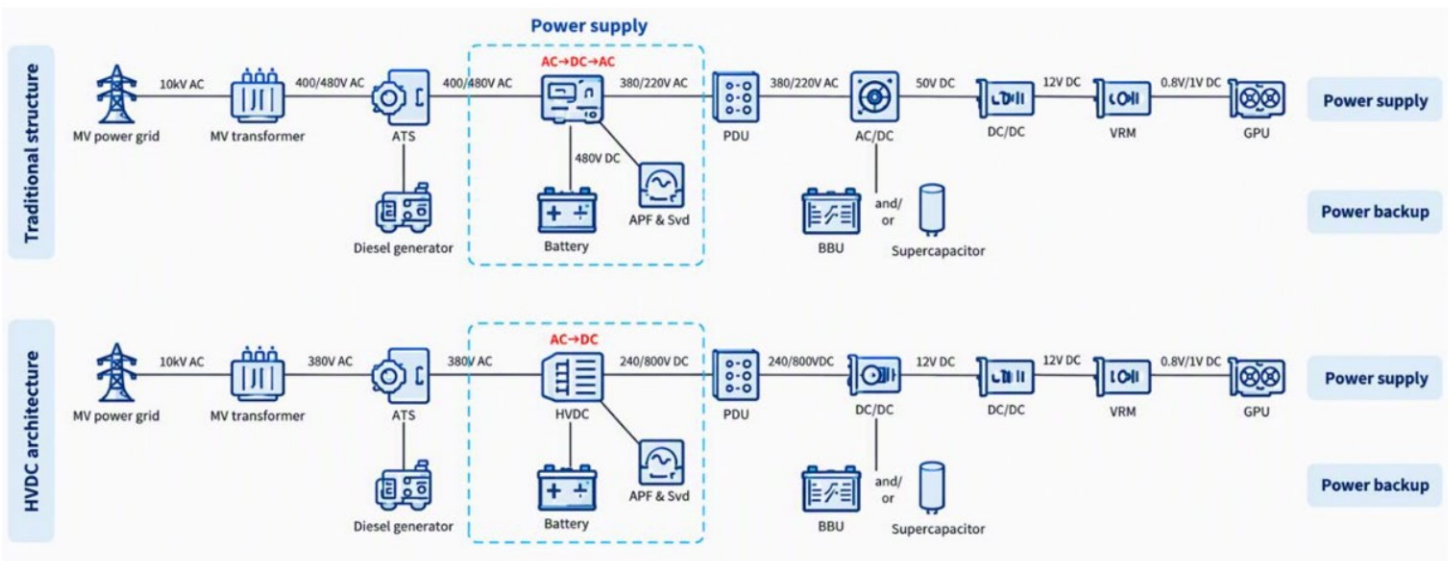
Exhibit 64: Illustration of traditional power supply architecture and HVDC architecture for AIDC
HVDC solution can achieve higher energy efficiency, thanks to less steps of power conversion



Source: BofA Global Research

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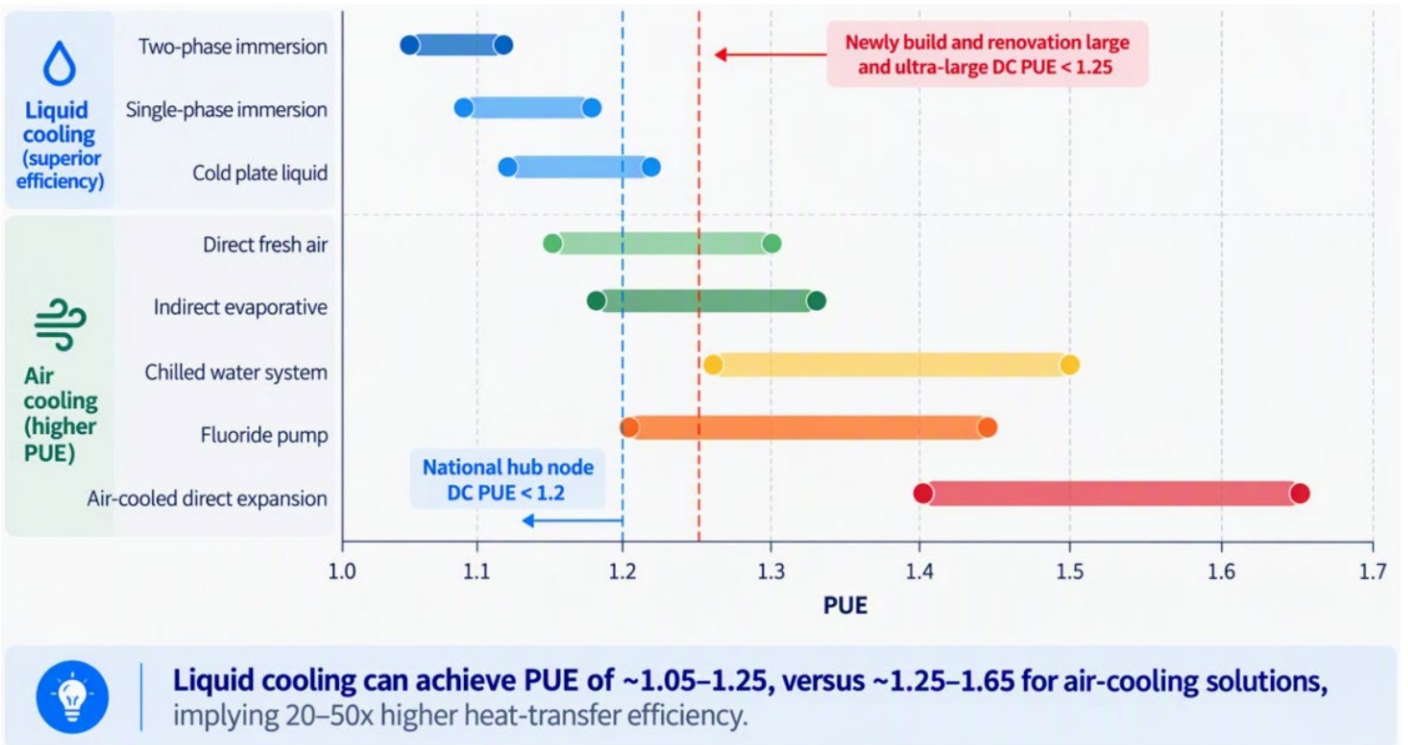
Cooling: rising power density drives liquid cooling demand

Liquid cooling is becoming increasingly important in AIDCs as rising AI computing demand increases electricity consumption and heat generation. As chip- and rack-level power densities rise, particularly beyond 40kW per rack, traditional air cooling is approaching its practical limits. Liquid cooling provides 20-50x greater heat-transfer efficiency, removes heat faster and consumes less energy than air cooling, potentially reducing PUE to c.1.1. Adoption is also supported by tighter energy-efficiency requirements and the shift toward liquid-cooling architectures in next-generation platforms, including NVIDIA's GB300 and Huawei's upcoming systems.

Exhibit 65: Liquid cooling has 20-50x higher heat-transfer efficiency than air
Liquid cooling has advantages in heat dissipation efficiency



Source: ZTE Liquid Cooling White Book, BofA Global Research

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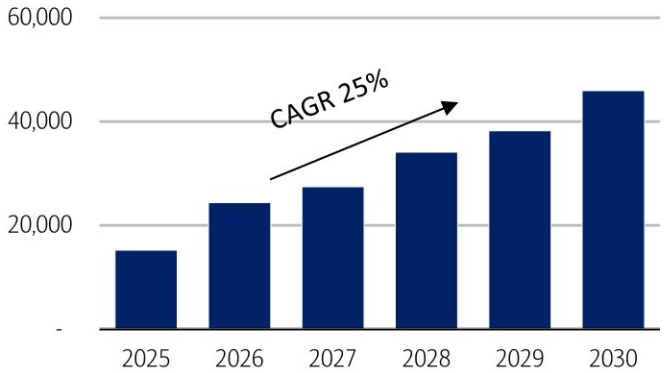
China cooling market to post 25% CAGR in 2025-30E

China's installed data center capacity is expected to rise from 29GW in 2025 to 78GW by 2030, driving the cooling market to RMB46bn by 2030. As AIDCs expand rapidly, we expect liquid-cooling penetration in new installations to increase from 30% in 2025 to 70% by 2030.

- Air-cooling demand is projected to increase from 3.2GW in 2025 to 4.4GW in 2030, representing a CAGR of 6%.
- Driven by high-density AI workloads, liquid-cooling demand is expected to surge from 1.4GW in 2025 to 10GW in 2030, representing a CAGR of 49%.

Overall data center capacity growth, combined with the structural shift toward liquid cooling, should make liquid cooling a key growth engine for China's data center infrastructure market.

Exhibit 66: China cooling market size to see 25% CAGR in 2025/30E
China cooling market demand and forecast (RMBmn)



Source: BofA Global Research estimates

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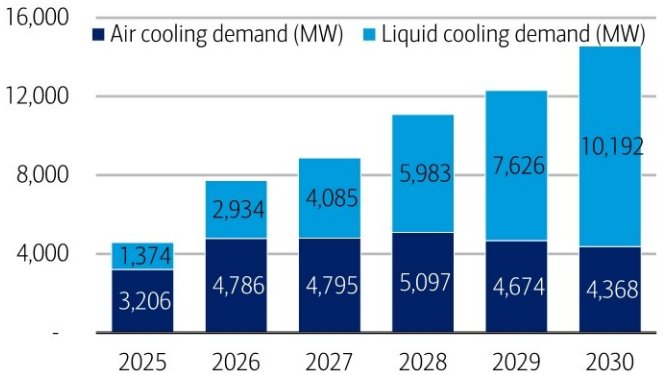
Exhibit 67: China liquid cooling penetration to grow to 70% in 2030E
China air cooling & liquid cooling demand



Source: BofA Global Research estimates

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Cold-plate liquid cooling currently accounts for over 90% of China's liquid-cooling market, supported by its compatibility with existing server architectures, mature supply chain and adoption in systems such as GB200. Immersion cooling remains at an early stage of commercialization, with a market share of c. 5-10%.

Immersion cooling offers greater heat-dissipation capacity and potentially simpler heat removal, but broader adoption remains constrained by material compatibility, equipment standardization, maintenance requirements and higher upfront costs. We estimate that its share of the liquid-cooling market will rise from 5% in 2025 to 9% by 2030, with the market reaching RMB6bn.

Exhibit 68: Comparison of major liquid-cooling technologies
Liquid-cooling systems vary by heat-transfer method and coolant phase

Criteria	Indirect		Direct		
	Cold plate type (冷板式液冷)		Immersion (浸没式液冷)		Spray (喷淋式液冷)
	Single-phase	Two-phase	Single-phase	Two-phase	
Main advantages	High compatibility; mature technology and complete industrial chain	Water in two-phase type does not enter the server	High heat dissipation capacity; energy efficient; quiet		Saving cooling liquid; precise cooling
Limitations	Leakage problem; requires air-cooling compensation	High working pressure; Difficult to control	Material compatibility; high load-bearing requirements	System sealing; High cost	Material compatibility; niche technology
Heat dissipation capacity	Medium	Medium-high	High	Highest	High
PUE	1.1-1.2	1.05-1.15	1.05-1.1	1.03-1.05	1.05-1.1
Maintainability	High		Medium		Medium
CAPEX	$	$-$$	$$-$$$	$$$	$$
OPEX	$$	$$	$$$	$$$	$$
Heat recovery potential	Medium	Relatively high	Relatively high	High	Relatively high
Maturity	★★★	★	★★	★	★
Application scenarios	Large-scale AIDC; Old DC upgrading		Small-scale AIDC; supercomputing, education, R&D		Small-sized high heat DC

Source: BofA Global Research

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FAQs

Question #1: Slower AIDC deployment and easing supply deficits
Recent downward revisions to AIDC power demand, driven by equipment shortages, tighter permitting and cautious capex. Expanding supply from alternatives gas gensets, SOFCs and emerging-market gas turbine suppliers, have raised expectations that power value-chain bottlenecks may gradually ease.

Our view: We believe investors have become overly selective amid concerns about delayed AIDC deployment and easing supply constraints. Demand is being deferred rather than destroyed, supporting a longer AI infrastructure capex cycle. Grid upgrades remain the key driver of transformer demand, while gas turbine supply is likely to remain tight through 2030, given sustained >1x book-to-bill. Investors are increasingly selective within "Powering AI," with key names including Sieyuan, Jereh and Weichai.

Question #2: Geopolitical uncertainty poses headwinds for Chinese exports
Geopolitical risks remain a key overhang for Chinese exports. Recent U.S. and European measures—including Executive Order 14420, U.S. FCC requirements, and European procurement and localization policies—have increased scrutiny of Chinese suppliers in critical infrastructure, raising compliance costs and barriers to market entry.

Our view: Potential exclusion from the U.S. market could pressure both the earnings and valuations of Chinese exporters. Risks are concentrated mainly in transmission and grid equipment, while generator suppliers face limited direct impact. Broader sentiment and peer de-rating could still weigh on the sector.

Question #3: Weak ESS demand visibility and potential inventory destocking
A widening gap between energy storage system (ESS) battery shipments and installations has raised concerns about inventory buildup and potential order cuts. China ESS battery shipments grew 95% YoY in 10M26, compared with 21% YoY growth in China installations in 9M26 and 30-40% YoY growth in overseas installations in 1H26.

Our view: China's monthly installations need to exceed 100GWh by end-2026 to absorb ESS battery inventory. Otherwise, the industry could face inventory destocking and a renewed oversupply cycle across the value chain.

Question #4: Weak earnings due to FX losses and Middle East tensions
Investors remain concerned that FX losses and ongoing Middle East tensions could weigh on near-term earnings. Currency volatility may create non-operating headwinds, while regional uncertainty could affect project execution, customer spending and order timing, reducing earnings visibility for exporters.

Our view: These are largely one-off, non-fundamental factors. Although near-term earnings may be affected, long-term demand and order backlogs remain intact. FX risk management will become an increasingly important focus for Chinese exporters.

Question #5: Delayed mass adoption of HVDC and SST
Investors are concerned that mass adoption of high-voltage direct current (HVDC) and solid-state transformers (SSTs) may take longer than expected because of technology validation, reliability requirements and lengthy customer approval processes.

Our view: Although adoption timelines may be pushed out, we see this as a timing issue rather than a structural setback. Over the longer term, rising AI rack power density and the industry's transition toward 800-volt direct current (800VDC) architectures should support demand. Although SST commercialization may take longer than expected, its superior efficiency should support long-term adoption.

Question #6: Risk of oversupply in China's AIDC liquid-cooling market
The growing number of companies announcing plans to enter the AIDC liquid-cooling market has raised concerns about potential overcapacity.

Our view: We see limited risk of overcapacity over the next two years. First, despite numerous market-entry announcements, only a small number of companies have begun generating meaningful revenue from AIDC liquid cooling. Second, liquid-cooling systems comprise a broad range of components, and some companies claim liquid-cooling exposure despite supplying only a small component within the overall system. Third, liquid-cooling suppliers face lengthy product-validation and customer-qualification cycles. Therefore, even if more companies begin investing across the full liquid-cooling value chain today, we believe the risk of industry-wide overcapacity will remain low over the next two years.

AI equipment: cherry picking

Compared with AI power and AI materials, we see Chinese companies have yet to establish a dominant position across the broader AI equipment market. However, we believe selected domestic suppliers are advancing rapidly in niche segments, supported by above-average execution, shorter delivery cycles and faster capacity ramp-up. Once technological breakthroughs are achieved and customer validation is secured, we see these strengths can accelerate import substitution and market-share gains for Chinese AI equipment suppliers. We highlight several promising segments and related Chinese equipment suppliers, including optical transceiver assembly and testing equipment, CPO testing equipment, PCB drill bits and PCB laser processing equipment.

Optical transceivers equipment TAM at CAGR 67%

We expect a multi-year optical equipment upcycle as pluggable transceivers migrate from 800G in 2025-26 to 1.6T in 2027-28 and 3.2T beyond 2028. Rising technical requirements are increasing demand for precision assembly and testing equipment, while CPO, NPO, Optical I/O and OCS should create additional opportunities in advanced packaging, alignment and high-speed testing.

We estimate that the global OT manufacturing equipment market will reach RMB55bn by 2028E, implying a 67% CAGR over 2025-28E. The migration from 800G to 1.6T/3.2T requires greater automation, coupling precision and testing sophistication, driving equipment investment. We estimate that coupling, die attach, wire bonding and testing represent 40%/20%/10%/30% of equipment spending, respectively. We see Chinese players are gaining share in optical coupling, die attach and testing equipment, supported by their proximity to leading Chinese OT manufacturers, shorter delivery lead times, as well as competitive pricing. As Chinese companies account for the majority of global OT capacity expansion, we believe rising equipment localization should create further import-substitution and market-share opportunities for domestic suppliers.

Exhibit 69: Optical interconnect technology migration to drive higher demand for optical assembly and test equipment
Pluggable optical transceiver speeds upgrade, CPO/NPO/Optical I/O and OCS adoption to drive demanding specifications, hence higher equipment demand

Source: BofA Global Research

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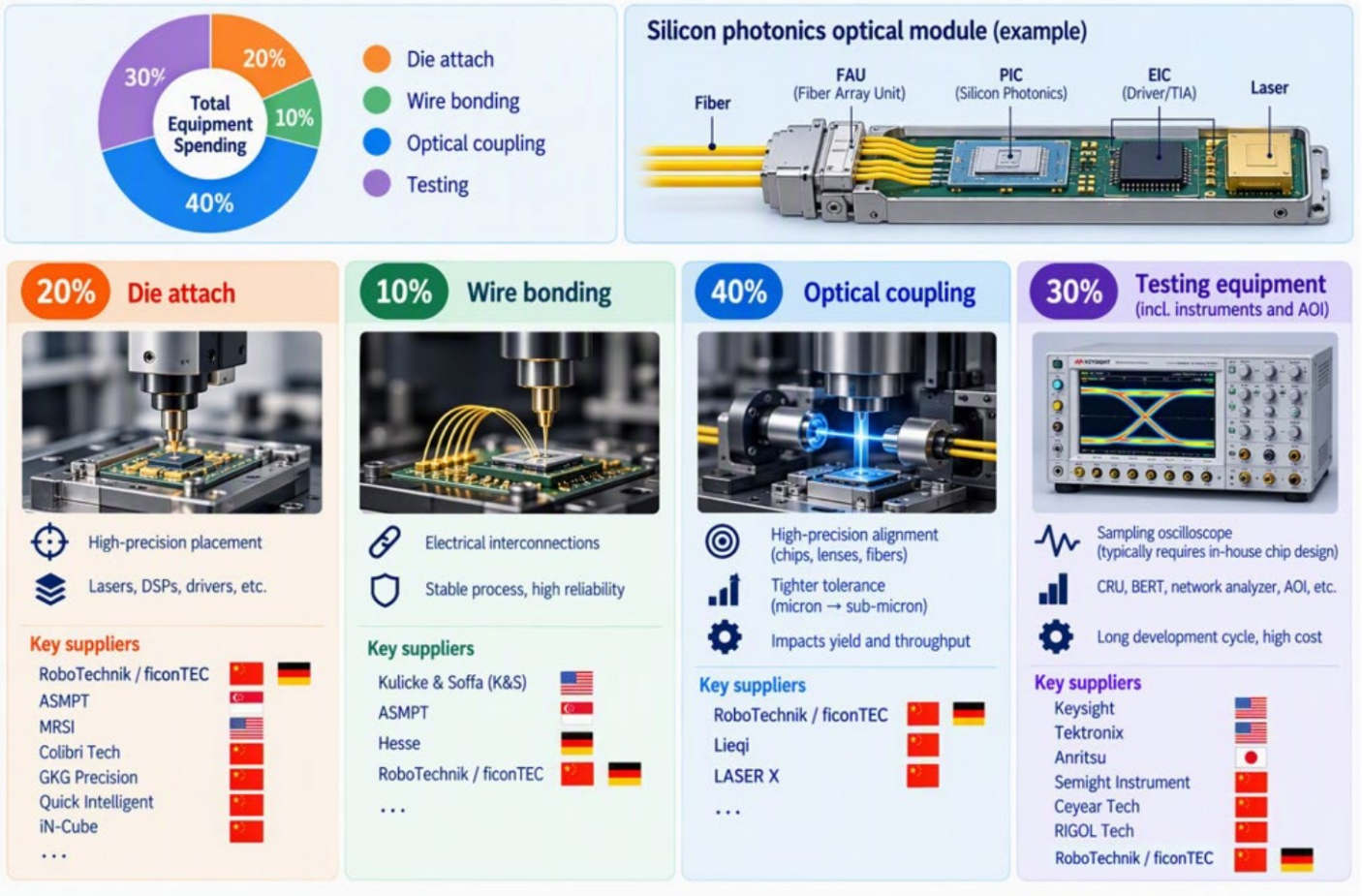
Exhibit 70: Optical coupling and testing account for 70% of total optical equipment spending; Chinese suppliers are gaining share
Optical transceiver manufacturing equipment value chain and key suppliers



Source: BofA Global Research estimates

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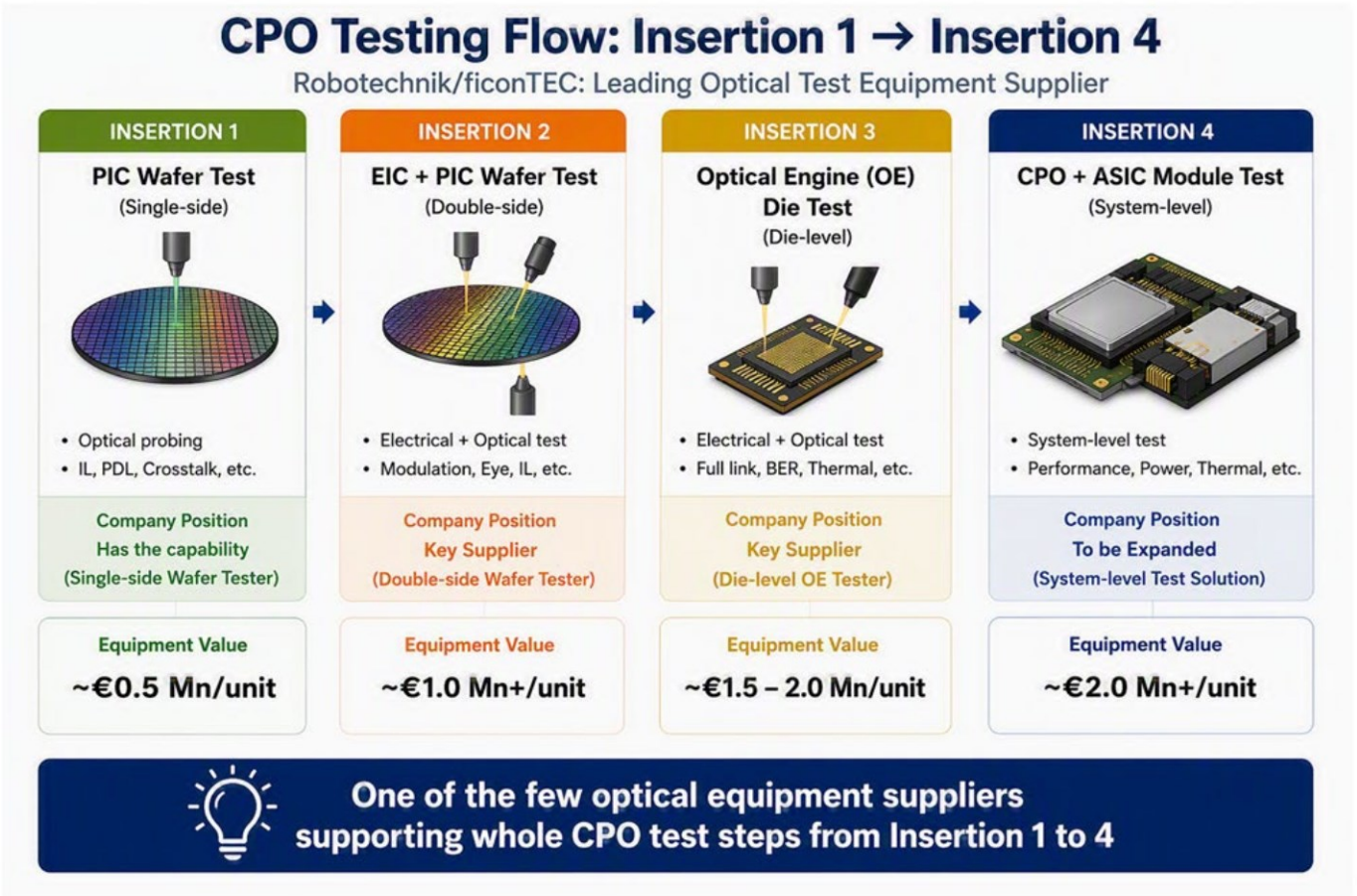
CPO adoption requires front-loaded CPO tester demand

Our Global Tech team (see report) forecasts CPO penetration in AI optical ports to reach 6%/12%/17% in 2028/29/30E, with CPO equipment demand 1-2 years front-loaded into 2026-27. We see RoboTechnik is an early beneficiary of front-loaded CPO tester deployment through its ficonTEC, who has established a leading presence in Insertion 2/3 testing equipment and is extending its product coverage into Insertion 4. The company has received CPO tester orders in 2026, and we see the CPO tester demand could accelerate in 2027–28 ahead of broader CPO production ramps. Given the stringent optical-loss tolerance and high cost of post-integration failures, CPO testing requires advanced capabilities across PIC, EIC and optical-engine testing. RoboTechnik's high-precision alignment technology, testing know-how and deep customer engagement should reinforce its first-mover advantage and support continued share gains.

Exhibit 71: RoboTechnik is the key optical testing equipment supplier in Nvidia/TSMC CPO test insertion 2 and 3, may further expand to insertion 4
CPO testing workflow: Ins1 (PIC single-side wafer test), Ins2 (PIC+EIC double-side wafer test), Ins 3 (OE die test), and Ins4 (CPO + switch ASIC module test)



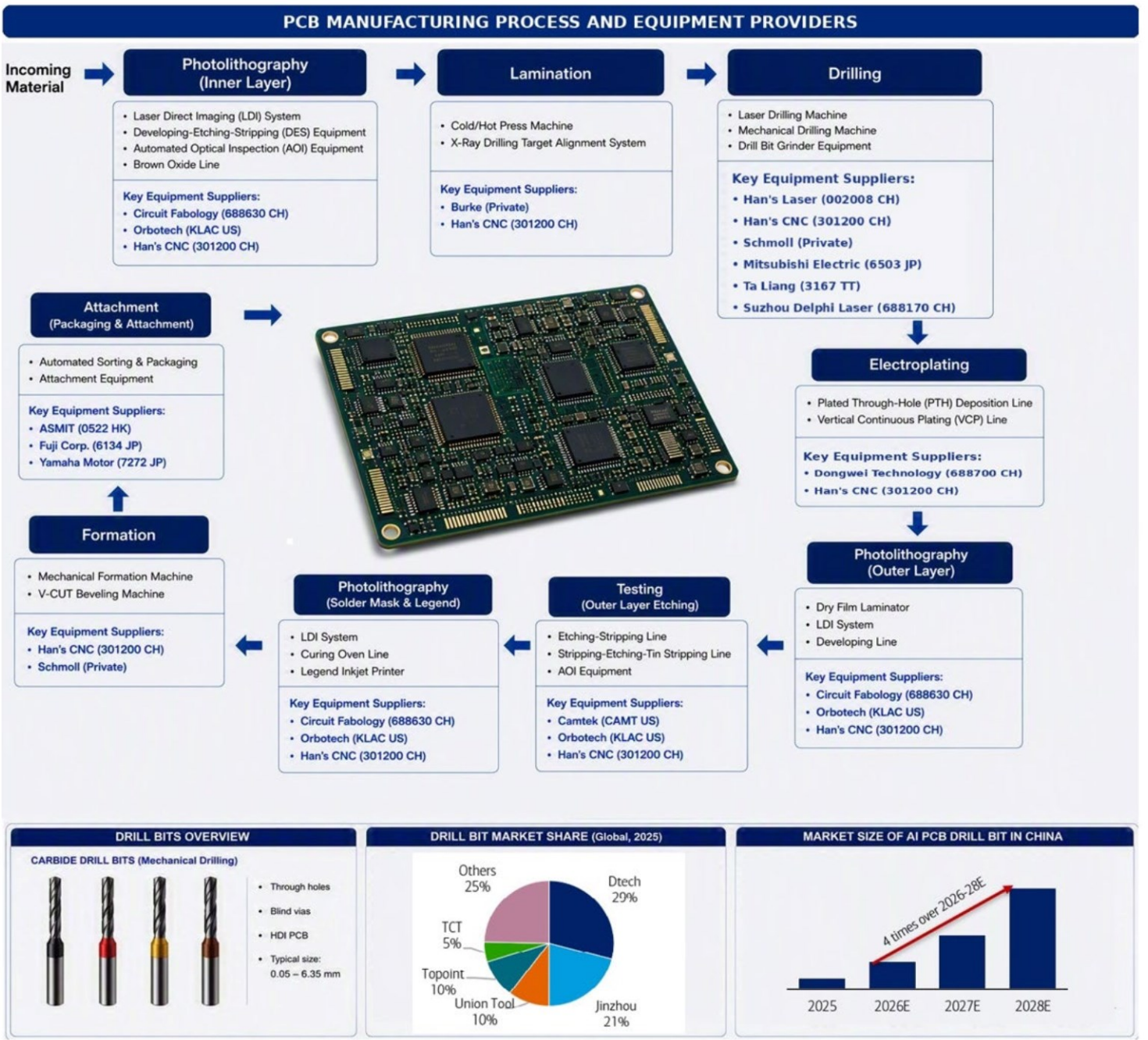
Source: BofA Global Research. CPO testing is conducted at four points to identify defects:
Ins1 (Insertion 1): single-side PIC wafer test, using optical probes to test the PIC wafer from one side, measuring IL (insertion loss, refers to optical power lost), PDL (polarization-dependent loss, refers to the variation in loss under different light polarizations), crosstalk (refers to unwanted signal leakage between optical channels)
Ins2 (Insertion 2): PIC and EIC bonded double-side wafer test, combining electrical probing with optical coupling to evaluate modulation performance, eye diagrams and optical loss
Ins3 (Insertion 3): testing the singulated optical engine (OE) at die level through a complete electrical and optical test solution, including full-link performance, bit error rate (BER) and thermal behavior, before the OE is integrated with the expensive switch ASIC
Ins4 (Insertion 4): testing the packaged CPO + ASIC module under system-level operating conditions, covering overall performance, power consumption and thermal stability

BofA GLOBAL RESEARCH

AI PCB drill bit market will deliver a 96% CAGR

We believe strong AI cloud capex and accelerating server upgrades are driving a faster PCB upgrade cycle, resulting in surging demand for PCB equipment and drill bits. Demand for PCB drilling equipment should be supported by resilient growth in mechanical drilling, reflecting increased use of HLC PCBs, as well as market share gains by Chinese OEMs in laser drilling equipment, underpinned by technological breakthroughs and greater capacity. We expect demand for PCB drill bits to grow exponentially, given their consumable nature. We estimate the AI PCB drill bit market (NIVDIA related) will deliver a 96% CAGR over 2026-28E, reaching RMB14bn by 2028E. Compared to global peers, we estimate drill bits manufacturers in China (>50% market share globally) to gain more upside from the surging demand, given their stronger execution in capacity expansion.

Exhibit 72: Manufacturing process of PCB and relevant equipment/drill bits providers
The PCB upgrade cycle triggers demand growth for PCB drilling equipment and drill bits



Source: Company data, BofA Global Research

BofA GLOBAL RESEARCH

Incremental usage of PCB=>more drill bits volume

We expect a wider variety of PCBs in next-generation AI servers. Nvidia's Rubin platform will include several additional PCB components: 1) a **midplane** (44-layer HLC with M8.5 CCL) replacing the overpass cable used in GB300; 2) an optional Rubin **CPX** board in the compute tray; and 3) additional PCBs in the **ConnectX** NIC/**BlueField** DPU. Into 2027-28, if backplane for Rubin Ultra is introduced, we are likely to see ongoing content value upside for PCB supply chain. Longer term, adoption of **CoWoP** could further increase PCB demand by using the PCB as the IC substrate within AI servers (see our report: Technology-APAC-Overview of AI PCB).

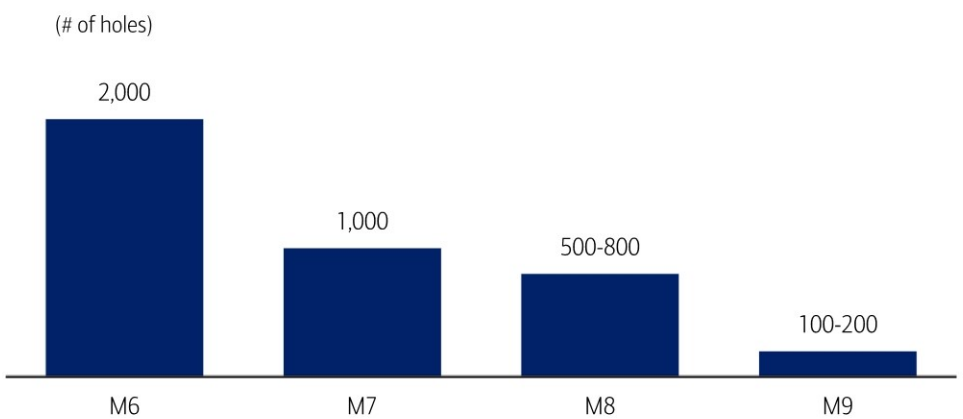
HLC require step-drilling=> surging demand for drill bits

We believe the rising penetration level as well as layer upgrade for HLC translate into more demand for step-drilling, given the rising thickness level (i.e., switch tray mainboard evolves to 32 layers for Vera Rubin & Rubin Ultra vs. 22 layers for GB300). The demand will surge significantly if backplane is introduced for Rubin Ultra, given its ultra thickness level of >70 layers. Step-drilling is a drilling technique used for thick multilayer boards and high aspect-ratio vias, where the drill bit does not penetrate the entire board thickness in a single pass. For example, if the thickness level of a PCB is >6mm, step drilling will firstly use 2mm long drill bits, then switch to a >4mm drill bits and eventually use >6mm drill bits to complete it. Compared to single-pass drilling, step drilling has lower drilling hit and smoother hole wall, and thus, ensures better quality. By adopting step-drilling, the consumption volume of drill bit surges as it require for more than one type of bits in drilling for one single hole.

Potential migration to M9 material=>more usage volume

We also see rising demand for drill bits with the upgrade of CCL materials within PCB. With the migration from M7/M8 to M9 materials, the adoption of harder electronic-grade glass-fabric materials and high-performance copper foil has significantly increased the breakage rate of drill bits. Notably, the Q cloth used in M9 materials (to be adopted in Rubin/Rubin Ultra AI servers from 2H26 onwards) has a much higher SiO_2 content (>99.9%). As such, the usable life of coated drill bits drops sharply to only 100–200 holes for M9 CCL, from 500-800 holes for M8 and c.1,000 holes for M7. This translates into c.5x demand for drill bits in the next-generation AI servers, vs. the GB200/GB300 era.

Exhibit 73: Usable life of coated drill bits drops sharply to only 100-200 holes for M9 CCL
Number of holes each drill bit produces by different types of materials



Source: BofA Global Research

BofA GLOBAL RESEARCH

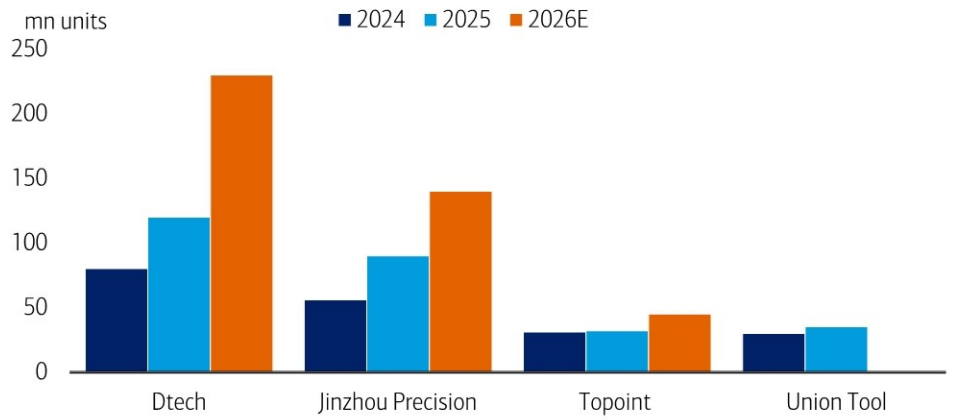
China drill bits manufacturers have faster capacity addition

The supply of drill bits has been tight in recent quarters, due to surging demand from AI servers coupled with limited supply. Against this backdrop, major drill bit producers have hiked prices because of the tight supply and upstream cost pressures: Dtech has raised PCB drill bit prices by >10% in 1H26; Shareate Tools has also implemented >20%/>35% price hikes for ordinary/high-end drill bits in 1H26. Looking ahead, we expect the tight supply-demand to persist in 2026-28, especially for the high-aspect-ratio drill bits (>30:1), given (1) exponential demand growth for PCB drill bits driven by the ramp-up of HDI/HLC and CCL material evolution; and (2) limited players and capacity for high-end PCB drill bits. This could further support the prices of drill bits in the coming 1-2 years.

In response to the surging demand from AI PCB, leading players in China in the PCB drill bit industry are rapidly expanding their production capacity, enabling them to reap more upside from the growing market. We estimate the total production capacity of the global

top-4 players to grow >60% YoY by end-2026. Of these, Dtech could see the fastest growth in production capacity (nearly double YoY) by end-2026, which would enable it to further gain market share from its peers.In addition to the top-4 suppliers, there are some emerging players entering into the PCB drill bit market, including Shareate (Good Team Electronics), Up-shine (Xiazhi Precision), and OKE (Yongxin). Some other players such as Worldia and SF Diamond are also entering into the market with polycrystalline diamond (PCD) drill bits.

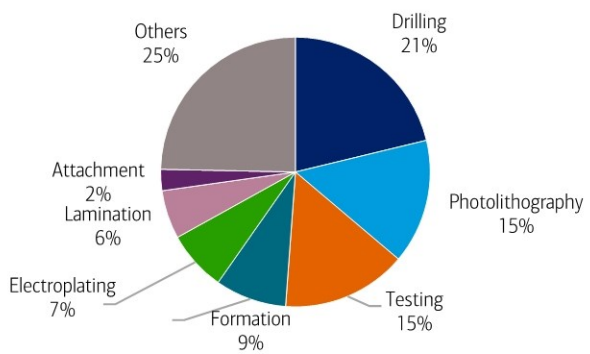
Exhibit 74: Production capacity of the global Top-4 drill bit suppliers
Dtech is expanding production capacity at a faster-than-peers' pace



Source: Companies, BofA Global Research estimates

BofA GLOBAL RESEARCH

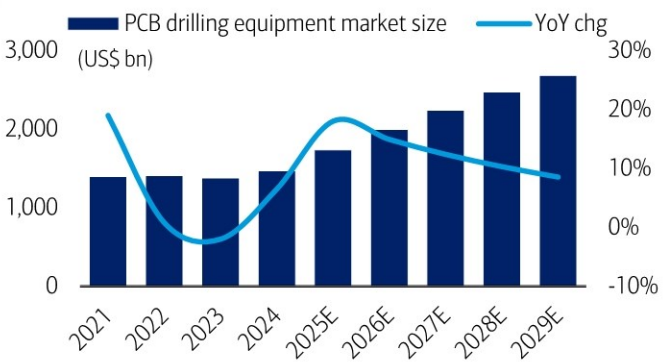
Exhibit 75: Breakdown of PCB equipment market (2025)
Drilling and photolithography represent higher market size



Source: Han's CNC prospectus

BofA GLOBAL RESEARCH

Exhibit 76: Market size and growth rate of PCB drilling equipment
The drilling equipment market will achieve 12% CAGR over 2025-29E



Source: Han's CNC prospectus

BofA GLOBAL RESEARCH

FAQs

Question #1: Whether automated equipment is essential for Chinese OT makers
Many Chinese optical transceiver manufacturers have traditionally relied on manual or semi-automated assembly, raising concerns that their preference for asset-light production could constrain demand for fully automated assembly and testing equipment.

Our view: We believe the ramp-up of 800G, 1.6T and 3.2T SiPh transceivers, along with adoption of NPO, CPO and OCS, will require tighter coupling tolerances, greater process consistency and higher testing throughput. In addition, overseas capacity expansion by Chinese OT makers should accelerate adoption of fully automated equipment, helping to mitigate high labor costs and improve production efficiency outside China. Demand for high-precision automated OT equipment is therefore growing.

Question #2: Fiercer OT equip. competition could pressure pricing and margins
Investors are concerned that intensifying competition could reduce industry-wide equipment pricing and profitability as more Chinese optical transceiver equipment vendors enter the market with products priced at lower ASPs than those of overseas suppliers.

Our view: We expect profitability for leading OT equipment vendors to remain stable or improve, with competitive pricing pressure offset by: 1) product-mix upgrades involving greater alignment precision and higher-speed testing, which should increase equipment dollar content; 2) higher shipment volumes, which should generate manufacturing scale benefits; and 3) recurring maintenance and upgrade services, which should provide additional margin support.

Question #3: Early-stage CPO adoption could delay equipment demand
Investors are concerned that CPO remains at an early stage of commercialization, with limited visibility on near-term deployment. This raises questions about whether meaningful demand for CPO assembly and testing equipment could take longer to materialize.

Our view: We expect CPO adoption to inflect meaningfully from 2028 and reach 17% penetration of total AI optical network ports in 2030, supported by the deployment of scale-up CPO switches on Nvidia's Feynman platform. Because equipment procurement and capacity preparation typically precede mass production, we expect CPO equipment deliveries to begin ramping in 2026-27. Our supply chain check suggests that CPO commercialization remains on track, with deliveries of CPO testers already underway to support both volume production and capacity expansion.

Question #4: The competitive landscape for CPO equipment remains unclear
Market share allocation in CPO test equipment remains under debate. RoboTechnik (ficonTEC) indicates that it has established positions in CPO Insertion 2 and Insertion 3 equipment and is validating Insertion 4 solutions. Taiwanese peers, including Chroma and MPI, have also reported progress across the Insertion 2/3/4 stages. This makes it difficult to identify a clear long-term winner at this stage.

Our view: We believe CPO equipment demand is emerging from a broad customer base, including fabless firms, foundries and OSATs, making it too early to determine eventual vendor market shares. We believe investors should focus on order intake, customer validation and shipment execution. As the market scales, multiple vendors with proven technology and first-mover advantages are likely to benefit, rather than the market developing into a winner-takes-all outcome.

Question #5: Changes in competitive landscape for PCB equipment and drill bits
Leading drill bit manufacturers are expanding capacity to meet rising demand from AI PCBs, while new entrants are seeking to gain market share. This has raised investor concerns about the future competitive landscape, pricing and margins.

Our view: We believe barriers to entry for drill bits remain high. First, switching costs are high. Drill bits account for only a low-single-digit percentage of total PCB manufacturing costs, but their performance directly affects drilling quality and production efficiency, making reliability a key purchasing consideration. In addition, precision manufacturing equipment—a key bottleneck for high-end drill bits—remains in tight supply globally. We therefore do not expect a significant capacity ramp-up in drill bits, particularly high-end products used in AI PCBs.

Question #6: Uncertainty over future PCB technology and material trends
Backplanes are the largest swing factor for drill bit demand because they are significantly thicker (>70 layers) and larger. However, the market remains cautious about the launch timeline, given that the relevant technologies and materials are not yet mature.

Our view: We have not incorporated any impact from backplanes into our Dtech model. Looking ahead, progress in technology development or material innovation, such as M9/PTFE hybrids, could provide an upside catalyst.

Key market participants

AI materials

AI metals

Zijin Mining (2899 HK; Buy)

Zijin is the largest copper and gold miner in China and remains one of our top picks, supported by: 1) resilient prices across lithium, copper and gold; 2) strong multi-year volume growth, with copper output expected to increase by 7%/11%, gold by 15%/6%, and lithium reaching 121kt/189kt in 2026/27E; and 3) an undemanding valuation compared with global copper/gold peers.

Jiangxi Copper (358 HK; Buy)

Jiangxi Copper is China's largest copper smelter. We maintain Buy, supported by: 1) high sulfuric acid prices; 2) resilient copper prices; 3) a long-term increase in self-mined copper concentrate volume; and 4) potential upside from the company's copper foil business.

CMOC (3993 HK; Buy)

We have a Buy rating, given: 1) resilient copper price, 2) its consistent tech reform should improve copper volume (+7% YoY), 3) cobalt price surge to offset the impact on likely extended cobalt export ban, and 4) solid B/S, qualified new set of management, more M&As (e.g. Lumina Gold, Leagold & Luna Gold in Brazil), and the expansion of TFM/ KFM to 0.8-1mnt of copper by '28 offering solid ground of development.

MMG (1208 HK; Buy)

We have a Buy rating on MMG, given: 1) its transition from a single asset story into a diversified copper growth platform, with Las Bambas stabilizing and Khoemacau + Kinsevere ramping, 2) 20% copper volume growth to 600kt by 2028, driven by Chalcobamba/Ferrobamba upgrades at Las Bambas, Khoemacau Phase II targeting 130ktpa, and Kinsevere sulphide expansion adding 80ktpa, 3) strong leverage to a tightening copper market.

Xiamen Tungsten (600549 CH; Not covered)

Xiamen Tungsten is one of China's leading tungsten material producers with diversified exposure across tungsten & molybdenum, rare earths and battery materials. Key subsidiaries include Xiamen Golden Egret (cutting tools), Xiamen Honglu (tungsten wire), Jinlong Rare Earth and XTC New Energy (688778 CH), providing broad exposure to advanced materials and new energy supply chains.

China Tungsten and Hightech Materials (000657 CH; Not covered)

China Tungsten is the flagship tungsten platform under China Minmetals Group, with operations spanning the entire tungsten value chain from mining and smelting to tungsten powder, cemented carbide, cutting tools and PCB micro-drills. Its major subsidiaries include Zhuzhou Cemented Carbide, Zhuzhou Diamond Cutting Tools and Shenzhen Jinzhou Precision Technology, making it one of China's most integrated tungsten manufacturers.

JL Mag (6680 HK, Buy)

JL Mag Rare-Earth is a leading Chinese manufacturer of high-performance NdFeB permanent-magnet materials for automotive, wind-power, appliance, and other applications. We maintain Buy, supported by: 1) accelerating demand for high-performance NdFeB magnets, particularly from emerging applications such as humanoid robots; 2) a clear capacity-expansion roadmap that reinforces its position as the industry's scale leader; and 3) China's tighter rare-earth export controls, which raise entry barriers and further concentrate market share among top-tier magnet producers, benefiting JL Mag as the global leader.

PCB materials

Defu Tech (301511 CH; Buy)

Defu Tech is a leading Chinese copper foil supplier, with PCB copper foil capacity expected to increase from 35kt to 85kt by end-2027. It targets HVLP shipments of 800t/month by end-2026 and 2,000t/month by end-2027, of which 50% will be HVLP4. In addition to equipment ordered from Mifune, its self-developed equipment can already produce HVLP4 foil. Buy Defu Tech on: 1) volume growth from the 50kt capacity expansion; 2) mix upgrades toward higher-ASP HVLP3/4; and 3) greater use of self-developed equipment to support the high-end product ramp-up.

Tongguan Copper Foil (301217 CH; Not covered)

Tongguan is one of China's leading copper foil suppliers, focusing on high-precision electronic copper foil. Its key products include HTE, RTF and HVLP copper foil for PCBs, as well as lithium-battery copper foil used as an anode current collector.

Sinoma Science & Technology (002080 CH, Buy)

We rate Sinoma Sci & Tech Buy on its transformation from a traditional materials producer with cyclical earnings into a cutting-edge AI materials supplier with AI-driven growth. It leads Chinese producers in commercializing special e-glass and has supplied bulk volumes to a leading CCL maker since 2025, capturing demand that Japanese producers cannot meet because of capacity constraints. Sinoma plans to nearly quadruple capacity from 24mn meters currently to 94mn meters by 2027 through three expansion projects. We expect the improving product mix to lift unit margins, as premium products such as low-Dk2, low-CTE, and Q-glass are at least 2-3x more expensive than low-Dk1, which currently accounts for most output.

China Jushi (600176 CH, Buy)

Historically, glass-fiber roving has contributed most of Jushi's earnings, while e-glass margins have been only marginally positive. However, consecutive monthly e-glass price increases should materially boost earnings. We forecast e-glass earnings to overtake roving earnings from 2027 and remain high amid multi-year supply tightness driven by sustained industry-wide product upgrades. Jushi's expansion into AI special e-glass offers another long-term growth catalyst that is not yet priced in. We view Jushi as an earnings-driven re-rating story. Buy.

Optical materials

Yangtze Optical Fibre and Cable (6869 HK; Not covered)

YOFC is a leading global supplier of optical fibre preforms, optical fibres and cables, with an integrated value chain extending into structured cabling, optical modules and communication network engineering. It is dual-listed on the Shanghai Stock Exchange and Hong Kong Stock Exchange.

Hengtong Optic-Electric (600487 CH; Not covered)

Hengtong is a leading Chinese provider of communication network and energy interconnection products and solutions, with offerings spanning optical fibre preforms, optical fibres and cables, submarine cables, power cables and offshore wind applications. The company has integrated capabilities across design, R&D, manufacturing, sales and services.

Jiangsu Zhongtian Technology (600522 CH, Buy)

We like Zhongtian for its exposure to higher fiber prices, a richer specialty-product mix, and an emerging demand uptrend that should outpace supply over the next 1-2 years. We expect growth in multi-fiber push-on (MPO) products, expanding capabilities with overseas cloud service providers (CSPs), and continued strength in subsea cable to support further re-rating. We estimate a c.112% earnings CAGR in 2026-27E, while valuation remains attractive at 8x 2027E P/E.

Yunnan Germanium (002428 CH; Buy)

We rate Yunnan Germanium Buy. The company is transitioning from a cyclical germanium producer into a strategic supplier of semiconductor materials. We expect InP substrates to become its primary profit driver from 2027, supported by accelerating demand for AI optical infrastructure, growing domestic-substitution requirements, and expanding production capacity. We forecast NPAT of RMB0.3bn/0.9bn/1.5bn in 2026/27/28, implying a multi-year earnings inflection and valuation re-rating.

Semi materials

Three-Circle Group (300408 CH; not covered)

Three-Circle is an advanced materials and electronic components manufacturer. Its main products include MLCCs, precision ceramic components, fiber-optic ceramic ferrules, ceramic packaging substrates and electronic materials.

Fenghua Advanced Technology (000636 CH; not covered)

Fenghua is a diversified passive components manufacturer. Its main products include MLCCs and other capacitors, chip resistors, inductors, varistors and related electronic materials.

Sinocera (300285 CH; not covered)

Sinocera is an advanced materials manufacturer with businesses spanning ceramic powders, catalytic materials, biomedical materials, precision ceramics and thermal-interface materials. Its MLCC-related products mainly include ceramic dielectric materials, including barium-titanate-based powders.

Peric Special Gases (688146 CH; Not covered)

Peric is a specialized supplier of electronic specialty gases and fluorinated chemicals. Its key products include NF_3, WF_6, C_4F_8, C_4F_6 and HCl, mainly used in semiconductor deposition, etching and chamber-cleaning processes.

Huate Gas (688268 CH; Not covered)

Huate is one of China's leading private suppliers of specialty gases and related equipment, with a broad portfolio of more than 200 products. Its offerings cover electronic specialty gases, industrial gases and gas mixtures, serving semiconductor, display, photovoltaic and other manufacturing applications.

Nata Opto-electronic Material (300346 CH; Not covered)

Nata is a leading Chinese supplier of high-purity electronic materials, with operations spanning advanced precursors, electronic specialty gases, photoresists and related materials. Its specialty-gas portfolio includes PH_3, AsH_3, NF_3 and SF_6 for semiconductor and compound-semiconductor manufacturing.

Jinhong Gas (688106 CH; Not covered)

Jinhong is an integrated gas supplier offering specialty gases, bulk gases and related gas services. Its products serve semiconductor, display, photovoltaic, LED and a broad range of industrial customers, positioning the company as a one-stop gas-solutions provider.

AI equipment: PCB and optical Powering AI & Cooling

Transformer and power equipment

Sieyuan (002028 CH, Buy)

Sieyuan is a leading, diversified grid-equipment provider. We like the company for: 1) its edge in high-voltage GIS and transformers, which have higher entry barriers and stronger margins; 2) market-share gains in SGCC tenders; 3) growth in higher-margin overseas markets and a diversified export footprint; and 4) margin improvement as high-end products and exports contribute more.

Nari (600406 CH, Buy)

Nari is a leading, diversified grid-equipment provider. We like the company for: 1) the industry's most comprehensive product line, which makes it less vulnerable to macroeconomic fluctuations; 2) its position as a late-cycle beneficiary of grid investment, with ~85% of revenue from secondary equipment; 3) a solid track record of delivering quality earnings growth; and 4) support from State Grid Corporation of China, its largest shareholder and the source of c.60% of revenue.

Jinpan (688676 CH, Buy)

Jinpan is a leading Chinese transformer supplier, with a higher share of overseas revenue than peers (29% in 2024) and an advantage in data-center orders. We estimate that data-center-related customers will account for 20% of revenue in 2026, up from 7% in 2024, driven by: 1) rising demand as ISPs and CSPs expand capex; 2) sufficient capacity; and 3) R&D investment in new products.

Huaming (002270 CH, Buy)

Huaming is China's largest producer of tap changers. We like the company because: 1) >90% of profit comes from tap changers, an oligopolistic industry with high entry barriers and margins; 2) demand for transformers and tap changers is rising; and 3) it is gaining share from MR, which has longer order lead times.

Gas turbines and engines

Jereh (002353 CH, Buy)

Jereh is an emerging gas-turbine packager for US AIDC. We like the company because: 1) its transformation from an O&G equipment provider into a gas-turbine packager creates a compelling growth path; 2) AI-led power demand, industry-wide turbine shortages, and strong execution and channel advantages should support sustained profit growth and re-rating potential; and 3) natural-gas equipment and services orders in MENA and APAC should offset weakness in the oil businesses.

Dongfang Electric (1072 HK, Buy)

Dongfang is a leading power-generation equipment maker in China. We like Dongfang Electric for: 1) its leadership in China's baseload power-generation equipment market and strong revenue recognition; 2) potential expansion in gas-turbine exports; 3) positive sentiment around the Yarlung Zangbo project's long-term earnings contribution; and 4) sustainable growth in the power-station services business.

Yingliu (603308 CH, Buy)

Yingliu is a leading Chinese maker of hot-section components. We like its strong position in the global aerospace and industrial gas-turbine (IGT) hot-section components market, which has high entry barriers and tight capacity. We expect Yingliu to benefit from strong growth in gas-turbine deliveries and gain share as it expands production capacity and its client base.

Weichai Power (2338 HK, Buy, 000338 CH, Buy)

We rate Weichai A/H Buy, supported by its fast-growing large-bore engine business. Weichai entered the segment in 2009 through its acquisition of the French brand Baudouin. Its current large-bore engine products for DC include the M33 (0.5-2.1MW) and M55 series (1.8-4.2MW). We expect large-bore engines to contribute >25% of earnings in 2028, versus <20% in 2025.

Yuchai International (CYD US, Buy)

We rate Yuchai Buy, supported by solid growth in data-center engines. Yuchai is China's leading ICE engine brand and sold more than 2k large-bore power-generation engines in 2025, including 1.3k units for DC. We expect power-generation engine revenue to post a 24% CAGR in 2026-28 and account for 42% of total revenue in 2028, up from 13% in 2025. We expect Yuchai-MTU sales to reach 1.3k/1.6k units in 2026/27E.

ESS and power supply systems

Sungrow (300274 CH, Buy)

We like Sungrow for: 1) our positive view of the global ESS market and the company's strong ESS shipment growth; 2) ROE of more than 30% and a net-cash position; 3) a solid global footprint, with higher margins and entry barriers; and 4) potential long-term upside from AIDC.

CATL (3750 HK, Buy)

CATL is the global market leader in EV and ESS batteries. We rate CATL Buy, supported by a solid growth outlook for its ESS battery business and a stable margin outlook in 2026, underpinned by its strong market position.

Megmeet (002851 CH, Buy)

We like Megmeet because: 1) its AI server power-supply business is likely to deliver strong sales growth in 2026-28 amid the AI uptrend; and 2) GPM should improve from 2027 as high-margin AI server power supplies contribute more to sales and the home-appliance control business recovers.

Kehua (002335 CH, Buy)

Kehua is China's No. 2 UPS supplier, behind Huawei, and the world's No. 2 PCS supplier, behind Sungrow. We like the company for: 1) its leading position in UPS, a critical power-supply component, and exposure to strong China AIDC capex growth; 2) strong ESS installations in China and potential export growth; and 3) an attractive valuation.

AIDC cooling

Envicool (002837 CH, Buy)

Envicool is China's leading cooling-equipment supplier, with a comprehensive portfolio ranging from key components such as UQDs, manifolds, and CDUs to integrated liquid-cooling systems. We rate the company Buy, supported by: 1) robust liquid-cooling demand as chip and rack power densities rise; 2) a favorable product mix and relatively high overseas exposure; and 3) a strong position in the global cooling value chain, underpinned by certified product quality and technological capabilities.

Shenling (301018 CH, Buy)

Shenling is China's second-largest pure-play cooling-equipment supplier, serving data centers, power grids, energy-storage facilities, and other specialized applications. We rate the company Buy, supported by: 1) solid liquid-cooling demand as advanced chips become more compute-intensive; 2) potential upside from a recovery in domestic computing investment, supported by its long-standing relationship with Huawei; and 3) an undemanding valuation relative to domestic liquid-cooling peers.

AI equipment:

PCB equipment and drill bit

Han's Laser (002008 CH, Buy)

Han's Laser is China's largest AI PCB equipment manufacturer, with products spanning almost all PCB manufacturing processes. Its core offering covers PCB drilling equipment, from conventional mechanical drilling to laser drilling. The company is also China's largest laser-equipment manufacturer, serving consumer electronics, lithium batteries, semiconductors, and general manufacturing. We expect Han's Laser to benefit from rising AI-related PCB equipment demand and from stronger demand for laser equipment as consumer-electronics designs are upgraded. We therefore rate the company Buy.

Dtech (301377 CH/1377 HK, Buy)

Dtech is the world's largest PCB drill-bit manufacturer. We believe it is among the most direct beneficiaries of the accelerating AI PCB upgrade cycle, driven by rapid changes in server architecture. We like Dtech for: 1) direct exposure to growth in AI PCB content; 2) high visibility on resilient earnings growth over 2026-28E; 3) faster capacity expansion

than peers, supported by its vertically integrated drill-bit manufacturing platform; and 4) a longer growth runway than PCB drilling-equipment suppliers, as drill bits are a key consumable in PCB manufacturing. We therefore rate the company Buy.

Optical equipment

RoboTechnik (300757 CH, Buy)

Through its subsidiary ficonTEC, RoboTechnik is the world's No. 1 SiPh manufacturing-equipment vendor and our top pick in optical equipment. Our view is supported by: 1) its leading position in optical-transceiver and CPO equipment, with advantages in high-precision alignment and automation; 2) a multi-year AI optics investment cycle, with optical-transceiver shipments expected to grow at a 46% CAGR over 2025-30, driving revenue/net-profit CAGRs of 139%/310% over 2026-28E; and 3) accelerating CPO adoption, with the company serving as a key supplier in Nvidia's and TSMC's insertion-2 and insertion-3 testing workflows. We also expect the shift from PV automation to higher-margin optical equipment to improve profitability and cash flow.

Appendix

Exhibit 77: Stocks mentioned
Stocks mentioned

BofA Ticker	Bbg. ticker	Company name	Price	Rating
XUJKF	002028 CH	Sieyuan Electric	CNY 110.81	C-1-7
XKZNF	688676 CH	Hainan Jinpan	CNY 54.68	C-1-7
XKTIF	002270 CH	Huaming Equipment	CNY 18.73	B-1-7
XLODF	600406 CH	NARI Technology	CNY 22.74	B-1-7
CTATF	3750 HK	CATL	HKD 364.8	C-1-7
XCBIF	300274 CH	Sungrow Power	CNY 72.3	C-1-7
XDSDF	002353 CH	Jereh	CNY 91	C-1-7
DNGFF	1072 HK	Dongfang Electric	HKD 16.18	C-1-7
XANHF	603308 CH	Anhui Yingliu	CNY 33.68	C-1-7
WEICF	2338 HK	Weichai Power Co. Ltd	HKD 25.16	C-1-7
CYD	CYD US	Yuchai International	USD 25.3	C-1-7
XXLAF	002335 CH	Kehua Data	CNY 26.41	C-1-7
XMXLF	002851 CH	Shenzhen Megmeet	CNY 97.98	C-1-7
ZIJMF	2899 HK	Zijin Mining	HKD 30.7	C-1-7
CMCLF	3993 HK	CMOC Group Ltd	HKD 13.65	C-1-7
JIAXF	358 HK	Jiangxi Copper	HKD 30.26	C-1-8
XJDDF	1208 HK	MMG Limited	HKD 7.47	C-1-9
XXTTF	301511 CH	Defu Tech	CNY 75.99	C-1-7
XCKLF	002080 CH	Sinoma Sci & Tech	CNY 42.36	C-1-7
XZRIF	600176 CH	China Jushi	CNY 34.59	C-1-7
XJFJF	6680 HK	JL Mag	HKD 14.58	C-1-7
XYLXF	600522 CH	Jiangsu Zhongtian Technology	CNY 28.3	C-1-7
CGNMF	002428 CH	Yunnan Germanium	CNY 84.06	C-1-7
XPTIF	002837 CH	Envicool	CNY 45.15	C-1-7
XSRLF	301018 CH	Shenling	CNY 76.11	C-1-7
XYICF	300757 CH	RoboTechnik	CNY 448.03	C-1-7
XHQTF	002008 CH	Han's Laser	CNY 64.1	C-1-7
XNISF	1377 HK	Dtech Technology	HKD 192.6	C-1-7

Source: BofA Global Research

BofA GLOBAL RESEARCH

Exhibit 78: Abbreviation
Abbreviation

Abbreviation	Full name
1.6T	1.6 Terabits per Second
3.2T	3.2 Terabits per Second
800G	800 Gigabits per Second
800VDC	800-Volt Direct Current
a/c	Account for
AC	Alternating Current
AI	Artificial Intelligence
AIDC	Artificial Intelligence Data Center
ALD	Atomic Layer Deposition
AOI	Automated Optical Inspection
APF	Active Power Filter
ASIC	Application-Specific Integrated Circuit
ASP	Average Selling Price
ATS	Automatic Transfer Switch
$BaTiO_3$	Barium Titanate
BBU	Battery Backup Unit
BESS	Battery Energy Storage System
BRET	Bit Error Rate Tester
CAGR	Compound Annual Growth Rate
CCGT	Combined-Cycle Gas Turbine
CCL	Copper-Clad Laminate
CoWoP	Chip-on-Wafer-on-Platform
CPO	Co-Packaged Optics
CPU	Central Processing Unit
CPX	Compute Processor Complex
CRU	Clock Recovery Unit

Exhibit 78: Abbreviation
Abbreviation

Abbreviation	Full name
CSG	China Southern Power Grid
CSP	Cloud Service Provider
CTE	Coefficient of Thermal Expansion
CVD	Chemical Vapor Deposition
DC	Data Center / Direct Current
Df	Dissipation Factor
Dk	Dielectric Constant
DPU	Data Processing Unit
DSP	Digital Signal Processor
EER	Electrical Energy Router
EIC	Electronic Integrated Circuit
EML	Electro-Absorption Modulated Laser
ENTSO-E	European Network of Transmission System Operators for Electricity
ESL	Equivalent Series Inductance
ESR	Equivalent Series Resistance
ESS	Energy Storage System
EV	Electric Vehicle
FAU	Fiber array unit
FYP	Five-Year Plan
GaN	Gallium Nitride
GIL	Gas-Insulated Line
GIS	Gas-Insulated Switchgear
GPU	Graphics Processing Unit
GT	Gas Turbine
GW	Gigawatt
GWh	Gigawatt-Hour
HBM	High-Bandwidth Memory
HDI	High-Density Interconnect
HLC	High-Layer-Count PCB
HV	High Voltage
HVDC	High-Voltage Direct Current
HVLP	Hyper Very Low Profile
IC	Integrated Circuit
InP	Indium Phosphide
IPP / IPPs	Independent Power Producer(s)
KFM	Kisanfu Mining S.A.R.L.
kt	Kilotonne
ktpa	Kilotonnes per Annum
kV	Kilovolt
kW	Kilowatt
LFT	Low-Frequency Transformer
M&A	Mergers and Acquisitions
ME	Middle East
MLCC	Multilayer Ceramic Capacitor
MV	Medium Voltage
MW	Megawatt
MWh	Megawatt-Hour
NDRC	National Development and Reform Commission
NdFeB	Neodymium-Iron-Boron
NEHDT	New-Energy Heavy-Duty Truck
NF_3	Nitrogen Trifluoride
NIC	Network Interface Card
NPO	Near-Packaged Optics
OCGT	Open-Cycle Gas Turbine
OCS	Optical Circuit Switch
OECD	Organisation for Economic Co-operation and Development
OEMs	Original Equipment Manufacturers
OSAT	Outsourced Semiconductor Assembly and Test
OT	Optical transceivers
PCB	Printed Circuit Board
PCD	Polycrystalline Diamond
PCS	Power Conversion System
PDU	Power Distribution Unit
PIC	Photonic Integrated Circuit

Exhibit 78: Abbreviation
Abbreviation

Abbreviation	Full name
PTFE	Polytetrafluoroethylene
PUE	Power Usage Effectiveness
RTF	Reverse-Treated Foil
SGCC	State Grid Corporation of China
SiC	Silicon Carbide
SiPh	Silicon Photonics
SMM	Shanghai Metals Market
SOFC	Solid Oxide Fuel Cell
SST	Solid-State Transformer
TAM	Total Addressable Market
TC/RCs	Treatment Charges and Refining Charges
TFM	Tenke Fungurume Mining
TW	Terawatt
TWh	Terawatt-Hour
UHVAC	Ultra-High-Voltage Alternating Current
UHVDC	Ultra-High-Voltage Direct Current
UPS	Uninterruptible Power Supply
USGS	United States Geological Survey
VCB	Vacuum Circuit Breaker
VLP	Very Low Profile
WF_6	Tungsten Hexafluoride
XPU	Generic Term for Different Types of Processing Units

Source: BofA Global Research

BofA GLOBAL RESEARCH

Price objective basis & risk

Anhui Yingliu (XANHF)
We derive our PO of RMB67 per share for Yingliu by using the average of P/E and DCF valuation methods.

We derive a fair value of RMB63.5 from our DCF calculation, with FCF perpetuity growth assumption of 3%. For WACC, we use 2.5% risk-free rate and 11% market return (both in line with what we use for Chinese industrial equipment companies under our coverage) and 1.15 Beta (historical 2-year Beta) to calculate and derive the discount rate (WACC) of 10.4%.

We derive a fair value of RMB70.2, based on 60x P/E multiple over the 2027E EPS. Our target PER of 60x P/E multiple is 0.25SD above the company's past 10-year 12-month forward trading P/E.

Upside risks: faster gas turbine capacity expansion by clients leads to stronger component demand, price hike due to tight capacity of hot-section components, development of CJ series aero engines faster than expected

Downside risks: raw material price hike, slower-than-expected capacity expansion, unsuccessful product verification with new clients

CATL (CTATF)
We derive our PO of HK$630 for CATL from the average of DCF and EV/EBITDA valuation.

DCF valuation: A fair value of HK$658 is derived from our DCF calculation, with assumptions of 2% free-cashflow (FCF) perpetual growth and 12.1% weighted average cost of capital (WACC). For WACC, we use 1.1x equity beta, 3.0% risk-free rate and 12% market return, to calculate and derive the discount rate (WACC) of 12.1%.

EV/EBITDA valuation: A fair value of HK$602 is derived from 15x target EV/EBITDA over its 2027E EBITDA, largely in-line with the average of global leading manufacturing companies due to similar expected growth.

Risks: (1) slower-than-expected EV sales growth, (2) raw material price hikes, (3) market share loss due to fierce competition, (4) uncertainties and risks in overseas manufacturing and operations, and (5) foreign exchange rate fluctuations, (6) US tariff uncertainties.

China Jushi (XCKLF)
We value China Jushi using a DCF approach. Our PO of RMB96 is based on a 6.8% WACC and a 2% terminal growth rate. The relatively low WACC is driven by a low post-tax cost of debt of only 1.7%, supported by its state-owned enterprise status and a favorable debt mix within its capital structure.

Given Jushi's strong positioning in E-glass and the expected medium-term supply tightness, we forecast its earnings to grow significantly. Our PO implies 30x FY27 P/E, representing a significant discount to c.60x for other Chinese AI infrastructure and component players, and 38x for other glass fiber names, and is therefore justified in our view.

Upside risks: Faster-than-expected demand growth driving stronger pricing, successful penetration into premium E-glass products such as second-generation low-Dk, low-CTE, and Q-glass, improved efficiency or yield leading to lower production costs.
Downside risks: Slower-than-expected demand growth, increased supply of weaving

machines enabling additional capacity expansion and weakening pricing, slower R&D progress in next-generation products, including second-generation low-Dk and beyond.

CMOC Group Ltd (CMCLF)
Our H-share price objective (PO) of HKD22 is based on discounted cash flow (DCF), with a weighted-average cost of capital (WACC) of 8.3% and terminal rate of 1%.

Upside risks: Stronger commodities price, US Fed rate hike slowing, stronger solar/ wind installation and NEV production, earlier project ramp-up.

Downside risks: Metal prices decline, slower TFM/ KFM project construction progress, oversupply in the industry, weaker NEV growth, changes in battery material technology, exchange rate fluctuation, geopolitical risk.

Defu Tech (XJDDF)
Our PO of RMB118 is based on a P/E valuation methodology. We apply a 38x FY27E P/E, slightly lower than the average multiple of China PCB materials peers of 42x, because LiB foil will still dominate earnings in 2027E and only in 2028 do we expect higher margin PCB copper foil to become the primary earnings driver.

Upside risks: 1) faster-than-expected PCB copper foil demand growth, and 2) faster-than-expected LiB copper foil processing fee recovery.

Downside risks: 1) slower-than-expected PCB copper foil demand growth, 2) slower-than-expected LiB copper foil demand recovery, and 3) delays in Defu's capacity expansion or product qualification.

Dongfang Electric (DNGFF)
Our SOTP-based PO for DEC H share is HKD56. We assign 45x average 2026/27 EPS to gas turbine business, in line with Chinese gas turbine component peers average due to similar growth and 30x average 2026/27 EPS to non-gas turbine segment to factor in China's demand on energy security and independency.

Upside risks: 1) higher-than-expected China baseload power capex amid with countercyclical investment and energy stabilities, 2) stronger-than-expected policy for coal-fired flexibility upgrade, 3) faster-than-expected progress of large size gas turbine R&D and export.

Downside risks: 1) weaker-than-expected China baseload power capex, 2) slower-than-expected ASPs and margins recovery, 3) slower-than-expected progress of large size gas turbine R&D and export.

Dtech Technology (XNISF)
Our PO of HK$430 is derived from the average of DCF and P/E valuation methodologies. We believe DCF is a suitable valuation methodology as it better captures the company's net profit, working capital management, capex, and cash-flow generation. On the other hand, we also believe P/E is another suitable methodology as it captures the sustained earnings growth we anticipate the company will achieve.

DCF valuation: we derive a fair value of HK$430 from our DCF calculation, with FCF perpetuity growth assumption of 3%. Our WACC of 13.2% is based on 13.3% total cost of equity (i.e., risk-free rate of 3.5%, expected market return of 12% and 1.15x equity beta) and an after-tax cost of debt of 2.6%.

P/E valuation: we assign a target P/E of 45x to its 2027E EPS and derive a P/E-based fair value of HK$430. Our target P/E reflects a PEG of 0.5x, given the company's estimated earnings CAGR of 90% over 2026-28. In line with peers due to a similar structure.

Risks: (1) Slower-than-expected AI capex, (2) Slower-than-expected spec upgrade of AI PCB, (3) Market competition in PCB drill bit, and (4) Fluctuation of raw material prices & availability of high-end tungsten steel

Envicool (XPTIF)
We adopt P/E ratio in our valuation. Our PO of RMB76 is based on 2027E EPS estimates with a P/E of 85x, which equates to a 50% premium compared to peers' average. We believe the premium is justified by its 1) superior EPS growth versus peers, 2) the company's greater AIDC exposure, 3) integration into NVIDIA's supply chain, 4) higher proportion of products compared to peers in the growing liquid cooling market, 5) advanced technology such as MCCP.

Upside risks: 1) Better-than-expected BESS installation, 2) better-than-expected IDC and 5G station construction, 3) higher-than-expected EV penetration, 4) less fierce market competition.

Downside risks: 1) Weaker-than-expected BESS installation, 2) weaker-than-expected IDC and 5G station construction, 3) lower-than-expected EV penetration, 4) more fierce market competition.

Hainan Jinpan (XKZNF)
We assign 75x average 2027 P/E to Jinpan's artificial intelligence data center(AIDC) segment (in line with AIDC infrastructure peer's average level) and 33x 2027 P/E to the non-AIDC segment (largely in line with power equipment peers due to similar expected growth).

Downside risks: higher US/Europe tariff and trade disputes, slower pace of launching new overseas factories, fierce competition in low voltage transformer market, higher material cost, slower booking schedule.

Upside risks: lower tariff level, higher margin level, and faster booking schedule.

Han's Laser (XHQTF)
Our new PO of RMB185 is based on the average of DCF-derived and P/E-based valuations.

(1) DCF-based fair value: A new fair value of RMB175 is derived from our DCF calculation, with assumptions of 2% FCF perpetuity growth rate and 12.4% WACC.

(2) P/E-based fair value: We derive a new fair value of RMB195 on higher earnings estimates. We apply target P/E of 50x over 2027E EPS, close to 2SD above its 10-year historical average level to reflect the Apple supply chain upcycle and growth rate for the PCB segment.

Downside risks: 1) A slowdown in demand growth in consumer electronic products, 2) slower-than-expected progress in upgrade of cell phone manufacturing techniques, 3) lower-than-expected market share in 3D print equipment, 4) slower growth rate in PCB segment, 5) lower margin level and 5) intensified competition from new entrants, 6) slower progress in ultra-fast laser drilling equipment.

Huaming Equipment (XKTIF)
Our PO is based on 33x P/E over its average 2026/27 EPS, same as target level for its domestic peers with edge in overseas market.

Downside risks: 1) higher than expected tariff level, 2) weaker demand, 3) lower margin, 4) lower share in China and overseas markets, 5) higher materials cost and 6) FX loss.

Upside risks: 1) lower tariff level, 2) stronger demand in domestic market and overseas, 3) higher-than-expected market share

Jereh (XDSDF)
Our SOTP-based PO is RMB168 and blended PE of 26x, at 1 standard deviation above its historical average, reflecting the rising earnings contribution from the higher-growth, higher-margin gas turbine business.

We value its gas turbine business at 40x 2027E EPS, in line with the average valuation of global gas turbine peers, supported by persistent overseas power shortages, ongoing gas turbine supply tightness, and the company's strong order momentum in the premium US market.

We assign 12x 2027E EPS to non-gas turbine segment, broadly in line with the average of Chinese oil equipment peers, reflecting a weaker global oil capex outlook and potential earnings headwinds stemming from ongoing tensions in the Middle East.

Upside risks: 1)stronger-than-expected GT order intake driven by robust U.S. AIDC power demand 2)stronger GT pricing and faster O&M business ramp-up, supporting higher margins and a more diversified revenue mix 3) stronger oil and gas prices.

Downside risks: 1)weaker-than-expected gas turbine orders due to softer U.S. AIDC demand, delayed project approvals or easing supply constraints 2) weaker gas turbine pricing from intensifying competition 3)limited engine access 4) lower oil prices 5) FX losses 6)Middle East-related execution delays and 7) potential U.S. sanctions given its exposure to U.S. data center clients.

Jiangsu Zhongtian Technology (XJFJF)
Our price objective of RMB65 is based on DCF valuation, forecasting cash flow from 2026-35E. Our weighted average cost of capital (WACC) is estimated at 9%, assuming 3% terminal growth and company beta of 0.98.

Upside risks to our PO: 1) Higher than expected offshore wind installation, 2) Higher than expected overseas demand and acceptance of made-in-China cables, 3) Raw materials like copper price drop faster than expected, 4) Higher than expected demand for optical fiber cable and energy storage system

Downside risks to our PO: Lower than expected offshore wind power installation, 2) High reliance on SOEs' investment strategy, 3) Increasing trade barriers among China and other countries, 4) Raw materials like copper, shipping cost rise more than expected, 5) Lower than expected demand for optical fiber cable and energy storage system, 6) Privacy and national security risks in telecom industry.

Jiangxi Copper (JIAXF)
Our price objective of HK$50 is based on DCF valuation methodology. We apply a WACC of 7.9% and a terminal growth rate of 2.0%.

Upside risks to our PO: 1. Loosening monetary policy translates into end consumption growth, driving up prices of base metals. 2. Further progress at the company's mines' ramp-up projects, increasing its copper concentrate to a certain extent. 3. Brighter outlook on the company's processed copper products (mainly used in electrical vehicles). Downside risks to our PO: 1. Further delay in the Co.'s domestic mines' ramp-up plan. 2. Potential saturation within the copper wire & rods market. 3. Larger-than-expected losses due to the Co.'s derivative positions.

JL Mag (XZRIF)
Our H-share PO of HK$24 is derived from a DCF valuation, based on an 8.5% WACC and a 4% terminal growth rate.

Upside risks are: 1) faster-than-expected growth in humanoid robot demand, 2) quicker capacity ramp-up.

Downside risks are: 1) slower-than-expected humanoid robot market development, 2) tighter-than-expected export controls on rare earth products, 3) elevated volatility in rare earth prices, (4) substitution risk.

Kehua Data (XXLAF)
We derive our P/E-based PO of CNY38 with 40x 2027 PER, at around 1 standard deviation above its historical average P/E of 27x to factor in a higher earnings proportion from IDC products with a better sector margin vs. other businesses.
Upside risks: better-than-expected China AIDC capex, better-than-expected China BESS demand, stronger overseas expansion of ESS products, potential breakthrough in developed market AIDC providers, etc.
Downside risks: worse-than-expected China AIDC capex, intensive competition, rising lithium prices, foreign exchange volatility losses, etc.

MMG Limited (MMLTF)
Our DCF-based PO of HK$12.0 uses a WACC of 11.2% and terminal growth rate of 2%. The WACC is computed based on 3% risk-free rate and 10% market premium (aligning with our China metals coverage), 1.24 beta, 4.1% post-tax cost of debt.

Upside risks: copper price strength, faster-than-expected project expansion or M&As, stronger-than-expected cost performance
Downside risks: Geopolitical and social risk (Peru), project delays, copper price volatility.

NARI Technology (XLODF)
Our RMB27 PO is based on 24x average 26/27E PE, close to its average P/E since 2012 and in line with target PER for other domestic grid equipment companies.

Downside risks: 1) higher tariff level in its overseas markets, 2) slower-than-expected booking schedule, and 3) slower-than-expected pace of order growth.

Upside risks: 1) Higher-than-expected growth rate in the smart grid segment, 2) higher-than-expected margin from UHV investment, and 3) higher-than-expected dividend payout ratio, which may result into higher dividend yield.

RoboTechnik (XYICF)
Our RMB664 price objective is based on 30x 2028E P/E, largely in line with peer average. We believe the 30x multiple is justified by RoboTechnik's unique positioning in the global AI optical equipment supply chain, particularly in CPO-related applications. The valuation is further supported by the company's superior growth profile, including a 310% EPS CAGR over 2026-28E, an attractive 0.25x PEG ratio, ROE expansion to above 50% by 2028E, and margin upside driven by a richer product mix and higher ASPs from CPO- and OCS-related equipment deliveries.

Risks are: 1) slower-than-expected adoption and commercialization of CPO and OCS, 2) weaker-than-expected optical transceivers producers' capex cycle, 3) fiercer competition in automated transceivers manufacturing equipment, 4) prolonged weakness in the PV equipment industry.

Shenling (XSRLF)
Our PO of RMB120 is based on 63x 2027E PER, in line with the average multiple of China AI infrastructure peers, which we believe is justified by: 1) the company's close linkage to its key customer, a key beneficiary of AI capex-driven domestic substitution, and 2) its ongoing overseas expansion.

Upside risks: 1) Stronger-than-expected China DC capex expansion, 2) Robust demand from DC/industrial/specialized sectors amid macro/policy uncertainties, 3) Higher liquid cooling penetration, 4) GPM improvement from lower material/labor costs

Downside risks: 1) Weaker-than-expected China DC capex expansion, 2) Softer demand from DC/industrial/specialized sectors amid macro/policy uncertainties, 3) Lower liquid cooling penetration, 4) GPM pressure from higher material/labor costs, 5) Intensified competition.

Shenzhen Megmeet (XMXLF)
Our PO of RMB164 is based on SOTP method, with (1) 26x 2027E P/E of its legacy business, largely in line with the sector average, and (2) 40x 2027E P/E for its AI server power supply business, higher than the sector average, considering the higher-than-peer earnings growth of its AI server power supply business in 2026-28.

Risks: (1) Geopolitical risks from US-China trade conflicts, (2) weaker-than-expected sales of Nvidia's GB300/Rubin AI server, (3) slower-than-expected market-share expansion in AI server power supply, (4) slower overseas sales growth of home appliance control, (5) fluctuation of raw material prices, (6) tariff hike in the overseas market.

Sieyuan Electric (XUJKF)
Our RMB198.0 PO is based on 33x average 2026/27E P/E, 1SD above historical average to reflect our positive outlook about its overseas revenue growth.

Downside risk: higher-than-expected tariff in US and other areas, share loss in SGCC tendering, slower order and revenue growth, lower margin from overseas market

Upside risk: faster new order growth, higher margin level from overseas market, faster share gain in SGCC tendering

Sinoma Sci & Tech (XXTTF)
We value Sinoma S&T using DCF. Our PO of RMB88 is based on 6.0% WACC and 2% terminal growth rate. The low WACC is driven by cheap post-tax cost of debt at only 2.0%, thanks to its state-owned enterprise status, and higher debt mix in the capital structure.

Upside risks: Faster-than-expected demand growth driving up pricing, greater market share gains in premium products such as 2nd gen low-Dk, low-CTE and Q-glass special glass, better efficiency or yield improving cutting cost.
Downside risks: Slower-than-expected demand growth, increased qualified special glass supply reducing the market tightness and weaken pricing, slower R&D progress into newer generation of products such as 2nd gen low-Dk and beyond.

Sungrow Power (XCBIF)
We derive our P/E-based PO of RMB186 on 22x average 2026/27E EPS, which is on par with mainland China peers' average due to similar expected growth. Risks: 1) US tariff, 2) weaker demand, 3) lower margin, 4) lower share in China and overseas markets.

Weichai Power Co. Ltd (WEICF)
Our PO of HK$47.5 for Weichai H is based on SOTP methodology. In our valuation, we use 14x EV/EBIT on its 2H26 & 1H27 EBIT to value its cyclical segments including

engine, automobile, and components (including gearbox and axle). We use 11.5x EV/EBIT on 2H26 & 1H27 EBIT to value KION (Europe's largest and the world's second largest manufacturer of forklifts) based on historical average multiple, per our BofA European Capital Goods Research's estimates. We use 6x EV/EBIT on its 2H26 & 1H27 EBIT to value its agricultural machinery segment. We also use 4x 2028 EV/sales on its 2028 sales to value its SOFC system solutions.

Upside risks to our PO are faster infra FAI growth, Weichai's market share expansion, tightened control on HDT over-loading issue, faster elimination of NES IV HDTs, and higher-than-expected margin and market share of its LNG HDT engine.

Downside risks to our PO are slower-than-expected of property construction, slower growth in demand from logistics, lower-than-expected market share for fiercer competition and slower-than-expected sales growth in KION.

Yuchai International (CYD)
Our PO of USD66 is derived from 6x 2026-2027E EV/EBIT (USD/CNY at 6.8). Our target EV/EBIT multiple of 6x is 15% lower than its Chinese engines peers average. In our view, the target multiple is justified by Yuchai's faster industry earnings growth but lower-than-industry ROE of 9% in 2026 vs industry at 12%.

Downside risks: slower-than-expected downstream demand, intensifying competition of power generation engine in the domestic market, potential price hikes and supply disruption for raw materials and core components, and changes in regulations on emissions in China.

Upside risks: faster-than-expected market share gain in China, stronger-than-expected demand for power generator engines for data center in China and overseas, quicker-than-expected capacity expansion in power generation engines

Yunnan Germanium (XYLXF)
Our RMB126 price objective is based on 55x 2028E P/E, reflecting Yunnan Germanium's transition from a cyclical rare-metal producer to a strategic semiconductor-material supplier. While the valuation represents a premium to traditional minor-metal peers, we believe it is supported by the company's superior earnings growth profile (>200% EPS growth in 2027E and 60% in 2028E), low PEG ratio, ROE improving sharply, and increasing exposure to AI-driven optical infrastructure through InP substrates. As InP becomes the dominant profit contributor, we expect investors to increasingly benchmark the company against compound semiconductor and optical-material peers rather than commodity producers.

Risks are slower-than-expected InP demand growth, delays in customer qualification or capacity ramp-up, weaker-than-expected AI optical infrastructure spending, declining InP substrate prices as industry supply expands, intensified competition from domestic peers, and lower germanium prices or demand. In addition, execution risk on new expansion projects and changes in export-control or industry policies could affect earnings growth and valuation re-rating.

Zijin Mining (ZIJMF)
We derive our price objective of HK$43, based on a discounted cash flow (DCF) valuation methodology. We apply a weighted-average cost of capital (WACC) of 8% and a terminal growth rate of 2.0%.
Upside risks: 1. Loosening monetary policy translates into end consumption growth, driving up prices of base metals. In addition, precious metals prices stay elevated when real interest rates remain low. 2. Production addition delivers more smoothly than expected. 3. More effective cost controls.
Downside risks: 1. Production contribution from the capacity ramp-up comes in lower

than the expectation. 2. Worse-than-expected cost controls of the newly acquired assets. 3. Hike in financial costs due to further M&A activities. 4. Disruption of Kamoa-Kakula Mine.

Analyst Certification

We, Matty Zhao, Edward Leung, CFA, Fiona Liang, Gary Tsang, Ming Hsun Lee, CFA, Miriam Chan, CFA, Peter Wang, Summer Wang, CFA and Yibing Xia, hereby certify that the views each of us has expressed in this research report accurately reflect each of our respective personal views about the subject securities and issuers. We also certify that no part of our respective compensation was, is, or will be, directly or indirectly, related to the specific recommendations or view expressed in this research report.

APR - Industrials Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	Aboitiz Equity Ventures, Inc.	ABOIF	AEV PM	Julian Roxas
	AirTAC	ATCXF	1590 TT	Ming Hsun Lee, CFA
	Anhui Yingliu	XANHF	603308 CH	Fiona Liang
	Ayala Corporation	AYYLF	AC PM	Russ Toribio
	Centre Testing International	XCTEF	300012 CH	Ming Hsun Lee, CFA
	Daifuku	DAIUF	6383 JP	Kenjin Hotta
	Daikin Industries	DKILF	6367 JP	Kenjin Hotta
	Daikin Industries	DKILY	DKILY US	Kenjin Hotta
	Dtech Technology	XDTXF	301377 CH	Summer Wang, CFA
	Dtech Technology	XNISF	1377 HK	Summer Wang, CFA
	Envicool	XPTIF	002837 CH	Yibing Xia
	GeekPlus	BGTCF	2590 HK	Summer Wang, CFA
	Goldwind	XJNGF	2208 HK	Gary Tsang
	GTCAP	GTHOF	GTCAP PM	Russ Toribio
	Guangdong Songfa	XDZGF	603268 CH	Amy Han
	Guizhou Aviation	XNSZF	688239 CH	Fiona Liang
	Hainan Jinpan	XKZNF	688676 CH	Matty Zhao
	Hangcha Group	XXCLF	603298 CH	Summer Wang, CFA
	Hangzhou Great Star	XNZLF	002444 CH	Fiona Liang
	Hankuk Carbon	HKUCF	017960 KS	KJ Hwang
	Han's Laser	XHQTF	002008 CH	Summer Wang, CFA
	Hanwha Aerospace	SSASF	012450 KS	KJ Hwang
	HD Hyundai Electric	XPELF	267260 KS	Sun Jung Lee
	HD Hyundai Heavy Industries	XHZXF	329180 KS	KJ Hwang
	Himile	XMSVF	002595 CH	Fiona Liang
	Hiwin Technologies	HITEF	2049 TT	Ming Hsun Lee, CFA
	Huaming Equipment	XKTIF	002270 CH	Matty Zhao
	Hyosung Heavy Industries	XKRUF	298040 KS	Sun Jung Lee
	IHI	IHICF	7013 JP	Kenjin Hotta
	Inovance Technology	XMYRF	300124 CH	Ming Hsun Lee, CFA
	Japan Steel Works	JPSWF	5631 JP	Yuri Nishizaki
	Jiangsu Hengli Hydraulic Co., Ltd	XHDCF	601100 CH	Ming Hsun Lee, CFA
	Kawasaki Heavy Industries	KWHIF	7012 JP	Kenjin Hotta
	Keyence	KYCCF	6861 JP	Kenjin Hotta
	Korea Aerospace	XAQHF	047810 KS	KJ Hwang
	LIG Defense & Aerospace	XLQUF	079550 KS	KJ Hwang
	LS Electric	LGXSF	010120 KS	Sun Jung Lee
	Makita	MKEWF	6586 JP	Kenjin Hotta
	Mingyang Smart Energy	XLARF	601615 CH	Charlotte Xia
	Mitsubishi Heavy Industries	MHVYF	7011 JP	Kenjin Hotta
	Mitsui E&S	MIESF	7003 JP	Yuri Nishizaki
	MNC Solution	XZXRF	484870 KS	Dongho Shin
	Nabtesco	NCTKF	6268 JP	Kenjin Hotta
	Namura Shipbuilding	NMRSF	7014 JP	Yuri Nishizaki
	NARI Technology	XLODF	600406 CH	Matty Zhao
	OMRON Corporation	OMRNF	6645 JP	Kenjin Hotta
	OMRON Corporation	OMRNY	OMRNY US	Kenjin Hotta
	RoboTechnik	XYICF	300757 CH	Matty Zhao
	Samsung Heavy Industries	SMSHF	010140 KS	KJ Hwang
	Sany Heavy Industry Co., Ltd	SHVYF	600031 CH	Ming Hsun Lee, CFA
	Sany Heavy Industry Co., Ltd	XHEVF	6031 HK	Ming Hsun Lee, CFA
	Sany International	SNYYF	631 HK	Summer Wang, CFA
	SATS Limited	SPASF	SATS SP	Nathan Gee, CFA
	Shenling	XSRLF	301018 CH	Yibing Xia
	Shenzhen Megmeet	XMXLF	002851 CH	Fiona Liang
	Sieyuan Electric	XUJKF	002028 CH	Matty Zhao
	Sinexcel	XSBEF	300693 CH	Summer Wang, CFA
	Sinotruk Hong Kong Ltd	SHKLF	3808 HK	Fiona Liang
	SM Investments Corporation	SVTMF	SM PM	Russ Toribio
	SMC	SMECF	6273 JP	Kenjin Hotta
	SMC	SMCAY	SMCAY US	Kenjin Hotta
	ST Engineering	SGGKF	STE SP	Jessie Lo
	Suzhou Maxwell Technologies	XSMTF	300751 CH	Gary Tsang
	TGOOD	XQGDF	300001 CH	Ming Hsun Lee, CFA

APR - Industrials Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	THK	THKLF	6481 JP	Kenjin Hotta
	UBTECH	UBTRF	9880 HK	Ming Hsun Lee, CFA
	Weichai Power Co. Ltd	WEICF	2338 HK	Fiona Liang
	Weichai Power Co. Ltd	XWPEF	000338 CH	Fiona Liang
	XCMG Construction Machinery Co Ltd	XVPZF	000425 CH	Ming Hsun Lee, CFA
	XJ Electric	XJECF	000400 CH	Matty Zhao
	Yangzijiang Shipbuilding	YSHLF	YZJSGD SP	Amy Han
	Yaskawa Electric	YASKF	6506 JP	Kenjin Hotta
	Yuchai International	CYD	CYD US	Fiona Liang
	Zhejiang Supcon Technology	XDXFF	688777 CH	Ming Hsun Lee, CFA
	Zoomlion Heavy Industry	XLRYF	000157 CH	Ming Hsun Lee, CFA
	Zoomlion Heavy Industry	ZLIOF	1157 HK	Ming Hsun Lee, CFA
NEUTRAL				
	Anhui Heli	AHUHF	600761 CH	Summer Wang, CFA
	D&L Industries, Inc.	DALQF	DNL PM	Philip Albert Felix
	Dongguan Yiheda Automation	XDXGF	301029 CH	Ming Hsun Lee, CFA
	Fanuc	FANUF	6954 JP	Kenjin Hotta
	Fanuc	FANUY	FANUY US	Kenjin Hotta
	Kubota Corp	KUBTF	6326 JP	Kenjin Hotta
	Kubota Corp	KUBTY	KUBTY US	Kenjin Hotta
	Neway Valve	XNNVF	603699 CH	Summer Wang, CFA
	Zhuzhou CRRC Times Electric	ZHUZF	3898 HK	Ming Hsun Lee, CFA
	Zhuzhou CRRC Times Electric	XWCTF	688187 CH	Ming Hsun Lee, CFA
UNDERPERFORM				
	Alliance Global Group, Inc.	ALGGF	AGI PM	Russ Toribio
	CKD	CKDOF	6407 JP	Kenjin Hotta
	ESTUN Automation	XNJNF	002747 CH	Ming Hsun Lee, CFA
	ESTUN Automation	XZMTF	2715 HK	Ming Hsun Lee, CFA
	Goldwind	XJGWF	002202 CH	Gary Tsang
	Hitachi CM	HTCMF	6305 JP	Kenjin Hotta
	Hongfa Technology	XQXAF	600885 CH	Fiona Liang
	Komatsu	KMTUF	6301 JP	Kenjin Hotta
	Komatsu	KMTUY	KMTUY US	Kenjin Hotta
	L.K. Technology	XBQXF	558 HK	Joey Yang, CFA
	Leader Harmonious Drive Systems	XLHNF	688017 CH	Ming Hsun Lee, CFA
	LT Group, Inc.	LTGIF	LTG PM	Philip Albert Felix
	OPT Machine Vision	XOTVF	688686 CH	Summer Wang, CFA
	SCG Packaging PCL	XAFLF	SCGP TB	Komsun Suksumrun
	Shanghai Bochu Electronic	XWUTF	688188 CH	Fiona Liang
	Shenzhen SC New Energy	XENYF	300724 CH	Ming Hsun Lee, CFA
	SK Bend	SUKWF	014620 KS	KJ Hwang
	Unitree / Yushu Technology	XYAJF	688836 CH	Ming Hsun Lee, CFA

APR - Energy Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	China Shenhua	CUAEF	1088 HK	Matty Zhao
	China Shenhua	XCUAF	601088 CH	Matty Zhao
	COSL	CHOLF	2883 HK	Yiming Wang
	COSL	XCNRF	601808 CH	Yiming Wang
	Dongfang Electric	DNGFF	1072 HK	Peter Wang
	Dongfang Electric	XDNFF	600875 CH	Peter Wang
	Formosa Plastics	FSAPF	1301 TT	Sun Jung Lee
	GAIL Limited	XGLAF	GAIL IN	Bharat Subramanian
	GAIL Limited	GAILF	GAID LI	Bharat Subramanian
	GS Holdings	GSHDF	078930 KS	Sun Jung Lee
	Indorama Ventures Public Company Limited	INDMF	IVL TB	Komsun Suksumrun
	Indraprastha Gas	XTNFF	IGL IN	Bharat Subramanian
	Jereh	XDSDF	002353 CH	Peter Wang
	Kehua Data	XXLAF	002335 CH	Matty Zhao
	Kumho Petrochemical	KKMHF	011780 KS	Sun Jung Lee

APR - Energy Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	L&F	XLNTF	066970 KS	Taekyoung Ha
	LG Chem	LGCLF	051910 KS	Sun Jung Lee
	LG Energy Solution	XQIEF	373220 KS	Taekyoung Ha
	Mahanagar Gas	XMGRF	MAHGL IN	Bharat Subramanian
	Muyuan Foods	XMFXF	2714 HK	Matty Zhao
	Muyuan Foods	XMYFF	002714 CH	Matty Zhao
	Oil and Natural Gas Corporation Limited	XOFOF	ONGC IN	Bharat Subramanian
	PetroChina	PCCYF	857 HK	Matty Zhao
	Petrochina Co Ltd-A	XPCLF	601857 CH	Matty Zhao
	PI Industries	XOQTF	PI IN	Charul Agrawal
	PTT Exploration	PTXLF	PTTEP TB	Komsun Suksumrun
	PTT Global Chemical PLC	XPGOF	PTTGC TB	Komsun Suksumrun
	PTT pcl	PTTPF	PTT TB	Komsun Suksumrun
	Reliance Inds	XRFLF	RELIANCE IN	Sachin Salgaonkar
	Santos	STOSF	STO AU	Cameron Needham
	Sinopec	SNPMF	386 HK	Matty Zhao
	Sinopec -A	SNPAF	600028 CH	Matty Zhao
	Sinopec Engineering	SENGF	2386 HK	Yiming Wang
	SK Innovation	SKOVF	096770 KS	Sun Jung Lee
	S-Oil	SOOCF	010950 KS	Sun Jung Lee
	Star Petroleum Refining	XSJSF	SPRC TB	Komsun Suksumrun
	Sungrow Power	XCBIF	300274 CH	Matty Zhao
	Thai Oil - L	TOIJF	TOP TB	Komsun Suksumrun
	Wanhua Chemical	YWNHF	600309 CH	Yiming Wang
	W-Scope Chungju Plant	XWSWF	393890 KS	Taekyoung Ha
	Yankuang Energy	YZCHF	1171 HK	Matty Zhao
	Yankuang Energy	YZCIF	600188 CH	Matty Zhao
NEUTRAL				
	China Coal	CCOZF	1898 HK	Matty Zhao
	China Coal	XAWLF	601898 CH	Matty Zhao
	Formosa Chemicals & Fibre	FMSFF	1326 TT	Sun Jung Lee
	Hanwha Solutions Corp.	HNWFF	009830 KS	Sun Jung Lee
	Indian Oil Corporation Limited	IOCOF	IOCL IN	Bharat Subramanian
	Kunlun Energy - H	KUNUF	135 HK	Yiming Wang
	Lotte Chemical Corporation	XLCCF	011170 KS	Sun Jung Lee
	Nan Ya Plastics	NNYPF	1303 TT	Sun Jung Lee
	Oil India Limited	XLCRF	OINL IN	Bharat Subramanian
	People & Technology	XDFYF	137400 KS	Taekyoung Ha
	Petronet LNG	POLNF	PLNG IN	Bharat Subramanian
	PTT Oil and Retail Business	XNINF	OR TB	Komsun Suksumrun
	Woodside Energy Group	WOPEF	WDS AU	Cameron Needham
	Woodside Energy Group	WDS	WDS US	Cameron Needham
UNDERPERFORM				
	Bharat Petroleum Corporation Limited	XBPCF	BPCL IN	Bharat Subramanian
	Ecopro BM	XKURF	247540 KS	Taekyoung Ha
	Formosa Petrochemical	FPTCF	6505 TT	Sun Jung Lee
	Gujarat Energy	XGGRF	GUJENERG IN	Bharat Subramanian
	Hindustan Petroleum Corporation Limited	XHTPF	HPCL IN	Bharat Subramanian
	IRPC	IRPSF	IRPC TB	Komsun Suksumrun
	Petronas Chemicals	PECGF	PCHEM MK	Sun Jung Lee
	POSCO Future M	XTAMF	003670 KS	Taekyoung Ha
	Sinopec Shanghai Petrochemical Co	SPTJF	338 HK	Yiming Wang
	Sinopec Shanghai Petrochemical Co	SINFF	600688 CH	Yiming Wang
	SKC	XSXOF	011790 KS	Taekyoung Ha
	Solus Advanced Materials	XSOAF	336370 KS	Taekyoung Ha

APR - Basic Materials Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	Anhui Conch	AHCHF	914 HK	Matty Zhao
	Anhui Conch -A	AHCAF	600585 CH	Matty Zhao

APR - Basic Materials Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	Banpu Pcl	XNORF	BANPU TB	Komsun Suksumrun
	Baosteel	BAOSF	600019 CH	Matty Zhao
	CGN Mining	CGNMF	1164 HK	Miriam Chan, CFA
	Chalco	ALMMF	2600 HK	Matty Zhao
	Chalco -A	XAMTF	601600 CH	Matty Zhao
	China Hongqiao	CHHQF	1378 HK	Miriam Chan, CFA
	China Jushi	XCKLF	600176 CH	Edward Leung, CFA
	CMOC Group Ltd	CMCLF	3993 HK	Miriam Chan, CFA
	CMOC Group Ltd	XCMUF	603993 CH	Miriam Chan, CFA
	CNBM	CBUMF	3323 HK	Matty Zhao
	Defu Tech	XJDDF	301511 CH	Yibing Xia
	Ganfeng Lithium	GNENF	1772 HK	Matty Zhao
	Ganfeng Lithium	XMOZF	002460 CH	Matty Zhao
	Hoa Phat Group	XPHTF	HPG VN	Jessie Lo
	Huayou Cobalt	XZHCF	603799 CH	Miriam Chan, CFA
	Jiangxi Copper	JIAXF	358 HK	Matty Zhao
	Jiangxi Copper	JGXCF	600362 CH	Matty Zhao
	JL Mag	XJMRF	300748 CH	Yibing Xia
	JL Mag	XZRIF	6680 HK	Yibing Xia
	Lee & Man Paper	LMPMF	2314 HK	Edward Leung, CFA
	Lynas Rare Earths	LYSDY	LYSDY US	Chen Jiang
	MMG Limited	MMLTF	1208 HK	Miriam Chan, CFA
	Nickel Asia Corporation	NCKAF	NIKL PM	Philip Albert Felix
	Nine Dragons Paper	NDGPF	2689 HK	Matty Zhao
	POSCO Holdings	PKXFF	005490 KS	KJ Hwang
	POSCO Holdings	PKX	PKX US	KJ Hwang
	Putailai	XVHGF	603659 CH	Edward Leung, CFA
	SD Gold	SGOLF	600547 CH	Miriam Chan, CFA
	SD Gold	SDGMF	1787 HK	Miriam Chan, CFA
	Sinoma Sci & Tech	XXTTF	002080 CH	Edward Leung, CFA
	Tianqi Lithium	TQLCF	002466 CH	Matty Zhao
	Tianqi Lithium	XYECF	9696 HK	Matty Zhao
	Tinci Materials	XCVHF	002709 CH	Edward Leung, CFA
	United Tractors	PUTKF	UNTR IJ	Jessie Lo
	Xinyi Glass	XYIGF	868 HK	Matty Zhao
	Yunnan Energy	XLGFF	002812 CH	Edward Leung, CFA
	Yunnan Germanium	XYLXF	002428 CH	Miriam Chan, CFA
	Zhaojin Mining	ZHAOF	1818 HK	Miriam Chan, CFA
	Zijin Gold Int'l	ZJNGF	2259 HK	Matty Zhao
	Zijin Mining	ZIJMF	2899 HK	Matty Zhao
	Zijin Mining	XZNMF	601899 CH	Matty Zhao
NEUTRAL				
	FLAT Glass	FGSGF	6865 HK	Edward Leung, CFA
	FLAT Glass	XFLCF	601865 CH	Edward Leung, CFA
	Maanshan Iron & Steel	MAANF	323 HK	Miriam Chan, CFA
	Maanshan Iron & Steel -A	MISIF	600808 CH	Miriam Chan, CFA
	Press Metal	PSSMF	PMAH MK	Jessie Lo
	PT Merdeka Battery Materials	XDMBF	MBMA IJ	Jessie Lo
	Shenzhen Dynanonic	XJDCF	300769 CH	Edward Leung, CFA
	Shenzhen Senior	XHYRF	300568 CH	Edward Leung, CFA
UNDERPERFORM				
	Angang Steel	ANGGF	347 HK	Miriam Chan, CFA
	Angang Steel -A	AGGSF	000898 CH	Miriam Chan, CFA
	Arafura Rare Earths	ARAFF	ARU AU	Chen Jiang
	Chengxin Lithium Group	XCNGF	002240 CH	Miriam Chan, CFA
	China Resources Building Materials	CJRCF	1313 HK	Matty Zhao
	Hangzhou First Applied Material	XMLMF	603806 CH	Miriam Chan, CFA
	Kibing Group	XIOCF	601636 CH	Edward Leung, CFA
	Oriental Yuhong	XSUUF	002271 CH	Edward Leung, CFA
	Siam Cement	SCVQF	SCC TB	Komsun Suksumrun
	Sichuan Yahua Industrial Group	XVHEF	002497 CH	Miriam Chan, CFA
	Tongwei	XGWTF	600438 CH	Miriam Chan, CFA
	Xinyi Solar	XNYIF	968 HK	Edward Leung, CFA

APR - Basic Materials Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
RVW				
	Semirara Mining and Power Corporation	SMIRF	SCC PM	Julian Roxas

APR - Utilities Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	Aboitiz Power Corporation	ABZPF	AP PM	Julian Roxas
	B.Grimm Power	XVSGF	BGRIM TB	Narumon Ekasamut
	Beijing Enterprises Holdings Limited	BJINF	392 HK	Cissy Guan
	CGN Power	XALQF	003816 CH	Gary Tsang
	CGN Power	CGNWF	1816 HK	Gary Tsang
	China Everbright Environment Group	CHFFF	257 HK	Cissy Guan
	China Yangtze Power Co., Ltd	CYGZF	600900 CH	Gary Tsang
	CK Infrastructure Holdings	CKISF	1038 HK	Cissy Guan
	CLP Holdings	CLPHF	2 HK	Cissy Guan
	CLP Holdings - A	CLPHY	CLPHY US	Cissy Guan
	Doosan Enerbility	DOHIF	034020 KS	Taekyoung Ha
	ENN Energy Holdings Ltd	XNGSF	2688 HK	Gary Tsang
	First Gen Corporation	FSGCF	FGEN PM	Julian Roxas
	Global Power Synergy	XWODF	GPSC TB	Narumon Ekasamut
	Guangdong Investment	GGDVF	270 HK	Cissy Guan
	Guangxi Guiguan Electric Power Co Ltd	XGPEF	600236 CH	Charlotte Xia
	Gulf Development	XKRIF	GULF TB	Narumon Ekasamut
	Jiangsu Zhongtian Technology	XJFJF	600522 CH	Gary Tsang
	Jinko Solar	XJGCF	688223 CH	Gary Tsang
	Manila Electric Co.	MERVF	MER PM	Julian Roxas
	Manila Water	MWTCF	MWC PM	Julian Roxas
	NHPC Ltd	XNTHF	NHPC IN	Amish Shah, CFA
	Ningbo Orient Wires & Cables	XORRF	603606 CH	Gary Tsang
	NTPC Ltd	NTHPF	NTPC IN	Amish Shah, CFA
	Power Assets	HGKGF	6 HK	Cissy Guan
	Power Grid Corporation of India Ltd	XPPWF	PWGR IN	Amish Shah, CFA
	SDIC Power Holdings	XLGVF	600886 CH	Charlotte Xia
	Sembcorp Industries	SCRPF	SCI SP	Mcrid Wang
	Synergy Grid & Development Phils., Inc.	XSLIF	SGP PM	Julian Roxas
	Tenaga Nasional Berhad	TNABF	TNB MK	Mcrid Wang
	Xizi Clean Energy	XXIZF	002534 CH	Cissy Guan
NEUTRAL				
	China Power International	CPWIF	2380 HK	Charlotte Xia
	China Resources Gas Group Limited	CRGGF	1193 HK	Cissy Guan
	China Resources Power	CRPJF	836 HK	Gary Tsang
	Electricity Generating Company Ltd.	EYGGF	EGCO TB	Sirichai Chalokepunrat
	Hong Kong & China Gas	HOKCF	3 HK	Gary Tsang
	Korea Electric Power Corp	KEPLF	015760 KS	Taekyoung Ha
	Korea Electric Power Corp	KEP	KEP US	Taekyoung Ha
	Ratch Group PCL	XEGEF	RATCH TB	Sirichai Chalokepunrat
UNDERPERFORM				
	ACEN Corporation	ACPIF	ACEN PM	Julian Roxas
	Beijing Enterprises Water	BJWTF	371 HK	Cissy Guan
	China Datang Renewable	XGDRF	1798 HK	Gary Tsang
	China Gas Holdings Limited	CGHOF	384 HK	Cissy Guan
	China Longyuan Power Group Corp. Ltd.	CLPXF	916 HK	Gary Tsang
	China Longyuan Power Group Corp. Ltd.	XXRYF	001289 CH	Gary Tsang
	China Three Gorges Renewables	XWRFF	600905 CH	Charlotte Xia
	Huadian Power International	HPIFF	1071 HK	Charlotte Xia
	Huadian Power Intl Corp-A	HPIZF	600027 CH	Charlotte Xia
	Huaneng Power International	HUNGF	902 HK	Gary Tsang
	Huaneng Power-A	HUNWF	600011 CH	Gary Tsang
	JA Solar	XJSTF	002459 CH	Gary Tsang
	LONGi Green Energy Technology	XAQZF	601012 CH	Gary Tsang

APR - Utilities Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	Tata Power Company Ltd	XNSGF	TPWR IN	Amish Shah, CFA
	Trina Solar	XVSLF	688599 CH	Gary Tsang

APR - Autos Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	ARCHION	XKBNP	543A JP	Shiro Sakamaki
	Ashok Leyland	XDBVF	AL IN	Gunjan Prithyani
	Astra International	PTAIF	ASII IJ	Jessie Lo
	Ather Energy	XWSZF	ATHERENE IN	Gunjan Prithyani
	Bajaj Auto	XBJBF	BJAUT IN	Gunjan Prithyani
	Bethel	XBETF	603596 CH	Joey Yang, CFA
	Bharat Forge	XUUVF	BHFC IN	Gunjan Prithyani
	BYD	BYDDF	1211 HK	Ming Hsun Lee, CFA
	BYD	XYMPF	002594 CH	Ming Hsun Lee, CFA
	BYD	BYDDY	BYDDY US	Ming Hsun Lee, CFA
	CALB	XCLSF	3931 HK	Ming Hsun Lee, CFA
	CATL	XMOQF	300750 CH	Ming Hsun Lee, CFA
	CATL	CTATF	3750 HK	Ming Hsun Lee, CFA
	CFMOTO	XCZOF	603129 CH	Joey Yang, CFA
	Chery Auto	XCAVF	9973 HK	Joey Yang, CFA
	DENSO	DNZOF	6902 JP	Shiro Sakamaki
	Desay SV	XGDFF	002920 CH	Joey Yang, CFA
	EVE	XEVEF	300014 CH	Ming Hsun Lee, CFA
	Fuyao Glass	XFGIF	600660 CH	Jessie Lo
	Fuyao Glass	FIGIF	3606 HK	Jessie Lo
	Geely	GELYF	175 HK	Ming Hsun Lee, CFA
	GS Yuasa	GYUAF	6674 JP	Shiro Sakamaki
	Hankook Tire	XHTRF	161390 KS	KJ Hwang
	Hero Motocorp	HRHDF	HMCL IN	Gunjan Prithyani
	Hesai	HSAI	HSAI US	Jessie Lo
	Hesai	XHGZF	2525 HK	Jessie Lo
	HL Mando	XMADF	204320 KS	KJ Hwang
	Honda Motor	HMC	HMC US	Shiro Sakamaki
	Honda Motor	HNDAF	7267 JP	Shiro Sakamaki
	Hyundai Motor India	XZHZF	HYUNDAI IN	Gunjan Prithyani
	Isuzu Motors	ISUZF	7202 JP	Shiro Sakamaki
	Joyson	XQMWF	600699 CH	Fiona Liang
	Joyson	XJYSF	699 HK	Fiona Liang
	Kia Corp	KIMTF	000270 KS	KJ Hwang
	Koito Manufacturing	KOTMF	7276 JP	Shiro Sakamaki
	Leapmotor	ZJLMF	9863 HK	Ming Hsun Lee, CFA
	Mahindra & Mahindra (M&M)	XKQRF	MM IN	Gunjan Prithyani
	Minth	MNTHF	425 HK	Fiona Liang
	Momenta	XJAOF	6880 HK	Joey Yang, CFA
	Nexteer	NTXVF	1316 HK	Joey Yang, CFA
	NHK Spring	NHKGF	5991 JP	Shiro Sakamaki
	Ninebot	XNLTF	689009 CH	Joey Yang, CFA
	Ningbo Tuopu	XNCGF	601689 CH	Joey Yang, CFA
	Pony AI	PONY	PONY US	Ming Hsun Lee, CFA
	Pony AI	XNPYF	2026 HK	Ming Hsun Lee, CFA
	RoboSense	RBSTF	2498 HK	Joey Yang, CFA
	Shenzhen Kedali	XWCSF	002850 CH	Jessie Lo
	Shuanghuan Driveline	XZSHF	002472 CH	Fiona Liang
	Sona Comstar	XBNXF	SONACOMS IN	Gunjan Prithyani
	Stanley Electric	STAEF	6923 JP	Shiro Sakamaki
	Sumitomo Rubber Industries	SMTUF	5110 JP	Shiro Sakamaki
	Suzuki Motor	SZKMF	7269 JP	Shiro Sakamaki
	Tata Motors CV	XTMCF	TMCV IN	Gunjan Prithyani
	Toyota Motor	TOYOF	7203 JP	Shiro Sakamaki
	Toyota Motor	TM	TM US	Shiro Sakamaki
	WeRide	WRD	WRD US	Ming Hsun Lee, CFA

APR - Autos Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	WeRide	WRDIF	800 HK	Ming Hsun Lee, CFA
	Wuxi Lead	XZSRF	300450 CH	Ming Hsun Lee, CFA
	Wuxi Lead	XLFDF	470 HK	Ming Hsun Lee, CFA
	XPeng Inc	XPEV	XPEV US	Ming Hsun Lee, CFA
	XPeng Inc.	XPNGF	9868 HK	Ming Hsun Lee, CFA
	Yadea	YADGF	1585 HK	Joey Yang, CFA
NEUTRAL				
	Aisin Corporation	ASEKF	7259 JP	Shiro Sakamaki
	Bridgestone	BRDCF	5108 JP	Shiro Sakamaki
	Bridgestone	BRDCY	BRDCY US	Shiro Sakamaki
	Eicher Motors	XEIMF	EIM IN	Gunjan Prithyani
	Escorts Kubota	XSCOF	ESCORTS IN	Gunjan Prithyani
	Great Wall Motor	GWLLF	2333 HK	Ming Hsun Lee, CFA
	Hyundai Mobis	HYPLF	012330 KS	KJ Hwang
	Hyundai Motor	HYMLF	005380 KS	KJ Hwang
	Li Auto	LI	LI US	Ming Hsun Lee, CFA
	Li Auto	LAAOF	2015 HK	Ming Hsun Lee, CFA
	Maruti	MUDGF	MSIL IN	Gunjan Prithyani
	Mazda Motor	MZDAF	7261 JP	Shiro Sakamaki
	NIO	NIO	NIO US	Ming Hsun Lee, CFA
	NIO	NIOIF	9866 HK	Ming Hsun Lee, CFA
	NIO	XLTWF	NIO SP	Ming Hsun Lee, CFA
	SAMIL (Motherson International)	XMSUF	MOTHERSO IN	Gunjan Prithyani
	Subaru Corp	FUJHF	7270 JP	Shiro Sakamaki
	Subaru Corp	FUJHY	FUJHY US	Shiro Sakamaki
	Toyo Tire	TOTTF	5105 JP	Shiro Sakamaki
	Toyoda Gosei	TGOSF	7282 JP	Shiro Sakamaki
	TVS Motors	XFKMF	TVSL IN	Gunjan Prithyani
	VOYAH	VOYHF	7489 HK	Ming Hsun Lee, CFA
	Xingyu Automotive Lighting	XLIXF	601799 CH	Joey Yang, CFA
	Yokohama Rubber	YORUF	5101 JP	Shiro Sakamaki
	Yutong Bus	ZHYUF	600066 CH	Fiona Liang
UNDERPERFORM				
	Aima Technology	XIHBF	603529 CH	Joey Yang, CFA
	BAIC Motor	BMCLF	1958 HK	Ming Hsun Lee, CFA
	Ehang	EH	EH US	Fiona Liang
	GAC	XGNHF	601238 CH	Ming Hsun Lee, CFA
	Great Wall Motor	XGWMF	601633 CH	Ming Hsun Lee, CFA
	Guangzhou Automobile Group	GNZUF	2238 HK	Ming Hsun Lee, CFA
	Huayu	XHASF	600741 CH	Fiona Liang
	Hyundai Wia	XWHYF	011210 KS	KJ Hwang
	Jardine Cycle & Carriage	JCYCF	JCNC SP	Jessie Lo
	Mitsubishi Motors	MMTOF	7211 JP	Shiro Sakamaki
	Motherson Sumi Wiring	XMWDF	MSUMI IN	Gunjan Prithyani
	Nextage	NXAGF	3186 JP	Shiro Sakamaki
	Nissan Motor	NSANF	7201 JP	Shiro Sakamaki
	Nissan Motor	NSANY	NSANY US	Shiro Sakamaki
	Ola Electric Mobility Limited	XOLAF	OLAELEC IN	Gunjan Prithyani
	SAIC Motor	XZNJF	600104 CH	Ming Hsun Lee, CFA
	Seres	XSRSF	601127 CH	Joey Yang, CFA
	Seres	XSGLF	9927 HK	Joey Yang, CFA
	Tata Motors Passenger Vehicles Limited	XTTSF	TMPV IN	Gunjan Prithyani
	Toyota Boshoku	TDBOF	3116 JP	Shiro Sakamaki
	Yongda	CYYHF	3669 HK	Joey Yang, CFA
	Zhejiang Sanhua	XZSIF	002050 CH	Jessie Lo
	Zhejiang Sanhua	ZSINF	2050 HK	Jessie Lo
	Zhongsheng Auto	ZHSHF	881 HK	Joey Yang, CFA

Disclosures

Important Disclosures

Equity Investment Rating Distribution: Autos Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	74	54.81%	Buy	46	62.16%
Hold	30	22.22%	Hold	21	70.00%
Sell	31	22.96%	Sell	18	58.06%

Equity Investment Rating Distribution: Building Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	46	54.12%	Buy	31	67.39%
Hold	14	16.47%	Hold	10	71.43%
Sell	25	29.41%	Sell	14	56.00%

Equity Investment Rating Distribution: Electrical Equipment Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	18	66.67%	Buy	10	55.56%
Hold	3	11.11%	Hold	2	66.67%
Sell	6	22.22%	Sell	2	33.33%

Equity Investment Rating Distribution: Electronics Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	41	65.08%	Buy	20	48.78%
Hold	9	14.29%	Hold	1	11.11%
Sell	13	20.63%	Sell	7	53.85%

Equity Investment Rating Distribution: Energy Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	73	59.35%	Buy	60	82.19%
Hold	22	17.89%	Hold	19	86.36%
Sell	28	22.76%	Sell	22	78.57%

Equity Investment Rating Distribution: Engineering Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	4	80.00%	Buy	1	25.00%
Hold	1	20.00%	Hold	1	100.00%
Sell	0	0.00%	Sell	0	0.00%

Equity Investment Rating Distribution: Engineering & Construction Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	9	37.50%	Buy	6	66.67%
Hold	10	41.67%	Hold	4	40.00%
Sell	5	20.83%	Sell	3	60.00%

Equity Investment Rating Distribution: Industrials/Multi-Industry Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	76	67.86%	Buy	45	59.21%
Hold	11	9.82%	Hold	9	81.82%
Sell	25	22.32%	Sell	12	48.00%

Equity Investment Rating Distribution: Machinery/Diversified Manufacturing Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	45	60.81%	Buy	21	46.67%
Hold	12	16.22%	Hold	8	66.67%
Sell	17	22.97%	Sell	11	64.71%

Equity Investment Rating Distribution: Non-Ferrous Metals/Mining & Minerals Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	75	66.37%	Buy	50	66.67%
Hold	22	19.47%	Hold	16	72.73%
Sell	16	14.16%	Sell	9	56.25%

Equity Investment Rating Distribution: REITs (Real Estate Investment Trusts) Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	50	60.98%	Buy	39	78.00%
Hold	21	25.61%	Hold	18	85.71%
Sell	11	13.41%	Sell	11	100.00%

Equity Investment Rating Distribution: Technology Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	255	62.04%	Buy	157	61.57%
Hold	82	19.95%	Hold	46	56.10%
Sell	74	18.00%	Sell	31	41.89%

Equity Investment Rating Distribution: Telecommunications Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	58	55.24%	Buy	49	84.48%
Hold	23	21.90%	Hold	17	73.91%
Sell	24	22.86%	Sell	21	87.50%

Equity Investment Rating Distribution: Utilities Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	91	54.49%	Buy	72	79.12%
Hold	38	22.75%	Hold	34	89.47%
Sell	38	22.75%	Sell	21	55.26%

Equity Investment Rating Distribution: Global Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	2057	57.03%	Buy	1386	67.38%
Hold	806	22.35%	Hold	563	69.85%
Sell	744	20.63%	Sell	446	59.95%

[R1] Issuers that were investment banking clients of BofA Securities or one of its affiliates within the past 12 months. For purposes of this Investment Rating Distribution, the coverage universe includes only stocks. A stock rated Neutral is included as a Hold, and a stock rated Underperform is included as a Sell.

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Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster[R2]
Buy	≥ 10%	≤ 70%
Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

[R2]Ratings dispersions may vary from time to time where BofA Global Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

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